LIGHT*Sheer* ET/ST

Diode Laser System

SERVICE MANUAL

LUMENIS™
Enhancing Life. Advancing Technology.

1249 Quarry Lane, Suite 100
Pleasanton, CA 94566 USA
(925) 249-8000 - (877) LUMENIS

P/N 10-04719-00.AA

Please note that while every effort has been made to ensure that the data given is accurate, the information, figures, illustrations, tables, specifications, and schematics are subject to change without notice.

Please direct all inquiries about this manual to:

Lumenis, Inc.
1249 Quarry Lane, Suite 100
Pleasanton, CA 94566 USA
(925) 249-8000 Phone
(925) 249-8010 FAX

Trademark Acknowledgements:

Lumenis, the Lumenis logo, LightSheer, and the LightSheer logo are trademarks or registered trademarks of the Lumenis group of companies.

Cavicide® is the property of Metrex Research Company.

Virex™ is the property of S.C. Johnson Commercial Markets, Inc.

FieldMaster and Coherent are the property of Coherent, Inc.

Molectron is the property of Molectron Detector, Inc.

ColdFire® is the property of Motorola, Inc.

10-04719-00.AA

DISCLAIMER

Lumenis service manuals are written specifically for use by Lumenis service engineers and trained representatives who have received formal training in the servicing of Lumenis equipment, and by customers who have taken and passed a Lumenis certification training course for the equipment being serviced. Information on certification training courses offered to system owners can be obtained by contacting the Customer Service Department at the Lumenis – Pleasanton, CA, location, phone (925) 249-8005.

Lumenis does not accept responsibility for personal injury or property damage resulting form the servicing of Lumenis equipment by its customers or by third parties, except where such injury or property damage is a direct result of Lumenis' negligence. Customers, by accepting the service manual, agree to indemnify Lumenis against any claims alleging personal injury or property damage resulting from the servicing of Lumenis equipment by the customer or by third parties, except where such injury or property damage is a direct result of Lumenis' negligence. These limitations include situations where Lumenis personnel are advising customers on the repair of Lumenis equipment over the telephone.

The servicing of Lumenis equipment by persons who have not passed a Lumenis certification training course for that equipment will void Lumenis' product warranty.

Revision Information

This is the Rev. AA release of the LightSheer ET/ST Diode Laser System Service Manual. Each page of this manual is marked with a part number and revision level at the bottom. A Technical Note (TN) or Field Change Order (FCO) placed in Section 7 typically describes minor changes or updates to the system, parts list, or service procedures. Major changes to the system or service information will roll the revision of the Service Manual. Contact the Lumenis Customer Service Department to determine if this is the most current release of this service manual and for a complete list of Technical Notes.

Cover page, copyright page, disclaimer page, this page, table of contents page(s) revision level is that of the manual, with the following sections of the manual:

Manual P/N	Revision:	Date:	ECO#:	Description of Change(s):
10-04719-00	Rev. AA	Apr. 2003	03-0058	Initial Release

Table of Contents

1.0 General Information

1.1	Use of this Manual	1-1
1.2	Conventions Used in this Manual	1-1
1.3	Serial Number Format	1-2
1.4	Certification Compliance and Approvals	1-2
1.5	Service Information and FDA Compliance	1-3
1.6	Safety Precautions	1-6
1.7	System Specifications	1-9

2.0 Installation and Operation Information

2.1	Introduction	2-1
2.2	Inspection	2-1
2.3	Unpack System	2-1
2.4	Assembly Procedure	2-1
2.5	System Operation	2-3
2.6	Remote Interlock Switch Installation Procedure	2-3
2.7	Operational and Safety Checkout (Service Technician)	2-4
2.8	Periodic Maintenance	2-6
2.9	Return Shipping	2-7

3.0 Alignment, Adjustment and Calibration

3.1	Overview	3-1
3.2	Touch Screen Calibration	3-3
3.3	ADC Loop Calibration	3-3
3.4	Main Heat Exchanger Temperature Calibration	3-4
3.5	EPI (ChillTip) Temperature Calibration	3-4
3.6	Verify Heat Exchanger TE Coolers Current	3-5
3.7	Verify EPI TE Coolers Current	3-6
3.8	Calibrate Voltage Overcharge	3-6
3.9	Verify Laser Current (Pump)	3-7
3.10	Verify Pulse Width	3-8
3.11	Verify Array Over Temperature Bit	3-8
3.12	Output Energy Calibration	3-9

4.0 Theory of Operation

4.1	Introduction		4-1
4.2	User Interface		4-3
	4.2.1	Touch Sensitive Color Display Panel Assembly	4-3
	4.2.2	Footswitch (and Pneumatic Foot Pedal)	4-3
	4.2.3	Remote Interlock	4-4
	4.2.4	Key Switch	4-4
	4.2.5	Emergency Stop Switch	4-5
4.3	Handpiece Calibration Port		4-5
	4.3.1	Pocket (Holster)	4-6
	4.3.2	Proximity Sensor	4-6
	4.3.3	Diffuser Glass	4-7
	4.3.4	Photodiode (Energy Detector)	4-7
4.4	Software		4-8
	4.4.1	User Software Routines	4-8
		4.4.1.1 START-UP	4-8
		4.4.1.2 CALIBRATION	4-9
		4.4.1.3 TREATMENT	4-11
		4.4.1.4 SLEEP MODE (Screen Saver, Power Saver)	4-16
		4.4.1.5 INTERLOCK MODE	4-17
		4.4.1.6 FAULT STATE	4-18
	4.4.2	Contents of Persistent Memory (NVRAM)	4-21
	4.4.3	User Software Screens	4-22
	4.4.4	Service Software	4-24
		4.4.4.1 Access to Service Software	4-24
		4.4.4.2 Set-up Screen	4-28
		4.4.4.3 Diagnostic Screen	4-30
	4.4.5	Demo Software	4-34
4.5	Electrical Connection and Conditioning		4-35
	4.5.1	Facility Electrical Outlet	4-35
	4.5.2	Power Cord	4-35
	4.5.3	Power Input Module	4-36
	4.5.4	Main Power Switch (Circuit Breaker)	4-36
4.6	Power Supply and Laser Drive		4-37
	4.6.1	Lambda, and FET (Diode Current Control)	4-37
	4.6.2	Martek Power Supply	4-39
4.7	Control Electronics		4-40
	4.7.1	CPU Controller Board	4-40
	4.7.2	PWA Main Board	4-41
	4.7.3	Energy Meter Board	4-41

4.7.4 PWA Fan Speed Control Board 4-41
4.8 Cooling System 4-41
4.8.1 Overview 4-41
4.8.2 Components 4-42
4.9 Delivery System 4-45
4.9.1 Overview 4-45
4.9.2 Components 4-46

5.0 Troubleshooting

5.1 Overview 5-1
5.2 Interior Access and Part Location 5-2
5.2.1 Top Cover Removal 5-2
5.2.2 Component Location, Base Plate 5-3
5.2.3 Component Location, Top Cover 5-5
5.2.4 Component Location, Handpiece Assembly 5-5
5.3 Error Codes and Fault Isolation 5-8
5.4 Test Point and Connector Arrangement 5-15
5.5 Remove and Replace Procedures 5-27

6.0 Selected Part Numbers

6.1 Miscellaneous Parts / Quick List 6-1
6.2 Special Service Tools 6-1
6.3 Top Cover Assembly 6-2
6.4 Base Plate Assembly 6-4
6.5 Heat Exchanger Assembly and Attached Parts 6-6
6.6 Pump Assembly 6-8
6.7 Cooling Fan Assembly 6-9
6.8 Handpiece Assembly 6-10
6.9 Cable Harnesses 6-12

7.0 Technical Notes (TN) and Field Change Orders (FCO)

7.1 Introduction 7-1
7.2 Index of TN and FCO 7-1

8.0 Drawings and Schematics

Index of Drawings and Schematics 8-1
Simplified System Block Diagram 8-3
System Wiring Diagram (for systems with Lambda Power Supply) 8-4
System Wiring Diagram (for systems with Martek Power Supply) 8-5
CPU Controller Board Schematic 8-6
PWA Main Board Schematic 8-14
PWA FET Driver Board Schematic 8-17
PWA Fan Filter Schematic 8-18
PWA Energy Meter Board Schematic 8-19

Figures and Tables

Figure 1-1: Manufacturer Identification Label 1-2
Figure 1-2: "Danger" Label 1-3
Figure 1-3: Date of Manufacture Label 1-4
Figure 1-4: Aperture Label 1-4
Figure 1-5: Emergency Stop Symbol 1-4
Figure 1-6: Key Switch Position Symbols 1-5
Figure 1-7: Footswitch Symbol 1-5
Figure 1-8: Remote Interlock Symbol 1-5
Figure 2-1: Remote Interlock Wiring Diagram 2-4
Figure 4-1: LightSheer ET/ST Diode Laser System 4-1
Figure 4-2: Display Panel Assembly Connections 4-3
Figure 4-3: Footswitch and Remote Interlock Wiring Diagram 4-4
Figure 4-4: Key Switch and Emergency Stop Switch Wiring Diagram 4-5
Figure 4-5: Proximity Sensor Wiring Diagram 4-6
Figure 4-6: Photodiode Wiring Diagram 4-7
Figure 4-7: Loading Program Screen 4-23
Figure 4-8: Startup Screen 4-23
Figure 4-9: Warning Screen 4-24
Figure 4-10: Service Mode Activated 4-25
Figure 4-11: Screen Zones 4-26
Figure 4-12: Paused Screen 4-26
Figure 4-13: Mode Selection Screen 4-27
Figure 4-14: Setup Screen 4-28
Figure 4-15: Diagnostic Screen 4-30
Figure 4-16: Power Input Module, Circuit Breaker 4-36
Figure 4-17: Lambda Power Supply and FET Driver Board Block Diagram 4-38

Figure 4-18: FET Connection Diagram 4-39
Figure 4-19: Martek Power Supply Block Diagram 4-40
Figure 4-20: Cooling System Diagram 4-42
Figure 4-21: Handpiece Optical System 4-46
Figure 4-22: Laser Diode Array Optical Detail 4-47
Figure 4-23: Laser Diode Array Construction Detail 4-48
Figure 5-1: Top Cover Removal 5-3
Figure 5-2: Component Location, Base Plate 5-4
Figure 5-3: Component Location, Top Cover 5-5
Figure 5-4: Handpiece Covers 5-6
Figure 5-5: Component Location, Handpiece Assembly, Left Side View 5-7
Figure 5-6: Component Location, Handpiece Assembly, Right Side View 5-8
Figure 5-7: CPU Controller Board 5-16
Figure 5-8: PWA Main Board 5-18
Figure 5-9: PWA FET Driver Board 5-22
Figure 5-10: FET Connections 5-23
Figure 5-11: Lambda Power Supply 5-24
Figure 5-12: Martek Power Supply 5-25
Figure 5-13: Energy Meter Board 5-27

Table 1-1: LightSheer ST System Specifications 1-10
Table 1-2: LightSheer ET System Specifications 1-11
Table 3-1: LightSheer ET/ST Diode Laser System
Energy Meter Calibration Form (page 1 or 2) 3-14
Table 3-2: LightSheer ET/ST Diode Laser System
Energy Meter Calibration Form (page 2 or 2) 3-15
Table 4-1: Default Parameters 4-12
Table 4-2: Laser Treatment States 4-13
Table 4-3: Fault Codes 4-21
Table 4-4: Contents of Persistent Memory (NVRAM) 4-22
Table 4-5 Digital Input Bit Definitions 4-31
Table 4-6: Digital Output Bit Definitions 4-32
Table 4-7: Analog Input Channels 4-33
Table 5-1: Error Code Troubleshooting Table 5-15
Table 5-2: Controller Board Component Identifiers 5-17
Table 5-3: PWA Main Board Component Identifiers 5-21
Table 5-4: PWA FET Driver Board Component Identifiers 5-23
Table 5-5: FET Connections 5-23
Table 5-6: Lambda Power Supply Signal List 5-24
Table 5-7: Martek Power Supply Signal List 5-26
Table 5-8: Energy Meter Board Component Identifiers 5-27

(this page intentionally blank)

1.0 General Information

1.1 Use of this Manual

This manual contains service instructions for the Lumenis LightSheer ET/ST Diode Laser System. The content of this manual is intended solely for use by Lumenis Field Service Engineers, authorized service representatives, and Lumenis trained and certified customer technicians. Lumenis, Inc. cannot be responsible for service or repairs attempted by untrained or uncertified persons, and the use of this manual by such persons is prohibited.

This manual is to be used in conjunction with the Operator Manual for the Lumenis LightSheer ET/ST Diode Laser System, P/N 10-03656-04. The operator manual contains important information regarding instrument description, location of controls, specifications and normal operating procedures.

As necessary, Lumenis' Technical Support Department releases Technical Notes (TN) and Field Change Orders (FCO) for the Lumenis LightSheer ET/ST Diode Laser System. As they are released, the TN or FCO becomes part of this manual (Section 7).

1.2 Conventions Used in this Manual

Within the text, logic signals that are active low ("not") will appear with an exclamation preceding the signal name, as illustrated below.

!RESET

A “not” signal is "active," or true, when the logic level is low. When the logic signal !RESET is low, the reset loop is active and the reset circuitry is performing a reset function. When the logic level signal !RESET is high, the reset is not active and thus no reset transpires. On the schematic diagrams, such active low signals typically appear with a solid line above the signal name, or with a slash (/) before the signal name as illustrated below.

$\overline{\text{RESET}}$ or /RESET

The schematics in this manual do not include individual numbers for logic elements or operational amplifiers within a single component. For example, U1, which is shown below, contains two buffers. The output of the left buffer is referred to as U1-2 while the output of the right buffer is referred to as U1-4.

1.3 Serial Number Format

A each system is identified by a unique serial number located on the rear of the laser console. The serial number for the LightSheer ET/ST Diode Laser System is in the following format:

8xxx, or 9xxx

Where:

'8xxx' or '9xxx' is a number that identifies the frame during the production sequence. The four digit number is a unique serial number.

1.4 Certification of Compliance and Approvals

The LightSheer ET/ST Diode Laser System is designed and tested in accordance with Lumenis' procedures for self-certification for a CE mark.

The CE mark is a certification label that allows Lumenis to sell the LightSheer ET/ST Diode Laser System throughout the European Community. This label certifies that the LightSheer ET/ST Diode Laser System meets all regulations set forth by various countries in Europe. It is important to ensure that this label is on the system cover.

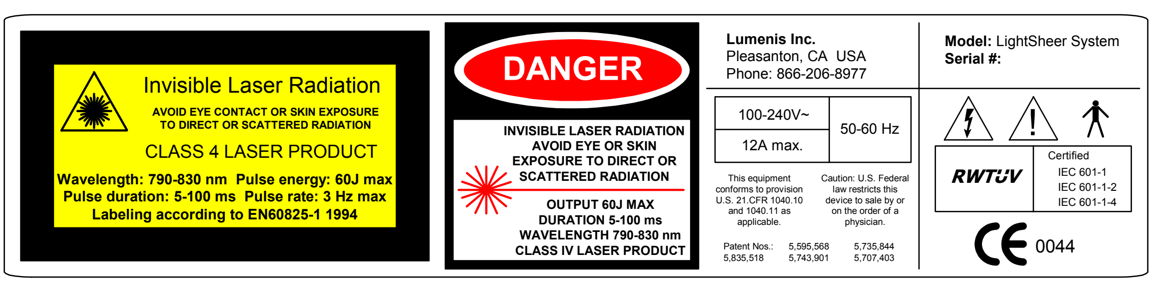


Figure 1-1: Manufacturer Identification Label (on rear panel)

The LightSheer ET/ST Diode Laser System complies with 21 CFR, Chapter I, Subchapter J, as administered by the Center for Devices and Radiological Health (CDRH) of the Food and Drug Administration (FDA). CE labeled devices comply with all appropriate performance standards as specified in Annex II of the Medical Device

Directive (MDD) 93/42/EEC. The LightSheer ET/ST Diode Laser System is classified as a Class IV laser by the CDRH and as a Class 4 laser by the European Standard EN 60825-1. **Caution – Use of controls or adjustments or performance or procedures other than those specified herein may result in hazardous laser radiation exposure.**

The system is available in one electrical configuration with a universal, 100-240 Vac, 50/60 Hz, single phase, 12 Amp (maximum) power input.

1.5 Service Information and FDA Compliance

In compliance to FDA regulations, brochures and specification sheets must include a reproduction of a complete warning logotype or warning statement as required on the product. Servicing information must contain the following:

- Procedures for service with appropriate warnings to avoid exposure (refer to Section 1.6 and 3.0).
- A schedule of maintenance to maintain the product in compliance (refer to Section 2.8).
- A list of controls that could increase the level of accessible radiation (refer to Section 4.4).
    - Fluence Up/Dn buttons on the LCD screen
    - Exposure Duration (Pulse Width Mode) Selection buttons on the LCD screen
    - [go to] Ready button on the LCD screen
- Identification of removable portions of protective housings (refer to Section 5.2).
- Procedures to avoid exposure (refer to Section 1.6).
- Reproductions of required labels and warnings (as illustrated on the following pages).

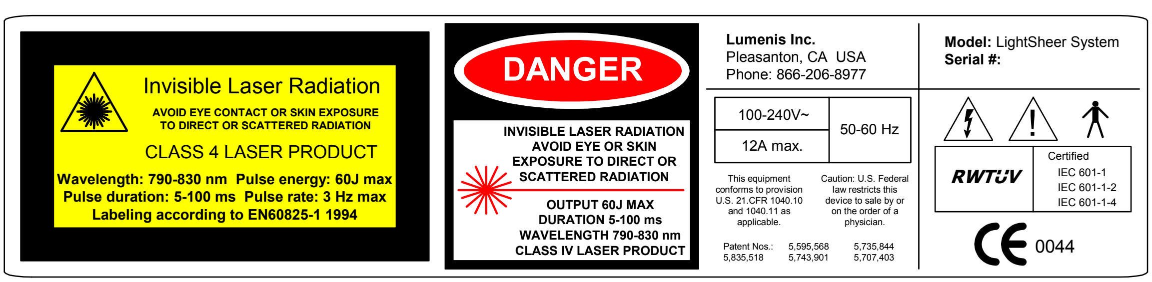


Figure 1-2: "Danger" Label
(on rear panel)

Figure 1-3: Date of Manufacture Label
(on bottom of base plate)

Figure 1-4: Aperture Label
(placed on handpiece near ChillTip)

Figure 1-5: Emergency Stop Symbol
(near emergency stop switch)

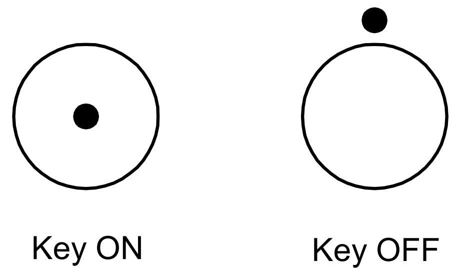


Figure 1-6: Key Switch Position Symbols
(near key switch)

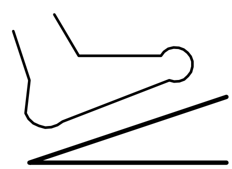


Figure 1-7: Footswitch Symbol
(placed near foot switch connector on rear panel)

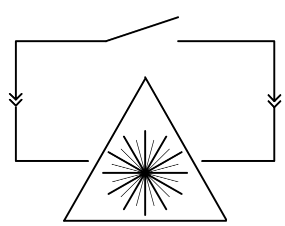


Figure 1-8: Remote Interlock Symbol
(placed near remote interlock connector on rear panel)

1.6 Safety Precautions

The following sections describe general safety issues regarding the use of the LightSheer ET/ST Diode Laser System.

The LightSheer ET/ST Diode Laser System is a medical instrument designed for safe and reliable operation. With proper operation and maintenance, trained, qualified practitioners can safely use the system. The Field Service Engineer and all other service personnel operating or maintaining this equipment should be familiar with the safety information provided in this chapter. The primary considerations should be for the safety of the patient, the operator and other personnel.

Note:

Restricted device! Federal law (USA) restricts the sales of this device by or on the order of a physician or any practitioner licensed by the law of the state in which he or she practices to use or order the use of the device.

Caution:

All LightSheer trained and certified service technicians operating or maintaining the LightSheer ET/ST Diode Laser System should read this manual thoroughly before attempting to operate or service this device.

Warnings:

- Any laser device in this classification can cause permanent injury or blindness if used improperly.
- High voltages and/or currents are present inside the console.
- Personnel operating laser systems should always be aware of hazards and possible dangers and should take appropriate safeguards as described in this manual.
- Use of controls or adjustments, or performance of procedures other than those specified herein may result in hazardous radiation exposure.

System Safety Measures

Considerable thought has been given during the design of the LightSheer ET/ST Diode Laser System, to maximize safety for both patient and operator. Following are some of the LightSheer ET/ST Diode Laser System safety features:

- A self-test of the electrical circuits takes place after the machine is turned on. The test circuits continuously monitor the operational system during treatment.
- The laser source is contained within the handpiece, which is normally in direct contact with the patient's skin. Laser light is emitted only at the aperture of the delivery device.
- The system can only deliver laser light at the activation of the “Ready” button, the depression of the foot pedal (activating an electronic shutter), and the triggering of the handpiece switch.
- Independent safety circuits disable the system to prevent under or over exposure treatments.
- An emergency shut-off knob expedites shutdown when necessary.
- A key switch prevents unauthorized activation of the device.
- A “beep” sounds with each treatment laser pulse.
- An emission indicator is displayed when the system is “Ready” to deliver treatment pulses.
- The inactivity timer feature switches the unit to “Standby” (idle) mode whenever the unit has not been used for five minutes.
- A current-limiting device (circuit breaker) is installed to protect the power source and console in case of an over-current event.

Optical Safety

- Laser light presents an eye hazard and a potential fire or burn hazard. Take all necessary precautions in areas where the LightSheer ET/ST Diode Laser System is being used.
- Exposure to direct or scattered radiation can result in permanent damage or blindness to the eyes and burns to the skin.
- The LightSheer ET/ST Diode Laser System emits intense laser light pulses. Prior to firing a pulse, always make sure that everyone in the room is protected against accidental exposure to this emission either directly from the delivery device (handpiece) or indirectly from a reflecting surface. To protect against eye damage and discomfort make sure that all personnel wear safety eyewear appropriate for the wavelength(s) in use.
- Never point the delivery device (handpiece) so that it discharges into free space. Make sure there is a suitable beam stop (wet towel) at or near the output aperture of the delivery device.

Eye Protection

The laser light emitted by the LightSheer ET/ST Diode Laser System is invisible to the human eye. Because the laser light energy cannot be seen, there is no visible indication of the primary or reflected beam. All persons in the area of the system must wear eye protection with Optical Density of 5.0 (OD 5) or greater at the wavelengths of 790-830 nm whenever the system is being used or serviced.

- **All persons potentially subject to laser exposure must wear appropriate eye protection whenever the main power and key switch are on.**
- **Never look directly into the laser aperture at the distal end of the handpiece, even if eye protection is being worn. Serious eye injury or blindness may result.**
- **Avoid directing the laser beam anywhere other than the calibration/storage port or intended treatment area. Stray laser light and reflection is always a potential hazard and may cause serious injury.**
- **Do not treat eyebrows, eyelashes, or other areas within the bony area surrounding the orbit. The light emitted by the LightSheer ET/ST Diode Laser System is capable of causing serious eye damage or blindness. For maximum safety, eye goggles (shields) must be worn by the patient for all facial treatments.**
- **Never permit reflective objects such as jewelry, watches, surgical instruments or mirrors to reflect the treatment light.**

Other Ocular Safety Considerations

- Identify the laser room clearly by posting approved laser safety signs in prominent locations.
- To prevent hazardous reflections within the treatment room, all mirrors and windows should be covered from the laser room. All windows must be covered to prevent laser light from escaping the laser room.
- Restrict entry to the laser room when the laser is in use. Allow access only to those personnel both essential to the procedure and well trained in laser safety.
- Ensure that the laser foot pedal is clean and working properly. Place the foot pedal where it will not be confused or mistaken for another piece of equipment.
- Do not attempt to remove the protective covers on the handpiece, which could allow exposure to high-intensity laser light.
- If unauthorized entry to the room is possible, use of the Remote Interlock is recommended. See Section 2.6, “Remote Interlock Switch Installation Procedure.”

Burn and Fire Hazard

- The absorption of optical energy raises the temperature of the absorbing material. Take precautions to reduce the risk of igniting combustible materials in the vicinity of the unit.
- If flammable solvents (alcohol or methanol) are used to clean any part of the system or delivery device, it must be allowed to dry thoroughly before use.
- Do not use the system in the presence of explosive anesthetics (such as ether) or other flammable materials.
- Never direct the output of the delivery device at anything other than a beam stop, the calibration port, or the targeted tissue.

Chemical Hazards

- The closed coolant loop contains approximately 240 ml (8 fluid ounces) of anti-freeze (a mixture of 1 part ethylene glycol to 4 parts distilled water). Ethylene glycol cannot be made non-poisonous. Immediately wipe up any leaks or spills, and dispose of contaminated materials properly.
- Flammable solvents (alcohol or methanol) may be used to clean parts of the system or delivery device. Use such solvents only in a well-ventilated area.

Electrical and Mechanical Safety

- No one other than Lumenis authorized service personnel may repair any part of the LightSheer ET/ST Diode Laser System within the protective cover. This includes making modifications or internal adjustments to the power supply, cooling system, power calibration, etc.
- Removing any cover creates an electrical safety hazard. The system contains high voltages and currents that are dangerous. Some components may retain an electric charge even after the power supply has been turned off. Always disconnect console from the power source before removing any cover.
- Never leave an operating or open unit unattended during system maintenance.
- The console is grounded through the grounding conductor in the power cable. This protective grounding is essential for safe operation.
- The console weighs approximately 27kg (60 pounds) and may cause injury if not properly handled when moved. The system is well balanced and is designed such that it can be moved without difficulty. Nevertheless, the unit should always be moved carefully and slowly.

1.7 System Specifications *(specifications subject to change without notice)*

(see following pages)

Laser

Laser Type	AlGaAs Laser Diode Array
Nominal Wavelength	800nm
Max. Peak Power (avg. during pulse)	1600W
Pulse Width	5-100ms
Repetition Rate	<= 1Hz
Pulse Energy Fluence	10-40J/cm^2
Spot Size	9mm x 9mm (0.35in. x 0.35in.)
Beam Divergence	20º x 20º nominal

Classifications

US FDA (Device) Classification	Class II Medical Device
CDRH Classification	Class IV Laser
MDD Classification	IIB
EN 60825-1 (Device) Classification	4
Operation Classification	Intermittent
Nominal Ocular Hazard Distance (NOHD)	50m (164 ft.)

Protective Eyewear

Optical Density @ 790-830nm	>=5

Electrical Power

Voltage	100-240 Vac, auto-ranging input
Phase	Single Phase, earth ground
Current	12 A Maximum
Frequency	50/60Hz

Cooling

Cooling Type	Forced Air (Convection)
Space Requirements	Minimum 15cm (6 in.) clearance around sides and rear. Console must be placed on hard, flat surface.

Physical Characteristics

Width	43cm (17 in.)
Depth	51cm (21 in.)
Height	38cm (15 in.)
Weight	27kg (60 lb.)

External Connections

Power Cord Length	2.4m (8 ft.)
Foot Pedal Hose Length	1.8m (6 ft.)
Operating Radius of Umbilical Cord	1.8m (6 ft.)

Environmental Requirements (Operating)

Pressure	90-110kPa (13-16 psi)
Temperature	15ºC to 27ºC (60ºF to 80ºF) - above dew point
Humidity	0-70% @ 27ºC (80ºF) non-condensing

Environmental Requirements (Non-Operating)

Pressure	Standard Commercial Shipping Altitude
Temperature	-10ºC to 55ºC (14ºF to 131ºF) - above dew point
Maximum Humidity	90% @ 55ºC (131ºF) non-condensing

Table 1-1: LightSheer ST System Specifications

Laser	
Laser Type	AlGaAs Laser Diode Array
Nominal Wavelength	800nm
Max. Peak Power (avg. during pulse)	1600W
Pulse Width	5-100ms
Repetition Rate	<= 2Hz
Pulse Energy Fluence	10-60J/cm^2
Spot Size	9mm x 9mm (0.35in. x 0.35in.)
Beam Divergence	20º x 20º nominal
Classifications	
US FDA (Device) Classification	Class II Medical Device
CDRH Classification	Class IV Laser
MDD Classification	IIB
EN 60825-1 (Device) Classification	4
Operation Classification	Intermittent
Nominal Ocular Hazard Distance (NOHD)	50m (164 ft.)
Protective Eyewear	
Optical Density @ 790-830nm	>=5
Electrical Power	
Voltage	100-240 Vac, auto-ranging input
Phase	Single Phase, earth ground
Current	12 A Maximum
Frequency	50/60Hz
Cooling	
Cooling Type	Forced Air (Convection)
Space Requirements	Minimum 15cm (6 in.) clearance around sides and rear. Console must be placed on hard, flat surface.
Physical Characteristics	
Width	43cm (17 in.)
Depth	51cm (21 in.)
Height	38cm (15 in.)
Weight	27kg (60 lb.)
External Connections	
Power Cord Length	2.4m (8 ft.)
Foot Pedal Hose Length	1.8m (6 ft.)
Operating Radius of Umbilical Cord	1.8m (6 ft.)
Environmental Requirements (Operating)	
Pressure	90-110kPa (13-16 psi)
Temperature	15ºC to 27ºC (60ºF to 80ºF) - above dew point
Humidity	0-70% @ 27ºC (80ºF) non-condensing
Environmental Requirements (Non-Operating)	
Pressure	Standard Commercial Shipping Altitude
Temperature	-10ºC to 55ºC (14ºF to 131ºF) - above dew point
Maximum Humidity	90% @ 55ºC (131ºF) non-condensing

Table 1-2: LightSheer ET System Specifications

(this page intentionally blank)

2.0 Installation and Operation Information

2.1 Introduction

The LightSheer ET/ST Diode Laser System is simple to install. The end user will typically perform the installation. Contact Customer Support if any questions or difficulties arise.

2.2 Inspection

Promptly inspect system packaging upon arrival. If visibly damaged, notify the shipping carrier immediately and contact Lumenis Customer Support. Allow package contents to reach room temperature before proceeding with the installation.

2.3 Unpack System

When ready to install the system, carefully open the shipping containers and check for missing or damaged items. Since the instrument is portable, it is not necessary to unpack and assemble the LightSheer ET/ST Diode Laser System in the treatment room.

Should service or repair be necessary, the compact design of the LightSheer ET/ST Diode Laser System allows for off-site (repair center) service by express shipping or for field repairs. Contact Lumenis Customer Support or your local Lumenis Service Representative for more information.

2.4 Assembly Procedure

Complete the following assembly steps:

1. Remove the straps that hold the box top to the bottom pallet. Lift the box top up and away. Resting on the pallet is a foam structure holding the console and the accessory box.

2. Open the accessory box and check the contents. The box should contain the foot pedal, a door sign, two in-service videos (different formats), several sets of protective eyewear, and the power cord.

3. Carefully remove the console from the foam. The handpiece must be uncoiled and placed in the handpiece pocket. Remove any residual packing materials and set the console upright on a sturdy, flat surface.

4. If desired, wire the remote interlock connector on the rear of the console to a customer supplied treatment door switch. Note that the LightSheer ET/ST Diode Laser System is shipped with a remote interlock jumper installed, allowing system operation. See Section 2.6 for remote interlock switch wiring information.

5. Verify electrical requirements have been met:

   100-240 Vac, single phase, 12 Amp (minimum), 50/60 Hz, grounded receptacle.

   Because electrical receptacles (outlets) are not internationally standardized, it may be necessary to rewire the LightSheer ET/ST Diode Laser System power cord to mate with the local wall receptacle. The ground wire (yellow/green) must be connected to earth (building) ground. Contact your Lumenis sales representative or Customer Support for any necessary assistance.

6. Locate the power cord and connect it to the input module socket on the rear of the console (below the main power switch). Connect the plug end of the power cord to a grounded receptacle.

7. Locate the foot pedal and firmly push the hose onto the footswitch nipple at the rear of the console. To test the foot pedal, a single audible "click" should be heard at the console when the pedal is depressed and held, and a second audible "click" should be heard when the pedal is released.

8. Locate the main power switch on the rear of the console. Turn the switch to the ON position and verify that the system starts and the color LCD display illuminates.

   If system does not start, reconnect power cord at console and receptacle and verify main power switch position. Verify that the electrical outlet is live by connecting a lamp or other small appliance. If necessary, refer to the Troubleshooting chart in the User Manual.

The installation procedure is now complete. Be sure to read, understand, and follow all instructions in the LightSheer ET/ST Diode Laser System User Manual before operating the system. Contact Lumenis Customer Support or your local Lumenis Service Representative if more information is required.

2.5 System Operation

For a complete description of System Operation, please refer to the LightSheer ET/ST Diode Laser System User Manual, P/N 10-03656-04.

2.6 Remote Interlock Switch Installation Procedure

The customer may elect to install a remote switch (door interlock device) to monitor the treatment room door where the LightSheer ET/ST Diode Laser System is used. If the interlocked door is opened, the system will not emit laser light. The remote interlock plug (a ¼ inch mono audio plug) must be connected to a customer supplied door switch, where the switch is closed when the door is closed. The door switch contact must be rated for at least 12Vdc/2A service, and the total length of cable from the door switch to the interlock plug should not exceed 5 m (16 ft).

If the LightSheer ET/ST Diode Laser System will be used in more that one room or at different sites, it will be necessary to wire each laser room door with a remote switch. Extra ¼ inch mono audio plugs are available for purchase.

The interlock plug must be wired as follows:

Tip: Connect the plug tip to the N.O. contact of the switch. When the door is closed, the N.O. switch contact to be connected to switch COM.

Shaft: Connect the plug shaft to switch COM.

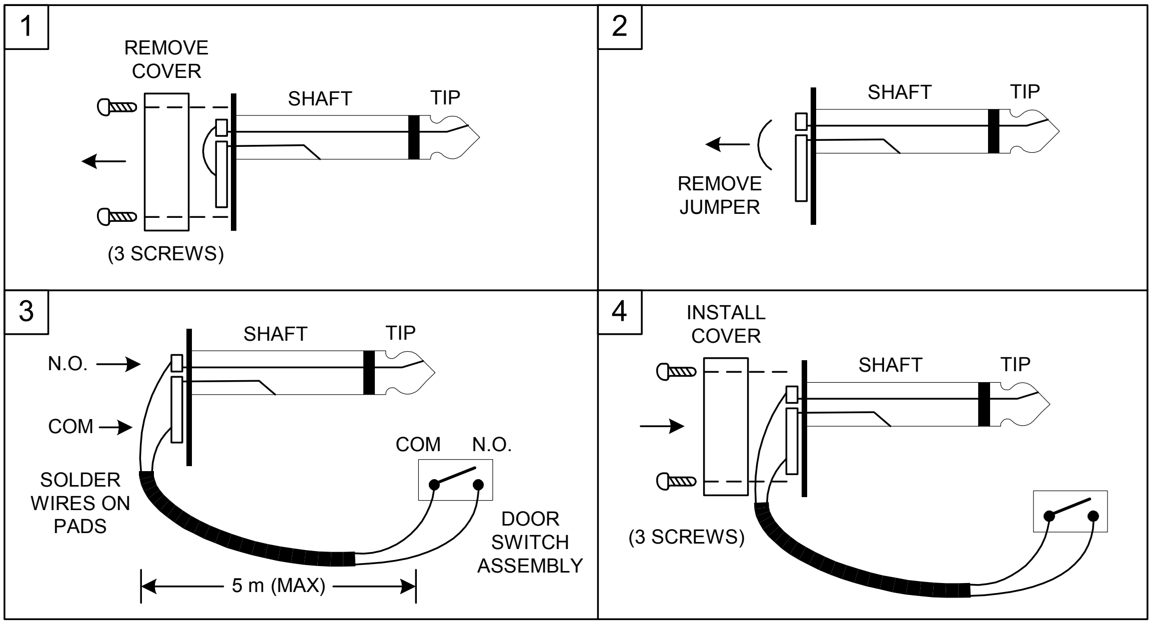


Figure 2-1: Remote Interlock Wiring Diagram

2.7 Operational and Safety Checkout (Service Technician)

Note: The following procedure is to be completed after any service work has been done to the LightSheer ET/ST Diode Laser System console or delivery device (handpiece).

1. Verify that the handpiece output lens and energy meter window are clean.

2. Turn the system on, and perform the User calibration.

3. Activate Service Mode software.

   Enter the Setup screen. Verify that the headroom for all modes is greater than or equal to 1.20 (for a new system or new handpiece only). If the headroom is less than 1.20, verify system calibration per procedure in Section 3.12.

4. Test external connections and switches. Turn system on and place in service diagnostics mode. To place system in diagnostic mode, refer to Section 4.4.2.1.

a. Press the emergency stop button and confirm that a check is placed in Digital Input box #1. Rotate the emergency stop button to release the interlock and verify the check mark disappears.

b. Turn the key to the OFF position and remove the key. Confirm that a check mark appears in Digital Input box #15. Insert the key, rotate to the ON position, and verify the key cannot be removed. Confirm that the check mark disappears.

c. Remove the remote interlock plug (at rear of console) and confirm that a check mark is placed in Digital Input box #9. Reinstall the remote interlock plug and confirm that the check mark disappears.

d. Connect the footswitch hose and depress the footswitch. Confirm that a check mark is placed in Digital Input box #3. Lift up on footswitch and confirm that the check mark disappears.

e. Place the handpiece in the calibration port and confirm that a check mark is placed in Digital Input box #4. Remove the handpiece from the calibration port and confirm that the check mark disappears.

6. Verify the voltage in Analog Input box #9 is 5.00 ± 0.10 Vdc.

7. Return system to “Normal Operation” software mode.

8. Turn system on and verify that normal User Mode software is shown. Verify that the Service “D” and “S” buttons are not displayed. Follow all on-screen prompts and verify that the user calibration runs without errors.

9. Verify Delivered Laser Energy

   a. Set up an external energy meter and detector head. Position the laser aperture (ChillTip) surface approximately 1cm from the detector surface.

   b. From User mode, turn the Chill Tip OFF. Verify that no condensation is on the outside of the sapphire lens.

   c. Set the Pulse Width mode to Auto. Select “Ready” and fire the laser into a calibrated energy detector. Working from low fluences up to the maximum, fire several shots at each setting and verify that the output, as measured by the external energy meter, is within ±10% of the (Requested Fluence * 0.81).

   d. Set the Pulse Width mode to 30ms. Select “Ready” and fire the laser into a calibrated energy detector. Working from low fluences up to the maximum, fire several shots at each setting and verify that the output, as measured by the external energy meter, is within ±10% of the (Requested Fluence * 0.81).

e. Set the Pulse Width mode to 100ms. Select “Ready” and fire the laser into a calibrated energy detector. Working from low fluences up to the maximum, fire several shots at each setting and verify that the output, as measured by the external energy meter, is within ±10% of the (Requested Fluence * 0.81).

f. If any pulse energy is outside of the ±10% window, perform the Output Energy Calibration procedure as described in Section 3.12.

10. Visually inspect exterior of console for damage or wear. Inspect power cord and footswitch tubing for damage or wear. Replace as required.

2.8 Periodic Maintenance

The only regular user-performed maintenance required for the LightSheer ET/ST Diode Laser System is periodic cleaning of the handpiece output lens, energy meter window, touch screen, and console.

To ensure proper system performance, it is required that a complete system inspection and energy meter calibration check be performed annually by a certified field service technician or by returning the system to the factory.

All service and repair should be performed only by the factory or by a Lumenis certified field service technician.

PM Check List:

1. Perform a visual inspection of the laser console exterior, footswitch, and power cord. Check for signs of wear, contamination, and visible damage.

2. Perform a visual inspection of the delivery system (umbilical cord and handpiece). Check for signs of wear, contamination, and visible damage. Repair or replace damaged parts.

3. With system turned off, clean …

   - handpiece output lens with lens tissue wetted with methanol
   - energy meter window with soft cloth wetted with methanol or commercial glass cleaner

* touch screen with soft cloth wetted with commercial glass cleaner (do not spray glass cleaner directly on display – spray glass cleaner on cloth)
* console with cloth wetted with Cavicide® or Virex™

4. Remove the console cover, being careful not to damage the Energy Meter and Proximity sensor wires and connectors, and perform a visual inspection of the internal components. Check for signs of wear, contamination, and visible damage. Repair or replace damaged parts. Remove any accumulation of dirt or dust (especially in and around the heat exchanger cooling fins).

5. Drain the main cooling loop, clean the coolant particle filter screen, and refill with fresh coolant.

6. Perform Section 2.7, Operational and Safety Checkout.

2.9 Return Shipping

The LightSheer ET/ST Diode Laser System has been designed for easy shipment in the event that off-site repairs are necessary. Please note that the packaging materials in which you received the LightSheer ET/ST Diode Laser System were intended for limited-use only. If you need to routinely transport or ship the system for any purpose, please contact Customer Support for new packaging containers. Lumenis is not responsible for damage due to mishandling or improper packaging.

(this page intentionally blank)

3.0 Alignment, Adjustment and Calibration

3.1 Overview

This section contains procedures for the field checks and adjustments required to maintain the LightSheer ET/ST Diode Laser System. The procedures assume the reader has successfully completed a Lumenis certification training course on the LightSheer ET/ST Diode Laser System. The information in this manual is subject to update by Technical Note (TN), Field Change Order (FCO), and service manual revisions.

Whenever any preventive maintenance is performed on the LightSheer ET/ST Diode Laser System, the system must pass all portions of the Operational and Safety Checkout (Section 2.7) before the system is returned to the user.

The procedures should be performed in the order presented, from beginning to end. If only a portion of the procedure is done, the servicing technician must consider the possible effect of doing just that portion of the procedure (i.e., there may be adjustments done earlier or later in the procedure that impact or are impacted by the portion performed).

Topic 3.2, Touch Screen Calibration – performed to establish two known positions on the touch panel from where all other positions can be resolved. Perform this procedure whenever the touch screen is replaced, or if the CPU Controller board is replaced.

Topic 3.3, ADC Loop Calibration – ADC Loop Calibration is performed to null undesired voltage offset(s) that may be introduced to the analog-to-digital converter by system electronics. Perform this procedure whenever the CPU Controller Board or Main Board is replaced.

Topic 3.4, Main Heat Exchanger Temperature Calibration – performed to establish proper set-points for the operation of the TEC devices, temperature sensors, and temperature control circuitry for the main heat exchanger loop. Perform this procedure whenever the main heat exchanger, temperature sensor(s), or Main Board is replaced.

Topic 3.5, EPI (ChillTip) Temperature Calibration – performed to establish proper set-points for the operation of the TEC devices, temperature sensors, and temperature control circuitry for the handpiece heat exchanger loop. Perform this procedure whenever the complete handpiece, Lockhart, or Main Board is replaced.

Topic 3.6, Verify Heat Exchanger TE Coolers Current – performed to verify proper set-points and operation of the TEC devices, temperature sensors, and temperature control circuitry for the main heat exchanger loop.

Topic 3.7, Verify EPI (ChillTip) TE Coolers Current – performed to verify proper set-points and operation of the TEC devices, temperature sensors, and temperature control circuitry for the handpiece heat exchanger loop.

Topic 3.8, Calibrate Voltage Overcharge – performed to establish an upper limit for the voltage overcharge comparator. Perform this procedure whenever the Main Board is replaced.

Topic 3.9, Verify Laser Current (Pump) – performed to verify that the current command, current drive, and current read back circuitry is functioning properly. Perform this procedure whenever the CPU Controller Board or the Main Board is replaced, or if the Martek power supply is replaced, or if the FET Driver Board is replaced on a system with a Lambda power supply.

Topic 3.10, Verify Pulse Width – performed to verify that the pulse width command, laser drive, and pulse width read back circuitry is functioning properly. Perform this procedure whenever the CPU Controller Board or Main Board is replaced, or if the Martek power supply is replaced.

Topic 3.11, Verify Array Over Temperature Bit – performed to verify that the diode array temperature measurement and comparator circuitry is functioning properly. Perform this procedure whenever the handpiece, CPU Controller Board or Main Board is replaced.

Topic 3.12, Output Energy Calibration – calibrates the internal energy detector and amplifier circuits in Volts per Joule (V/J) as compared to an external energy meter. Perform this procedure whenever the Energy Meter Board, Main Board, CPU Controller Board, or Handpiece is replaced.

REQUIRED TEST EQUIPMENT:

Digital Voltmeter with true RMS capability (such as Fluke 77) with probes
Temperature Probe (such as Fluke 80TK with surface sensor)
Current Probe (such as Fluke 80i-110s)
Power Meter Fixture (handpiece pocket assembly), P/N 90-04658-00
Firing Box (enclosed space to trap laser light)
Dummy Load (call Technical Support)
Frequency Counter
20 MHz Oscilloscope with probes
Laser Energy Measurement Equipment:
    Coherent FieldMaster GS with LM-40J Detector Head, or

Molectron EPM-1000 with PM150-50C Head.

3.2 Touch Screen Calibration

1. Verify that the display screen is clean. Clean as required.
2. Activate Service Mode software, as described in Section 4 (the service switch method is recommended).
3. Select the Diagnostic Screen, then touch the “CALIBRATE TOUCH” button at the bottom center of the screen.
4. The display will present an “X” as a target in the upper left corner of the display. Carefully touch the center of the “X” on the screen with a fingertip (not with fingernail, PDA stylus, pencil eraser, etc.).
5. The display will present an “X” as a target in the lower right corner of the display. Carefully touch the center of the “X” on the screen with a fingertip (not with fingernail, PDA stylus, pencil eraser, etc.).
6. If the screen calibration was successful, the Diagnostic Screen will reappear. Check the calibration by touching the screen in various locations and verify that the proper window/button responds appropriately.

3.3 ADC Loop Calibration

1. Connect voltmeter positive lead (red) to TP4 on the Main Board and the negative lead (black) to TP16 on the Main Board. Set voltmeter to read DC voltage.
2. Read the voltage displayed on the voltmeter. If V_{actual} is not 5.15V ± 0.15V, troubleshoot and repair the problem.
3. Set A/D Calibration Factor, $C_{A/D}$, to 0.000 in the diagnostic screen.
4. Record the value of 5V PS ($V_{computer}$) in the diagnostic screen. The voltage must be within V_{actual} ± 0.5V. If not, troubleshoot and repair the problem.
5. Calculate the A/D Calibration Factory using the following formula:

$$C_{A/D} = 17 * |V_{actual} - V_{computer}| / V_{computer}$$

$C_{A/D}$ must be between 0.70 to 1.30. If not, troubleshoot and repair the problem.

6. Set $C_{A/D}$ in the diagnostic screen to the calculated value.

7. Confirm that $V_{computer(new)}$ is calculated correctly by dividing $V_{computer(new)}$ by V_{actual}. If the result is not between 0.95 and 1.05, troubleshoot and repair the problem.

8. Disconnect the voltmeter leads.

3.4 Main Heat Exchanger Temperature Calibration

1. From the Diagnostic screen, verify to be off or turn off the EPI Cooler and EPI Set Point. Verify to be on or turn on the Heat Exchanger and wait a minimum of 5 minutes for the temperature to stabilize in the handpiece.

2. Slowly adjust the potentiometer R212 on the Main Board until the Heat Ex. Temp and Diode Temp are within 0.5°C of each other. Let the system stabilize for 2 minutes and recheck. Redo this step if they are outside of the limit. If the temperatures cannot be properly adjusted, troubleshoot and repair problem.

3. Allow the heat exchanger to run for an additional three minutes at that set point. Do not adjust the potentiometer during this time.

4. Confirm that both the Heat Ex. Temp and Diode Temp are still within 0.5°C of each other. If not, repeat the previous two steps.

5. Confirm that both the Heat Ex. Temp and the actual coolant temperature, as measured with an external temperature probe on the heat exchanger just above the bottom water fitting), are within 2°C of each other. If the temperatures are out of range, reposition the external temperature probe and re-measure.

6. If the software version is 4.01 or greater, double touch “Get Δ Temp” on the Diagnostic screen. This temperature difference must be 0 ± 1°C.

3.5 EPI (ChillTip) Temperature Calibration

1. Turn on the EPI Cooler. Wait for Heat Ex. Temp on the Diagnostic screen to stabilize.

2. Verify the Heat Exchanger and EPI cooler are on.

3. Turn on EPI Set Point and watch “EPI Cooler Hot” bit in the Digital Input section. As the EPI temperature in the ‘Analog In’ section decreases below 9.0°C, the EPI Cooler Hot bit should turn off. If the bit does not turn off, troubleshoot and repair the problem.

4. Verify EPI Cooler and EPI Set Point are on.

5. Measure the temperature of the ChillTip with and external temperature probe. Make certain the probe is flat against the ChillTip and the temperature has stabilized.

6. Slowly adjust the potentiometer R150 on the Main Board until the displayed EPI temperature and the measured ChillTip temperature are within 0.5°C of each other. Let system stabilize for 2 minutes and recheck. Redo this step if the temperatures are outside of the limit. If the displayed temperature cannot be properly adjusted, troubleshoot and repair the problem.

3.6 Verify Heat Exchanger TE Coolers Current

1. Turn off the heat exchanger, EPI cooler, and EPI Set Point.

2. Connect oscilloscope: Set channel 1 on the oscilloscope for a 1X probe and set trigger to auto mode. Set the channel 1 range to 50mV/div and sweep to 500µs/div. Set the current probe to 10 mV/A and check its zero adjustment. Using these settings will scale the scope display at 5.0A/div.

3. Clamp the current probe, with arrow pointing toward heat exchanger, to the three wires coming from the “HTEC+” J16 connector. Adjust the scope settings to obtain the best fit.

4. When the displayed heat exchanger temperature is greater than 20.0°C, turn on the heat exchanger and note the peak current draw (on oscilloscope). If the current is not within 15.0A to 18.0A, troubleshoot and repair the problem. Note that initially the TEC current is DC and becomes a pulse waveform when the temperature is close to or at the setpoint temperature.

5. Disconnect the current probe.

3.7 Verify EPI TE Coolers Current

Note: The heat exchanger must be on when the EPI TE cooler is operated.

1. Connect oscilloscope: Set channel 1 on the oscilloscope for a 1X probe and set trigger to auto mode. Set the channel 1 range to 100mV/div and sweep to 500µs/div. Set the current probe to 100mV/A and check its zero adjustment. This will scale the scope display at 1.0A/div.

2. Clamp the current probe (with arrow pointing toward handpiece) to the yellow wire from pin 1 of the "TO HAND PIECE" J17 connector. Adjust the scope settings to obtain the best fit.

3. Verify the Heat Exchanger is on. With EPI cooler already ON, enable EPI Set Point bit.

4. When the displayed EPI Temp is 19.0°C ±1.0°C, note the peak current. If the current is less than 3.25A, troubleshoot and repair the problem.

5. Turn off the EPI cooler, EPI Set Point bit, and Heat Exchanger.

6. Disconnect the current probe and switch it off.

3.8 Calibrate Voltage Overcharge

Measure the voltage between TP10 and ground symbol test points on the Main Board. Adjust the potentiometer R79 per the following chart. If the voltage is out of specification, troubleshoot and repair the problem.

System	Max Rated Fluence	TP10	Tolerance
ST	40	3.51V	± 0.05V
ET	60	4.96V	± 0.05V

Note: After any potentiometer adjustment, cover the adjustment screw with potentiometer dope or nail polish.

3.9 Verify Laser Current (Pump)

1. Connect the Dummy Load to J6 on the power supply. If a dummy load is not available, use the handpiece placed in a firing box and wear protective eyewear.

2. From the diagnostic screen, turn on the High Voltage, pulse width limit to 7 ms, pulse width to 5 ms and set the current to 10A. Without the footswitch depressed, verify on the Diagnostic screen that the shutter is closed.

3. Turn on the key switch and step on the footswitch. Verify on the Diagnostic screen that the shutter is open. Place the handpiece in a firing box, set the Fire Enable switch to the Enable position, and then press the trigger. The buzzer should sound.

4. Observe the monitor currents on the Diagnostic Screen and note the readings. If the currents are not within 8.6A to 11.4A, troubleshoot and repair the problem.

5. In the Diagnostic screen, set the current to 35A, pulse width limit to 7ms, and pulse width to 5ms. Press the trigger. The buzzer should sound.

6. Observe the monitor currents in the Diagnostic Screen and note the readings. If the currents are not within 32.6A to 37.4A, troubleshoot and repair the problem.

7. In the Diagnostic screen, set the current to 26A, pulse width limit to 32ms, and pulse width to 30ms and then press the trigger. The buzzer should sound.

8. Observe the monitor currents in the diagnostic screen and note the readings. If the currents are not within 23.0A to 28.1A, troubleshoot and repair the problem.

9. Turn off the high voltage bit, set Fire Enable to Disabled, and verify that the clamp is removed from the foot switch.

10. Disconnect the dummy load from the 6 pin connector on the power supply.

WARNING! DURING THE FOLLOWING STEPS, HIGH VOLTAGE MAY BE PRESENT ON THE 6 PIN CONNECTOR!

11. Connect the 6-pin connector from the handpiece to the “TO LASER” power supply connector from the power supply.

12. In the diagnostic screen, set the current to 10A, pulse width to 10ms, and pulse width limit to 12ms. Turn on the high voltage and heat exchanger bits, set the Fire Enable to the Enable position.

13. Put on the protective laser eyeglasses. Insert the handpiece into a firing box, press the foot switch down, and press the trigger until a single beep is heard. Release the trigger and let up on the footswitch, in that order.

14. Note the Current Mon A and B from the Analog In section on the Diagnostic screen. If both currents are not between 8.6A and 11.4A, troubleshoot and repair the problem.

3.10 Verify Pulse Width

1. Connect a frequency counter to TP22 on the Main Board and its ground to any test point with the triangular ground symbol. Set the counter to measure pulse width. Set the current to 5A and turn the high voltage on. Turn the key switch on, and then clamp the footswitch in the pressed position or step on the footswitch and hold it down.

2. In the Diagnostic screen, set pulse width limit to 7ms, pulse width to 5ms, and then press the trigger. The buzzer should sound and the frequency counter should register a new reading. The reading should be equal to the setpoint ± 0.05ms. Note the pulse width on the frequency counter. If the reading is out of specification, troubleshoot and repair the problem.

3. Repeat the previous step with pulse width limits of 22ms, 52ms, and 102ms and pulse widths of 20ms, 50ms, and 100ms, respectively.

3.11 Verify Array Over Temperature Bit

Attention: Technician must attend this test to prevent the array back plane cooler from overheating.

1. From the diagnostic screen, set current to 6.0A, pulse width to 30ms, and pulse width limit to 32ms.

2. Verify to be OFF or turn OFF the Heat Exchanger, EPI cooler, and EPI Set Point. Switch on High Voltage, turn key to the on position, and set Fire Enable to the Enable position.

3. Put on the protective laser eyeglasses. Place the handpiece in a firing box. While monitoring the diode temperature, clamp or otherwise hold the footswitch in the pressed position, press and hold the handpiece trigger. After approximately 75 shots,

the diode temperature should increase beyond 31°C. When the diode temperature reaches 31°C, the Array Overtemp bit (digital input bit #11) should turn on and the laser should stop firing. If the temperature exceeds 35°C without the Overtemp bit turning on, release the trigger, troubleshoot and repair the problem. If the temperature does not increase beyond 31°C, contact Technical Support.

4. Turn the heat exchanger ON.

5. When the Heat Ex. Temp has returned to 19°C ± 1.0°C, measure the thermistor voltage at test point TP17 on Main Board. If the voltage is not between 2.81V and 3.11V, troubleshoot and repair the problem.

3.12 Output Energy Calibration

The instructions to calibrate, confirm calibration, and perform a system calibration (automatic) are described below. A Calibration Form is attached at the end of this section, which may be copied, completed and filed with the LightSheer ET/ST Diode Laser System records. To accurately calibrate the LightSheer ET/ST Diode Laser System, verify that all test and measurement equipment used is currently in calibration.

The words "External Meter" are used to describe either a Coherent FieldMaster GS with LM-40J Detector Head, or a Molectron EPM-1000 with PM150-50C Head, or equivalent.

Verify that proper safety eyewear is worn by all present during the entire procedure.

1. Diagnostic Screen Settings (Set-up):

   a. Use glass cleaner and a clean soft cloth to clean the handpiece tip. Verify that the tip is clean.

   b. Remove calibration pocket and thoroughly clean the protective window above the energy meter with glass cleaner and a clean soft cloth. Blow out any loose particles with compressed air or nitrogen. Re-install the calibration pocket.

   c. Turn on the external energy meter.

   d. Turn on the LightSheer and activate Service software. Then turn the key on and acknowledge the warning screen to reach the Calibrate screen.

   e. Press the [S] button to enter the setup screen. Set the detector calibration value to $10J/cm^2/V$. Reset to $10J/cm^2/V$ if needed. Press [Return].

f. Press the [D] button to enter the Diagnostic screen. Turn ChillTip off. Set the current by pressing the [Current SP] field and entering the value from the following chart into the pop-up menu. Enter the pulse width of 0.020s. Set the pulse width limit timer to 0.022s. Set Fire Enable to "Enabled."

Handpiece	ET/ST
70-Bar	32.3A

g. Wait until the "Diode Temp" is between 17.0°C and 22.0°C.

2. Internal Energy Meter (Fire Laser Shots)

   a. Verify the security of the Class 4 laser controlled area and put on approved laser safety eyewear.

   b. Ensure that the handpiece is fully seated in the calibration port on the LightSheer system.

   c. Fire a single pulse into the calibration port while keeping the handpiece fully seated and read the fluence on the energy meter entry on the shot data line in the shot data window. The energy meter fluence is the last data item on the diagnostic shot line. Record the value.

   d. Repeat for (4) more shots, recording the energy meter fluence after each shot.

   e. Calculate and record the average by taking the sum of the five shots divided by 5. The fluence average should be between 32.0J/cm^2 to 36.0J/cm^2. If not, adjust R138 on the Main Board so that the fluence values are displayed as 34.0 ± 1.0J/cm^2. Repeat from Step c.

   Note: After any potentiometer adjustment, cover the adjustment screw with potentiometer dope or nail polish.

   f. Calculate and record the ±5% deviation limits. To calculate limits, multiply the average fluence by 0.95 (for lower limit) and 1.05 (for higher limit).

   g. Confirm that all five fluences fall within the limits.

3. External Energy Meter (Fire Laser Shots)

   a. Record the serial numbers of the external energy meter and probe in the External Energy Meter section of the Energy Meter Calibration Data Sheet.

   b. Verify the security of the Class 4 laser controlled area and put on approved laser safety eyewear.

c. Ensure that the handpiece is fully seated in the firing box, or centered over the detector head, with approximately 13mm (½ inch) space between the detector surface and the ChillTip.

d. Repeat the following steps five times:

1. Zero the external energy meter display by pressing the "Zero" button.

2. Fire a single pulse into the external energy meter probe while keeping the handpiece steady.

3. Record the pulse energy in Joules from the external energy meter display.

NOTE: If using a Molectron energy meter, after firing each pulse, wait until the trigger light extinguishes before resetting the meter to zero and firing additional shots.

e. Calculate and record the average by taking the sum of the five shots divided by 5. The energy average should be between 22.0J to 38.6J.

f. Calculate and record the ±10% deviation limits. To calculate limits, multiply the average fluence by 0.90 (for lower limit) and 1.10 (for higher limit).

g. Confirm that all five energies fall within the limits.

4. Calibration Coefficient (Calculation)

a. Calculate and record the calibration coefficient from the following formulas:

(Average Pulse Energy in Joules) / (0.081) / (Average Fluence)

b. Confirm that the calibration coefficient calculated is between 7.6J/cm^2/V and 14.5J/cm^2/V.

c. Press the [Return] button to exit the Diagnostic screen. Press the [S] button to enter the Setup Screen.

d. Enter the calibration coefficient (obtained in step 10.3.12) into the Detector field after pressing on the field to pop-up the numeric entry menu.

5. Calibration Current Adjustment

a. Set the four occurrences of Fmin to 1J/cm^2 (minimum setting).

b. Set the four occurrences of Fmax to 99J/cm^2 (maximum setting).

c. Press [Return] to exit the Setup Screen. Turn ChillTip off (in User screen).

d. Follow the screen prompts to perform a system calibration. If prompted for a cleaning, lift handpiece from pocket, release footswitch, replace handpiece in pocket and depress footswitch. Follow the screen prompts for another calibration.

e. From the treatment screen, press the [S] button to enter the Setup Screen. Adjust the current (I) for each of the calibration shots, such that the fluence cD[0] reads as close as possible but not less than 10.0J/cm^2 for the first shot, 40.0J/cm^2 for the second shot, 10.0J/cm^2 for the third shot, and 40.0J/cm^2 for the fourth shot of calibration. Adjust the current in 0.1A increments for the four shots.

f. Verify ChillTip is OFF, and repeat Step e (above). Repeat as required to get as close as possible to the 10J/cm^2 and 40J/cm^2 fluence values.

g. Set the Fmin and Fmax values according to the following table:

Shot	Fmin	Fmax
1	5.0	30.0
2	25.0	60.0
3	5.0	30.0
4	25.0	60.0

h. Confirm that cD[0] shot 1 and shot 3 values are greater than or equal to 10.0J/cm^2 and that shot 2 and shot 4 values are greater than or equal to 40.0J/cm^2. Record the fluence cD[0] values and current (I) values.

6. Headroom

<u>For new system or new handpiece only</u>: Confirm that “Headroom Auto” is greater than 1.20, “Headroom 30ms” is greater than 1.20, and “Headroom 100ms” is greater than 1.20. Record all Headroom values.

<u>For used system</u>: for any “headroom” value that is less than 1.00, the LightSheer is still functional, however, maximum fluences will not be available for those settings. To restore all available fluences, it is generally required to replace the handpiece.

7. Clean up

a. Press [Return] to return to the Startup Screen. Turn off the key. Press the Quit button and follow the screen prompts to turn off power to the system.

b. If Service mode was activated by switch setting, clear (turn OFF) the service switch (SW1) on the Main Board and reinstall the console cover.

c. Remove laser safety glasses. Calibration is complete.

Energy Meter Calibration Data Sheet (Page 1 of 2)

S/N:	Model:	Bars:	Technician:	Date:

Diagnostic Screen Settings

1. Set Current SP ☐ 32.3A for 70 bar ET/ST handpiece
2. Set Pulse Width to 0.020s ☐ (check to confirm)
3. Set Pusle Width Limit to 0.022s ☐ (check to confirm)
4. Set Fire Enable to "Enabled" ☐ (check to confirm)
5. "Diode Temp" within 17-22ºC ☐ (check to confirm)

Internal Energy Meter

Measurements

Pulse 1 Fluence: ☐ J/cm^2 (xx.xx)

Pulse 2 Fluence: ☐ J/cm^2 (xx.xx)

Pulse 3 Fluence: ☐ J/cm^2 (xx.xx)

Pulse 4 Fluence: ☐ J/cm^2 (xx.xx)

Pulse 5 Fluence: ☐ J/cm^2 (xx.xx)

Calculations

☐ J/cm^2 (xx.xx, avg fluence)

☐ J/cm^2 (xx.xx, low limit=0.95*avg fluence)

☐ J/cm^2 (xx.xx) hi limit=1.05*avg fluence)

Acceptance Criteria

1. 32.0J/cm^2 < avg fluence < 36.0 J/cm^2 ☐ yes
2. All 5 pulse fluences > low limit ☐ yes
3. All 5 pusle fluences < hi limit ☐ yes

External Energy Meter

External Energy Meter S/N: ☐

External Energy Probe S/N: ☐

Measurements

Pulse 1 Energy: ☐ J (xx.xx)

Pulse 2 Energy: ☐ J (xx.xx)

Pulse 3 Energy: ☐ J (xx.xx)

Pulse 4 Energy: ☐ J (xx.xx)

Pulse 5 Energy: ☐ J (xx.xx)

Calculations

☐ J (xx.xx, avg energy)

☐ J (xx.xx, low limit=0.90*avg energy)

☐ J (xx.xx) hi limit=1.10*avg energy)

Acceptance Criteria

1. 22.0J < avg energy < 38.6 J ☐ yes
2. All 5 pulse energies > low limit ☐ yes
3. All 5 pusle energies < hi limit ☐ yes

Calibration Coefficient

Calculation

Calibration Coefficient: ☐ J/cm^2/V (xx.xxx) Coefficient = avg energy / 0.081 / avg fluence

Acceptance Criteria

7.6 J/cm^2/V < Calibration Coefficient < 14.5 J/cm^2/V ☐ yes

Table 3-1: LightSheer ET/ST Diode Laser System
Energy Meter Calibration Form (page 1 of 2)

Energy Meter Calibration Data Sheet (Page 2 of 2)

S/N:	Model:	Bars:	Technician:	Date:

Calibration Current Adjustment

1. Set Fmin and Fmax: ☐ (check to confirm)
2. Perform System Calibration ☐ (check to confirm)
3. Turn ChillTip OFF ☐ (check to confirm)
4. Adj. currents to achieve target cD[0] fluence values ☐ (check to confirm)
5. Perform 2nd System Calibration ☐ (check to confirm)
6. Repeat current adjustment to target cD[0] values ☐ (check to confirm)
7. Perform 3rd System Calibration ☐ (check to confirm)
8. Reset Fmin and Fmax values to: ☐ (check to confirm)

Shot	Fmin	Fmax
1	5.0	30.0
2	25.0	60.0
3	5.0	30.0
4	25.0	60.0

9. Record cD[0] values and current (I) values:

Shot:	Current (A):	cD[0] fluence (J/cm^2):
1		
2		
3		
4		

__Acceptance Criteria__

1. Shot 1 cD[0] fluence >= 10.0 J/cm^2 ☐ yes
2. Shot 2 cD[0] fluence >= 40.0 J/cm^2 ☐ yes
3. Shot 3 cD[0] fluence >= 10.0 J/cm^2 ☐ yes
4. Shot 4 cD[0] fluence >= 40.0 J/cm^2 ☐ yes

Headroom

__Record cD[0] values__

Headroom Auto Mode: ☐ (x.xx)

Headroom 30ms Mode: ☐ (x.xx)

Headroom 100ms Mode: ☐ (x.xx)

__Acceptance Criteria__

1. Headroom Auto Mode > 1.20: ☐ yes
2. Headroom 30ms Mode > 1.20: ☐ yes
3. Headroom 100ms Mode > 1.20: ☐ yes

Table 3-2: LightSheer ET/ST Diode Laser System
Energy Meter Calibration Form (page 2 of 2)

(this page intentionally blank)

4.0 Theory of Operation

4.1 Introduction

The LightSheer ET/ST Diode Laser System, a portable, computer controlled, air-cooled semiconductor laser product, is intended to perform aesthetic treatments in private offices and clinics. Intense infrared laser light (principal wavelength of 800 nm) is delivered through a lightweight handpiece that is connected to the console via an umbilical cord. The handpiece contains an array of laser diodes and an optical system that deliver laser light to the laser aperture, a cooled sapphire lens, with a fixed spot size of 9 mm by 9mm. The console contains the computer, user interface, power supply, control electronics and a calibration port to measure delivered treatment energy.

Figure 4-1: LightSheer ET/ST Diode Laser System
(shown on optional accessory cart)

Note that while the laser output specifications are slightly different between the LightSheer ST and LightSheer ET models, the physical description, construction, parts list and service procedures are nearly identical. See Section 1 for model specifications.

The LightSheer ET/ST Diode Laser System is cleared for procedures that include permanent hair reduction, treatment of pseudofolliculitis barbae, leg veins, and benign pigmented lesions on skin types I – VI, including tanned skin, where wavelength specific absorption, small spot size, and high energy densities are required. Both models are designed for worldwide sales.

The LightSheer ET/ST Diode Laser System is comprised of the following functional components that are described in this section:

- Main Console
    - User Interface
        - Touch Sensitive Color Display Panel Assembly
        - Footswitch (with attached pneumatic foot pedal)
        - Remote Interlock
        - Key Switch
        - Emergency Stop Switch
    - Handpiece Calibration Port
    - Software
        - User
        - Service (Diagnostic)
        - Demo
    - Power Input Module and Main Power Switch (Circuit Breaker)
    - Power Supply (for computer and laser)
    - Control Electronics
        - CPU Controller Board
        - Main Board
        - FETs and a FET Driver Board (power supply dependent)
    - Cooling System
        - Coolant and Filter
        - Heat Exchanger
        - Pump
        - TEC (Thermo-Electric Cooler)
        - Temperature Sensors
- Delivery System (Handpiece)
    - Laser Diode Array
    - Optical System
    - Sapphire Lens (ChillTip)
    - Cooling System
    - Trigger (Fire Button)

4.2 User Interface

The display panel, footswitch (and attached pneumatic foot pedal), remote interlock, key switch, and emergency stop switch components make up the controls by which the user interfaces with the system. A brief description of each component follows.

4.2.1 Touch Sensitive Color Display Panel Assembly

The display panel consists of five main components: interface electronics, ultrasonic overlay (the touch screen membrane), a color LCD panel, a backlight, and an angled mount.

Electronics power, backlight power and pixel data are sent from the CPU Controller Board to the display panel assembly, while touch position data is sent from the overlay to the interface electronics back to the CPU Controller Board. The backlight is on whenever the computer is powered up.

The ultrasonic overlay (touch screen membrane) is calibrated by touching two computer-generated targets, much like a typical PDA display. The electronics decode the pressed overlay location and convert to an X-Y coordinate. Software then equates the X-Y coordinate with button, switch, or data entry field associated with the coordinate.

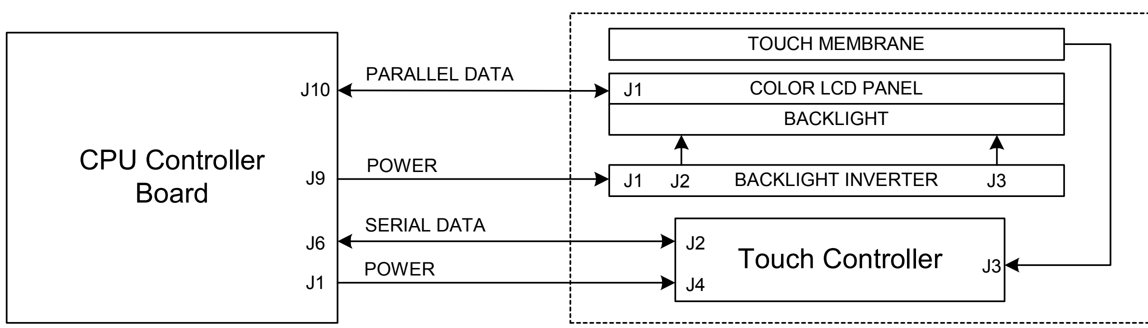


Figure 4-2: Display Panel Assembly Connections

4.2.2 Footswitch (and Pneumatic Foot Pedal)

The foot pedal assembly contains a pneumatic bulb, housed between two plastic covers, which develops high-pressure air carried through a small diameter flexible plastic tube (hose) to the footswitch, and a pressure-actuated diaphragm and micro-switch located at the rear of the LightSheer console.

The pressure-actuated switch is in a N.O. (normally open) condition until high pressure against a diaphragm closes a micro switch. Since the footswitch operation is entirely dependent upon air pressure, any leak within the system will appear to the circuit as a "pedal up" condition.

The footswitch terminals (N.O., N.C., and COM) are combined with the Remote Interlock connection in a single cable harness that is connected to the PWA Main Board.

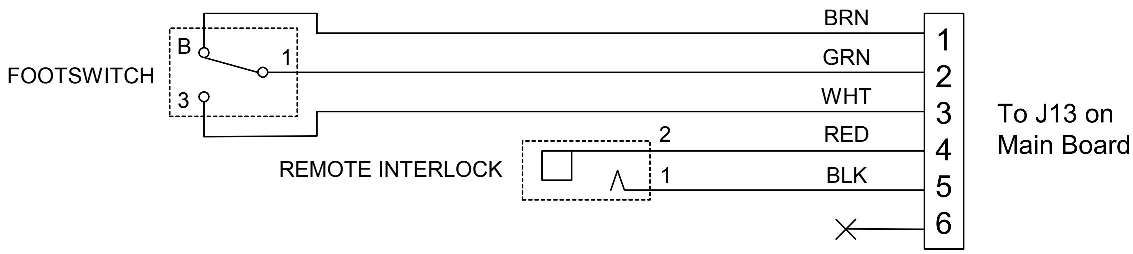


Figure 4-3: Footswitch and Remote Interlock Wiring Diagram

4.2.3 Remote Interlock

The remote interlock connection provides the user with the means to provide additional safety to those outside the laser treatment room by interlocking the treatment room door with a door-activated switch.

While the remote interlock connection typically contains a shorted ¼" phono plug, the short (wire) may be removed and connected across a door-activated switch. When the door is opened, the door-activated switch is sensed as open, interrupting the interlock circuit and disabling the electronic shutter preventing any current from driving the laser diodes in the handpiece.

See Section 2.6 for more information on installing a door-actuated remote interlock switch.

The Remote Interlock connection is combined with the footswitch terminals (N.O., N.C., and COM) in a single cable harness that is connected to the PWA Main Board.

4.2.4 Key Switch

The key switch provides the user with the means to prevent unauthorized use of the system. The key should be removed from the system when the system is not in use. The key cannot be removed except when in the Key OFF position (fully counter-clockwise). The system may be on, but without the key in the tumbler and rotated fully clock-wise to the Key ON position, the system software will not advance beyond the key switch icon screen and laser activation is disabled.

If the key switch is in the Key ON position at system turn-on, the software will require that the key position be cycled from ON to OFF and back to Key ON again, to verify proper operation of the switch.

The key switch terminals (N.O., N.C., and COM) are combined with the Emergency OFF switch terminals in a single cable harness that is connected to the PWA Main Board.

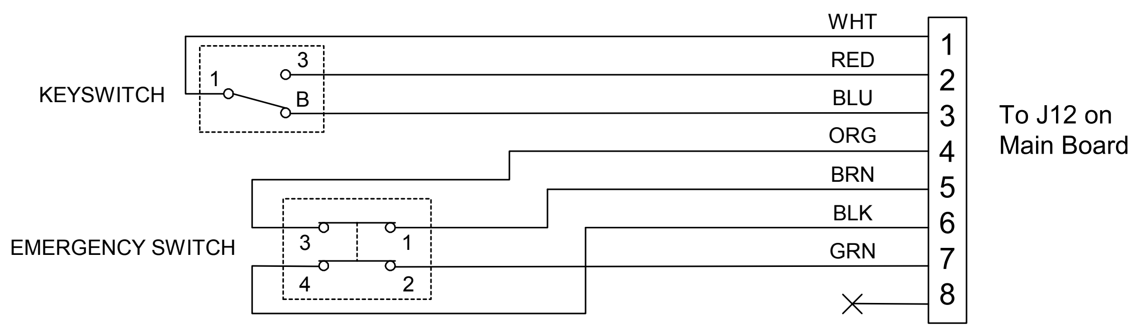


Figure 4-4: Key Switch and Emergency Stop Switch Wiring Diagram

4.2.5 Emergency Stop Switch

Normally the recommended system shut down procedure is followed when finished using the device. During and between treatments, releasing either the foot pedal or the handpiece trigger disables laser emission. In the event of an emergency, pushing the emergency stop switch, located at the front right of the console, will immediately stop laser emission.

The emergency stop switch contains two N.C. (normally closed) contacts. When pressed, the emergency stop switch will latch in the pressed position with the contacts open. To resume normal operation, rotate the red switch cap in the clock-wise direction until the red switch cap pops out.

The Emergency OFF switch terminals are combined with the key switch terminals (N.O., N.C., and COM) in a single cable harness that is connected to the PWA Main Board.

4.3 Handpiece Calibration Port

The handpiece calibration port is contained within the console cover and is located at the top center of the system. The calibration port serves two functions: chiefly as a safe location to perform the treatment energy calibration, and also as a rest or storage location for the handpiece. The handpiece pocket, the proximity sensor, the diffuser glass, and the photodiode (energy detector) are the components that make up the handpiece calibration port. The electrical signals from the calibration port (handpiece sense and energy signal) are connected to the PWA Main Board via separate cables.

4.3.1 Pocket (Holster)

The pocket is molded from a rubberized material and is removable from the top cover for cleaning, and to gain access to the calibration port diffuser glass so that it can be cleaned. The pocket contains an open aperture at the center that positions the handpiece laser aperture directly over the diffuser glass (for measurement by the photodiode) for calibration. The pocket also provides a convenient location for storing the handpiece between treatments.

It is possible to position the pocket in the top cover such that the pocket does not properly mate with the top cover. Always verify that the top of the pocket at the perimeter is flush with the top of the console cover.

4.3.2 Proximity Sensor

A proximity sensor is mounted to the top cover just above and to the right of the diffuser glass and looks for the presence of the handpiece when resting in the pocket. The proximity sensor emits light and can sense light reflected off the handpiece. The computer continually monitors the proximity sensor signal so that anytime the handpiece is detected to be in the pocket (sensor sees reflected light) and both the footswitch and handpiece trigger are depressed, the software begins the calibration routine.

The proximity sensor is connected to the PWA Main Board at J6. Power to the sensor is indicated by a steady glow of a green LED on the sensor body, and detection of reflected light is indicated by a steady glow of a yellow LED on the sensor body.

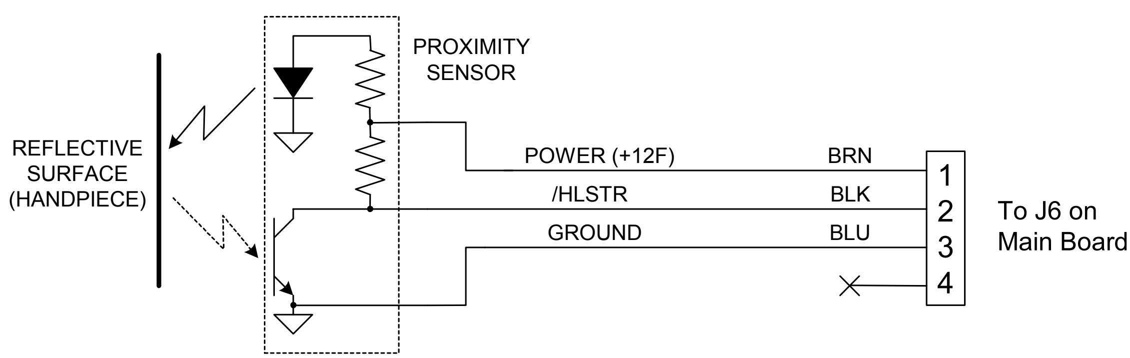


Figure 4-5: Proximity Sensor Wiring Diagram

4.3.3 Glass Window and Diffuser Glass

A disk shaped glass window is located at the bottom center of the top cover, just under the calibration aperture in the pocket and immediately above the diffuser glass covering the photodiode. The window must be clean for an accurate energy measurement. The surface of the glass may be cleaned with a soft towel wetted with commercial glass cleaner. Broken, pitted, or scratched glass must be replaced. The window is held in place with RTV.

A disk shaped diffuser glass is located just below the glass window. The diffuser glass is held in place by backside screws, and performs an integrating function as the treatment light that penetrates the glass is scattered in a small area underneath for measurement by the photodiode. Since each plate of diffuser glass has slightly different transmission characteristics, any time the diffuser glass is replaced, the photodiode (energy detector gain adjustment) calibration must be performed.

4.3.4 Photodiode (Energy Detector)

The photodiode, or energy detector, is highly sensitive to IR light and is used to measure the amount of light exiting the handpiece in order to calibrate the diode laser drive current calibration table. The photodiode is soldered directly to the Energy Meter Board Assembly, which is mounted below the diffuser glass. The photodiode passes a current proportional to the amount of light that is absorbed by the surface of the device.

Since each photodiode has slightly different electrical characteristics, any time the photodiode is replaced, the photodiode (energy detector gain adjustment) calibration must be performed.

The photodiode is connected to the PWA Main Board at J8. The photodiode converts the pulsed laser light into a current that is proportional to the intensity of the light. Amplifier circuits on the Main Board convert the current to a voltage, then integrate the pulse into a DC level. The DC level is sent to a sample/hold circuit, with an output of 0.1V/J.

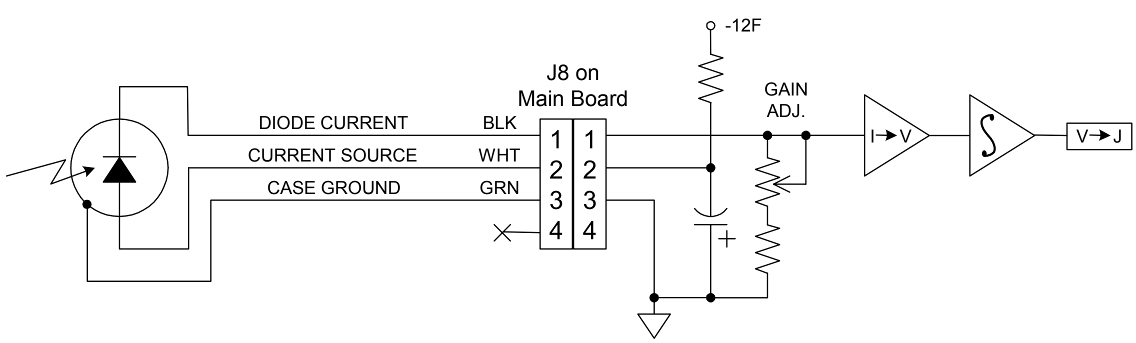


Figure 4-6: Photodiode Wiring Diagram

4.4 Software

System software is stored in the flash EPROM on the CPU Controller Board. The flash EPROM is programmed at the factory. The EPROM contains User Software, Service Software (diagnostic screens), and Demo Software.

4.4.1 User Software Routines

As the LightSheer System is turned on, the microprocessor will begin to boot-up. As the microprocessor boots, it will begin to execute the software stored in the flash memory. A visual indication is seen on the LCD Color Display and is typically displayed within 5 seconds of turning on the system. If the microprocessor does not boot successfully, and the application does not start properly (i.e., DRAM is defective, loose connector and etc.), the system will not be capable of the firing. If the computer boots successfully, the software will begins its initialization and the following tasks:

4.4.1.1 START-UP

1. Verify the checksum of persistent data (data stored in NVRAM).

   If the checksum of the primary persistent data does not match, the secondary persistent data is read. A fault flag will be generated if the checksum of the secondary data does not match. If the checksum of the primary data passes, the data is loaded from it and the data in the primary persistent data is copied to the secondary. Otherwise, if the secondary data is read and its checksum passes, the data is loaded from it.

2. The Main Heat Exchanger is turned on.

   This allows the system time to come into thermal equilibrium prior to calibration.

3. The LCD Color Display will display the Start-Up screen.

4. Check the version voltage.

   The microprocessor will check the version voltage provided by the upper console electronics and verify that the console electronics is a supported version level. If the reported console electronics version is not supported, the microprocessor will set a fault.

5. Initialize the I/O.

6. Check to see if a dated enable code is in effect.

If a dated code is in effect, software will begin to display an expiration warning screen when 7 days remain. The warning screen will inform the user of the number of days of usage remaining. Display buttons to permit the user to enter a code or proceed with startup.

7. Check to see if the dated enable code is expired.

   If the dated enable code is in effect and has expired, the display screen will inform the user that the code has expired. When the user acknowledges, the display will show the Enter code Screen, which will require a valid code to be entered before proceeding with Startup.

8. Display buttons to permit the user to quit or to proceed with the rest of the start-up routine.

   If the user selects "Quit", the display will show a message to turn off the laser. If the user selects the "Start" button, the system will prompt the user to turn the key to the ON position.

   If the key is ON, prompt the user to turn it OFF and then back ON again. The user may also turn the key without selecting the Start button. The software will not perform further initialization until the key has been cycled to the ON position. Key actions prior to this point are ignored. Responses to key actions after this point are defined by the Sleep protocol.

9. Display the Warning (glasses) screen.

10. Display the Calibration screen.

At this point, initialization is complete.

4.4.1.2 CALIBRATION:

Calibration will not take place until all of the following three items are sequentially true. If before firing the calibration pulses, any of these items become false, the software will return to the first item that is not true. The system should detect the occurrence of each of the following events.

1. Check to see if the handpiece is the holster.

   If the handpiece is already in the holster at the start of the calibration sequence, the software will require the user to remove and reinsert the handpiece. A "lift handpiece" screen is shown to prompt the user to perform this task.

2. Prompt the user to depress the footswitch.

   If the footswitch is depressed at the time that the handpiece is in the holster, the software will require the user to cycle the footswitch. A "release foot pedal" screen is shown to prompt the user to perform the task.

3. Prompt the user to depress the Handpiece trigger.

   If the handpiece trigger is depressed at the time that the footswitch is depressed, the software will require the user to release and depress the trigger again. A "release trigger" screen is shown to prompt the user to perform this task.

4. The system will prompt the user to fire four shots into the energy meter.

   The software will prompt the user to fire four pulses into the energy meter per the calibration procedure described in the "LightSheer Calibration Procedure" and display a calibration progress bar. A "pull trigger" screen will prompt the user to depress the trigger during this time. If either the holster switch, footswitch, or trigger position changes state during this firing sequence, firing is discontinued, and the software will return to the first item in the above order that is not true. Firing is disabled after the 4 calibration pulses until either treatment or calibration is restarted.

5. The software checks the four pulses.

   After each of the four pulses, the software checks that the measured pulse energy is within bounds as specified in the LightSheer Calibration Procedure. If the pulse energy is too high, a fault is generated. If the pulse energy is too low, the user is prompted to clean the Handpiece ChillTip and recalibrate. The Calibration Inconsistent description below defines the behavior for this screen. If the energy is too low on a second attempt, a screen will appear asking, "Are you sure you have cleaned the e-meter glass." If a YES response is entered, a fault is generated.

6. The software performs the calibration calculations.

   The software will perform the calibration calculations and error checking as described in the Light-Sheer Calibration Procedure. Pass and fail criteria and procedures are defined in that document with six possible outcomes:

   a. Fail. In this case, the software will enter the fault state.

   b. Calibration inconsistent (first time). In this case, the High Voltage enable bit is cleared and the application will prompt the user to clean the ChillTip. This prompt will be displayed until the user has removed the handpiece from the holster, reinserted the handpiece, and cycled the footswitch. At this point, a calibration will set the high voltage enable bit and return to the Calibration screen.

c. Calibration inconsistent (second or more times). If the second set of calibration results is different from both the set currently being used and the immediately previous results, the software will again clear the high voltage enable bit and then query the user to the following: "Are you sure you cleaned the tip?" This prompt will be displayed until the user has answered either "Yes" or "No." Answering "Yes" will generate a fault due to inconsistent calibrations. Answering "No" will prompt to clean the window and allow the user to perform an additional recalibration attempt per the Calibration Inconsistent (first time) description.

d. Pass with disabled auto mode and limited maximum fluence. In this case, the software will progress to Treatment but auto mode will not be available to the user and the maximum fluence selectable by the user (in 30 ms and 100 ms modes) will be less than the full fluence set in the set up screen. Before progressing to the Treatment screen, the user must acknowledge a message screen alerting the user that auto mode is not available and the fluence is limited.

e. Pass with disabled auto mode only. In this case, the software will progress to Treatment but auto mode will not be available to the user. The maximum fluence selectable by the user (in 30 ms and 100 ms modes) will be the full fluence set in the setup screen. Before progressing to the Treatment screen, the user must acknowledge a message screen alerting the user that auto mode is not available.

f. Pass with full functionality. In this case, the software will progress to Treatment with all pulse width modes available and the full fluence range available.

7. If the software enters into Sleep, Interlock, Diagnostic modes, or Setup screens during calibration, the software will return to the beginning of calibration.

8. If the high voltage relay opens, such as when the key switch is being turned off, the system will return to the start screen.

4.4.1.3 TREATMENT:

1. Software enters into Treatment mode.

   As the system enters into treatment, it displays a screen that enables the user to select the fluence, pulse width mode, open the options screen, and zero the shot counter. The Treatment screen also has indicators that display the ChillTip status and current PRF.

2. If entering Treatment after passing calibration, the ChillTip is turned on.

3. If entering treatment after passing calibration for the first time after the power has been turned on, the footswitch must be cycled before the laser may be fired.

4. If entering Treatment after passing calibration, the trigger must be cycled before the laser may be fired again.

5. If calibration passed with full functionality, the allowable fluence range is from 10 to maximum fluence limit set in Service Mode inclusive. This limit can be changed, in Service Mode, independently for each pulse width mode. The factory default settings are given in Table 4-1.

Fluence Settings	10 J/cm^2
Pulse width Mode	AUTO
Shot Counter	0
ChillTip	ON
Max. Fluence Limit	60 J/cm^2 (40 J/cm^2 on ST)
Pulse Repetition Frequency	0.5 Hz

Table 4-1: Default Parameters

Depending on the results of the calibration, the maximum fluence the user is allowed to select may be less than the limit, but will never be less than 10.

6. If the calibration passed with full functionality, the user may select between three allowable pulse width modes, AUTO, 30 ms and 100 ms. (Depending on the results of calibration, the user may not be permitted to select the AUTO setting). The software will persist this setting.

7. The treatment screen will have a status indicator. The indicator will change appearance to reflect three states of the laser in Treatment: STANDBY, READY and COOLING. These states are logically described in Table 4-2:

State (Indicated)	ChillTip	Shutter Open	EPI Cooler Hot	Out of Holster
COOLING	ON	TRUE	TRUE	TRUE
READY	ON	TRUE	TRUE	FALSE
READY	ON	TRUE	FALSE	TRUE
READY	ON	TRUE	FALSE	FALSE
COOLING	ON	FALSE	TRUE	TRUE
COOLING	ON	FALSE	TRUE	FALSE
STANDBY	ON	FALSE	FALSE	TRUE
STANDBY	ON	FALSE	FALSE	FALSE
READY	OFF	TRUE	TRUE	TRUE
READY	OFF	TRUE	TRUE	FALSE
READY	OFF	TRUE	FALSE	TRUE
READY	OFF	TRUE	FALSE	FALSE
STANDBY	OFF	FALSE	TRUE	TRUE
STANDBY	OFF	FALSE	TRUE	FALSE
STANDBY	OFF	FALSE	FALSE	TRUE
STANDBY	OFF	FALSE	FALSE	FALSE

Table 4-2: Laser Treatment States

8. The STANDBY message changes to READY as the shutter is opened to provide feedback to the user that the footswitch has been depressed.

9. ChillTip OFF message appears when the ChillTip has been turned off.

10. To fire the laser in Treatment, the software must perform the following tasks in sequence:

    a. Wait indefinitely for the system to come to READY status.

    b. Wait indefinitely for the trigger to be pressed. Note that if the trigger is pressed before the laser is in READY status, such as before the shutter is open, the software must ignore that event. The system can enter READY status from STANDBY if the trigger is depressed, but the user has to cycle the trigger before a shot will happen.

    c. After receiving a valid trigger pull, the software will write the current control values based on the treatment settings and calibration parameters out to the hardware via the I/O card, program the pulse width register on the I/O card, and trigger the pulse. The pulse should be repeated at the selected repetition rate for as long as the following conditions are true:

        1) The trigger is continuously depressed.

2) The foot pedal is continuously depressed.

3) No interlock condition has occurred.

4) No system fault has occurred.

5) No key switch has occurred.

d. The user can re-initiate the trigger sequence at any time, but the application will not allow pulses faster than the pulse rate frequency set in the Options window, even if manually triggered at a faster rate.

e. PRF (Pulse Rate Frequency, in Hz) control operation:

1) When 0.5 or 1.0 Hz pulse rates are selected in the options screen, the pulse rate will be a fixed rate of 0.5 or 1.0 Hz, depending on which rate is selected.

2) (ET Only) When 1.5 Hz is selected in the options screen, the pulse rate is fixed at 1.5 Hz over the fluence range of 10 - 25 J/cm^2, as selected on the treatment screen. For fluence settings between 40 and 60 J/cm^2 on the treatment screen the pulse rate is as follows, not exceeding 1.5 Hz:

- The pulse rate is decreased linearly from 1.5 Hz at 40 J/cm^2 to 1.0 Hz at 60 J/cm^2.

3) (ET Only) When 2.0 Hz is selected in the options screen, the pulse rate is fixed at 2.0 Hz over the fluence range of 10 - 25 J/cm^2, as selected on the treatment screen. For fluence settings between 25 and 60 J/cm^2 on the treatment screen the pulse rate is as follows:

- The pulse rate is decreased linearly from 2.0 Hz at 25 J/cm^2 to 1.5 Hz at 40 J/cm^2.

- For fluence settings between 40 – 60 J/cm^2 on the treatment screen, the pulse rate is decreased linearly from 1.5 Hz at 40 J/cm^2 to 1.0 Hz at 60 J/cm^2.

4) Maximum pulse rate frequency in 100 ms mode is limited to one-half of maximum pulse rate frequency in AUTO or 30 ms mode.

5) The treatment screen has a shot counter, which has two modes. The first is a counter that will display a maximum of 9,999 shots, after which the count will wrap to 0. This mode has a “reset button” so the user can force the count to 0.

The second mode is a cumulative shot counter that displays the total number of shots that have been fired since the unit was reset in service mode by a technician. The reset button for the cumulative shot counter is not available to the end user. The user can toggle between counter modes by touching the counter numbers. Both shot counter values will persist across sessions. The cumulative shot counter can only be reset in Service Mode by using the "Set Shots" button in the Diagnostic screen.

6) As defined in the Calibration Procedure, the system forces a calibration upon awakening from screen saver if at least 25,000 shots have occurred since the last calibration.

7) None of the controls on the treatment screen is active while the laser is firing.

8) The treatment screen has a button that will open the options screen.

   - The options screen has buttons to turn the Chill Tip ON and OFF. The laser status message on the treatment screen will be Cooling if the ChillTip is ON but not cold.

     Note that the system cannot fire when the "Cooling" status message is displayed. A warning screen will be presented if the user turns the ChillTip OFF. The ChillTip will be turned OFF in hardware only after the user confirms in the warning screen that the ChillTip should indeed be turned OFF. A status message on the treatment screen indicates "ChillTip OFF" if the ChillTip has been turned off in the options screen.

   - The options screen has a button (AutoPace) to change the pulse repetition rate of the laser. The available pulse repetition rates are indicated by settings under the button.

   - The options screen has a button to change the language used for the GUI. The available languages are displayed as settings below the button. The indicator for the currently selected language will change color to highlight the selection. The language change will take effect immediately, and the setting will be stored in persistent memory.

9) Calibration can be initiated from the Treatment screen by placing the handpiece in the holster, stepping on the footswitch, and pulling the handpiece trigger (firing into the calibration port). The order of the holster and footswitch sequence does not matter, but the user must follow the standard firing sequence, which requires the trigger to be cycled <u>after</u> the shutter is fully open (footswitch depressed). At that point, the Calibration screen is displayed and the calibration proceeds exactly as described above.

10) When the unit is in Demo mode, an indicator below the Opti-Pulse control will display DEMO; when not in demo mode, the indicator is invisible.

11) If the high voltage is present, as when the key switch is being turned off, the system will return to the startup screen.

12) The Treatment screen will have an indicator displaying the pulse repetition rate, in Hz.

13) When firing from the Treatment screen, there should not be a delay longer than 0.100 seconds from the time the trigger is pulled until the start of the laser pulse.

4.4.1.4 SLEEP MODE (Screen Saver, Power Saver):

1. The system enters Screensaver from any screen except the Diagnostic screen if there has been more than 15 minutes of inactivity. Activity is defined as one of the following events, which results in a visible change to the current screen.

    a. Touch-screen press.

    b. State change of the footswitch.

    c. State change of the trigger switch.

    d. State change of the key switch.

    e. State change of the holster switch.

    f. State change of the interlock switches.

2. As the software enters Screensaver, it performs the tasks in the order listed below:

    a. Display the Screensaver.

    b. Clear the high voltage bit.

    c. Turn off the ChillTip.

    d. Turn off the heat exchanger after 5 minutes.

3. The software awakens from Screensaver by any activity as defined above. The software processes the awakening event just as if it had occurred outside of Screensaver.

4. As the software awakens, it performs the tasks in the order listed below:

   a. Check the diode temperature sensor within tolerance of the main heat exchanger tolerance. If not within tolerance, the system will go into a interlock condition.

   b. Display the Glasses screen until the user has acknowledged the warning.

   c. Display the appropriate return screen.

   d. Turn on the heat exchanger.

   e. Set the high voltage enable bit.

   f. Turn on the ChillTip if return to treatment.

4.4.1.5 INTERLOCK MODE:

1. The software enters Interlock whenever it detects any of the following conditions from any screen, except Fault and Diagnostic (unless otherwise noted):

   a. The diode arrays are over-temperature.

   b. The FET heat sink is over-temperature.

   c. The remote interlock is open.

   d. The e-stop has been engaged.

   e. The heat exchanger and diode temperature differ more than ±4.0°C of the ΔTemp value only immediately prior to any fire event or upon returning from sleep.

2. When the software enters the Interlock state it should perform the following tasks in the order listed:

   a. Clear the high voltage enable bit.

   b. Display an Interlock screen stating the interlock message.

3. The priority of the interlocks and the interlock code that should be displayed for each interlock is as follows, from the highest to lowest:

a. Emergency stop, P1.

b. Remote interlock, P2.

c. Diode over-temperature, P3.

d. FET over-temperature, P4.

e. Temperature sensor, P5.

4. If two or more interlocks are present at the same time, only the message of the highest priority interlock is displayed.

5. The software will remain in the Interlock State until all of the interlock conditions have been cleared. Once the interlock conditions have been cleared, the user will be prompted to cycle the key. Once the key has been cycled, the software will display the Warning screen and when acknowledged, will set the high voltage enable bit and then return to the appropriate return screen.

6. When a diode array over-temperature interlock occurs, the software turns off the ChillTip.

7. When the diode array over-temperature interlock occurs during startup, it will display the interlock message. However, the heat exchanger is turned on and ChillTip is turned off. Once startup is complete, the interlock message is displayed for diode array over-temperature interlocks that occur.

8. The software ignores remote interlocks during system startup. Once startup is complete, remote interlocks are processed.

4.4.1.6 FAULT STATE:

1. The software will enter the fault state whenever it detects any of the following conditions:

   a. An over-current A or B, over-charge A or B, over-power A or B, over-voltage A or B or pulse width error after a shot in Cal, Treatment, or Diagnostic.

   b. Calibration has failed due to fluence out-of-range, measurements not self-consistent, inconsistent calibrations, or insufficient maximum fluence during calibration.

c. Electronic shutter has failed to open (enable) within 1.0 second of the footswitch being depressed during Calibration, Treatment or Diagnostic.

d. Electronic shutter has failed to close (disable) within 1.0 second of the footswitch being released during Calibration, Treatment or Diagnostic.

e. I/O board fault detected during initialization of the I/O board in Initialization.

f. Current is out of tolerance as specified in the “LightSheer Calibration Procedure” after a shot in Calibration, Treatment or Diagnostic.

g. Checksum failure (which includes missing file) detected when the persisted data file is read in Initialization.

h. The loop back test on the current setting is not within ±5% of the specified value prior to a shot in Calibration, Treatment or Diagnostic.

i. High Voltage Power Supply voltage is out of tolerance just prior to a shot during Calibration, Treatment or Diagnostic.

j. TEC Power Supply voltage is out of tolerance.

k. When a shot is attempted while firing is locked out by the hardware.

l. The pulse width read back is out of tolerance.

m. The laser diodes are connected while in Demo mode.

n. The PS Low Line is high.

o. If the Laser driver is set high.

p. If an unsupported version of the console electronics is detected.

2. When the software enters the Fault State, it performs the following tasks in the order listed except for the Diagnostic screen. In the Diagnostic screen, the fault is shown on the appropriate digital output check box and the fault screen appears after pressing “Return” on the Diagnostic screen.

   a. Display a Fault screen with a Quit button and Fault code.

      - If a checksum fault occurs, hide the “D” and “S” buttons.

- The LightSheer fault codes and the fault condition they indicate are listed in Table 4-3. A detailed explanation of each fault code is located in Section 5.

- Clear the high voltage enable bit.

- Turn off the ChillTip.

b. The software will remain in the Fault state until the Quit button is pressed and the power is turned off. As for all screens, the screensaver will come on after 15 minutes of inactivity from the Fault, but will return to the Fault screen upon awakening.

E1	BIOS Memory Failure
E2	I/O Board (Computer Failure)
E3	Persistent Memory (NVRAM) Failure
E4	Shutter (Electronic) Failed to Open
E5	Shutter (Electronic) Failed to Close
E6	Over Current A Array
E7	Over Current B Array
E8	Over Charge A Array
E9	Over Charge B Array
E10	Over Power A Array
E11	Over Power B Array
E12	Over Voltage A Array
E13	Over Voltage B Array
E14	Pulse Width Error
E15	Calibration Out of Range
E16	Calibration Energy Value Not Expected
E17	Calibration Inconsistent
E18	Current Tolerance Error
E19	Pulse Energy Too Low
E20	DAC Read-back Port A Error
E21	[NOT USED] DAC Read-back Port B Error
E22	Pulse Width Limit Error
E23	Laser Driver (/DRV CW) Bit Error
E24	Low Voltage Power Supply Voltage Out of Tolerance
E25	Laser Diodes Connected in Demo Mode
E26	High Voltage Power Supply Voltage Out of Tolerance
E27	TEC Power Supply Voltage Out of Tolerance
E28	Pulse Lockout Error
E29	Board Revision (Compatibility) Error
P1	Emergency Stop Button is Depressed

P2	Door Interlock (Remote Interlock) is Open
P3	Laser Diode Too Hot (Over Temperature)
P4	FET Too Hot (Over Temperature)
P5	Delta Temperature >4°C (Main Heat Exchanger Temp. and Diode Temp.)

Table 4-3: Fault Codes

4.4.2 Contents of Persistent Memory (NVRAM)

The software stores data in non-volatile memory so that it can persist across different sessions. As a reliability feature, a backup data set is also stored in NVRAM that is used if the primary data cannot be used. The content of Persistent Memory is shown in Table 4-4.

1. The data contains a checksum. If the stored checksum does not agree with that calculated as the data is read, the program reads the backup data. If the stored checksum of the backup data also does not agree, a fault is generated. Note that a checksum fault is also generated if the primary data fails and the backup data is missing.

2. The persistent data is written at startup, shutdown, after a treatment fire trigger release, after a fluence setting change, after a pulse width mode change, after a reset of the shot counter, upon entry to screensaver, after a successful calibration, after a numeric entry, after a shot count entry, after a code entry, after a touch screen calibration, after a language change, after a rate change, after a model change, and after a service mode change.

3. If the data is read in successfully from the primary data file, the primary data is copied to the backup data file.

DATA	VALUE
Improper Shutdown Flag	True or False
Service Mode Flag	Normal, Service, Demo, or Demo/Service
Resetable Shot Counter	Integer: 0 – 9,999
Resetable Cumulative Shot Counter	Integer: 0 – 9,999,999 (Reset in Service Mode)
Cumulative Shot Counter	Integer: 0 – 99,999,999
Number and Data of Last Event in Log File	Shot count, pulse width setting, diode current setting, calculated fluence (if fired in calibration port)
Fluence Set Point	Current Fluence Setting

Pulse Width Mode	Current Pulse Width Setting
Calibration Setup Parameters	I1, I2, I3, I4, T1, T2, T3, T4, Fmin1, Fmin2, Fmin3, Fmin4, Fmax1, Fmax2, Fmax3, Fmax4, Imax Auto, Imax 30ms, and Detector Calibration Factor
Calibration Data	CD0, CD3 (i.e. F1, F2, F3, F4, Headroom)
Checksum	Checksum value of the software revision
Base Frequency	Maximum pulse rate
Current GUI Language	English, German, French, etc.
Model Selection	LightSheer ET or LightSheer ST
Derating Fluence	Fluence above which the pulse rate is reduced in order to limit average power
High Voltage Power Supply Tolerance Setpoint	A value for a comparator used to determine if the output voltage is within acceptable tolerance
TEC Power Supply Tolerance Setpoint	A value for a comparator used to determine if the output voltage is within acceptable tolerance
Passed First Calibration	True of False
Pulse Rate Selection	Last Pulse Rate Selected by User
Touch Screen Calibration Data	Actual pixel location (in X/Y coordinates) for the two on-screen calibration targets
Analog-to-Digital Converter Calibration Factor	A software multiplier to compensate for any offset voltage(s) in the ADC loop.
Δ Current Threshold for Temp Sense	0.90 for Opto Diode Bars (most typical) 0.70 for Coherent Diode Bars (atypical)
Normalized Temperature	Default Value = 1.00.
Tau	Default Value = 1.80.

Table 4-4: Contents of Persistent Memory (NVRAM)

4.4.3 User Software Screens

The User software is set up as the "boot" program so that upon system turn on, the controller will grab the program files from the EPROM and load them into RAM memory. The user program automatically runs at system turn on.

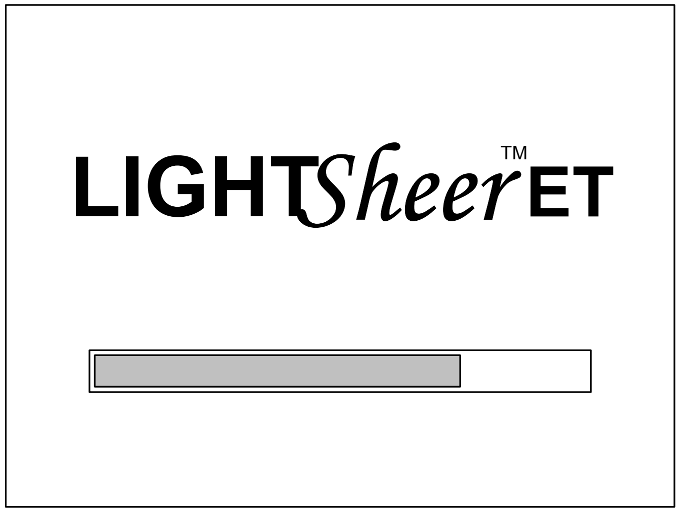


Figure 4-7: Loading Program Screen
(for ST models, the screen will display “LIGHTSheer ST”)

A Startup screen is shown once the system has been turned on or if the key has been turned off. The Startup screen displays either the LightSheer ET or ST logo along with Quit and Start buttons. To shut down the system, the Quit button is pressed, and the user is instructed to turn off the main power switch (circuit breaker) at the rear of the console. To prepare the system for use, the Start button is pressed. Pressing the start button prompts the user to turn the key, then displays the warning screen (to wear eye protection and to keep the tip clean during use). After acknowledging the warning, the software continues with internal safety tests and if all tests pass, the user is instructed to perform the calibration sequence.

Figure 4-8: Startup Screen
(for ST models, the screen will display “LIGHTSheer ST”)

Figure 4-9: Warning Screen

After a successful calibration, the controller displays the main treatment screen that allows the operator to select and view treatment parameters, change system status between Standby and Ready, and to choose additional display options. Please refer to the LightSheer ET/ST Diode Laser System User Manual (P/N 10-03656-04) for a complete description of user mode screens and controls.

4.4.4 Service Software

The Service software is nested within the User software and by default, is not activated. System and laser parameters are accessible from the service screens. Access to and use of the Service routines is to be strictly limited to trained service personnel.

Caution - Do not instruct users or non-trained technical personnel to access or utilize these service routines. Improper calibration, system damage, injury to user and/or patient, and/or a non-functional system may result from improper settings of the service screens!

4.4.4.1 Access to Service Software

There are two methods to activate Service software, 1) via switch setting, and 2) via touch panel:

1. Switch Setting (Top Cover Removed)

Upon system turn on, the controller looks at the Main Board service switch, SW1, to see if it is open (OFF, not set) or closed (ON, set). If the service switch bit is set (switch ON), the service routines will be activate as system power up, as indicated by the letters [D] and [S], and located in the top left and right sides of the display. Selecting [D] or [S] will bring up two useful service screens. If the switch method is employed to enter Service software, verify that the switch position is returned to its default position (OFF) before replacing the top cover so that there is no chance the user can get into the service software screens.

Figure 4-10: Service Mode Activated
(for ST models, the screen will display “LIGHTSheer ST”)

2. Touch Panel

Note: It is very important, when using the touch screen to access the service mode that the code is entered slowly and accurately. Failure to do so can enable the LightSheer service access codes. If an access code screen is enabled, contact Lumenis Customer Service for the access code.

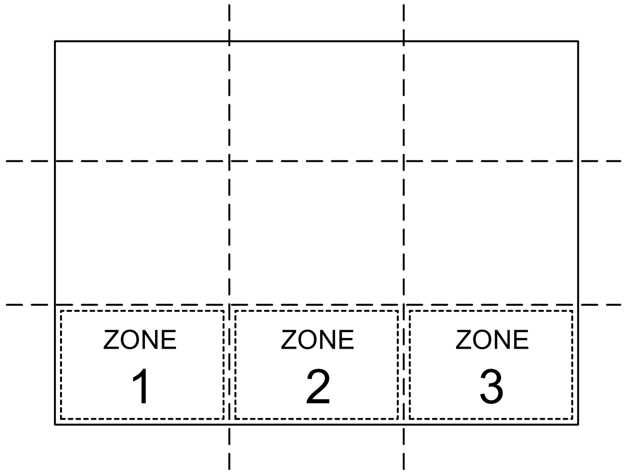


Figure 4-11: Screen Zones

Mentally divide the screen in thirds as shown above. After the system displays the "Start/Quit" screen, press the Emergency Stop switch, wait for the PAUSED screen, then slowly touch the three zones, in order, 3-2-1-2-3-3-3, in the bottom third of the display and wait one second. A new screen with the options "Normal Operation", "Service", "Demo", and "Service & Demo" will appear on the display. Select (touch) the "Service" button and verify the button is highlighted.

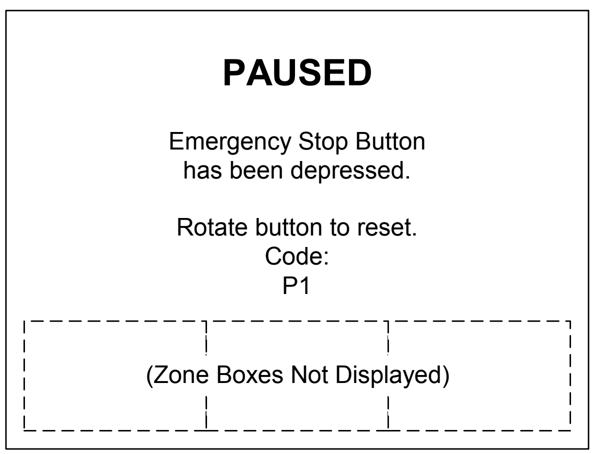


Figure 4-12: Paused Screen (Emergency Stop button depressed)

Once "Service" has been selected, slowly touch the three zones, in order, 1-1-2-2-3-3-2. Do not touch the "Cancel" button during this step! The Start-up screen will return with the letters [D] and [S] displayed as described above. Rotate the Emergency Stop button clock-wise to continue.

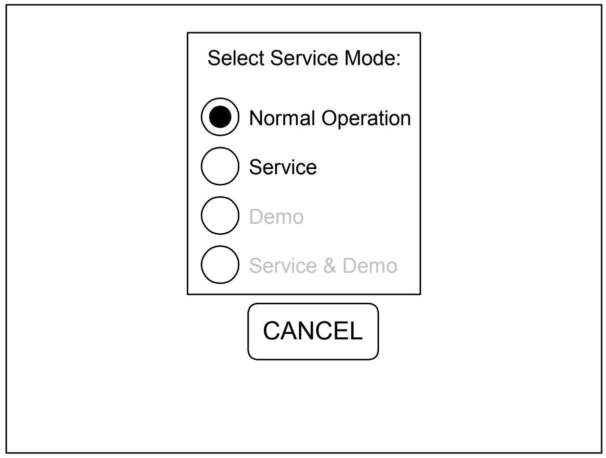


Figure 4-13: Mode Selection Screen

To leave Service software if the screen method was employed, either press the "Quit" button and turn off the system (the normal shut down procedure) or press the emergency stop button, and repeat the screen touch sequence as described above. Slowly touch the three zones, in order, 3-2-1-2-3-3-3, and then select "Normal Operation." Slowly touch the three zones, in order, 1-1-2-2-3-3-2, and wait one second. Rotate the Emergency Stop button clock-wise to continue. The letters [D] and [S] should no longer be displayed, and the system will return to the normal User mode.

The [S]etup Screen and [D]iagnostic Screen with a description of fields and default values are shown in the following figures.

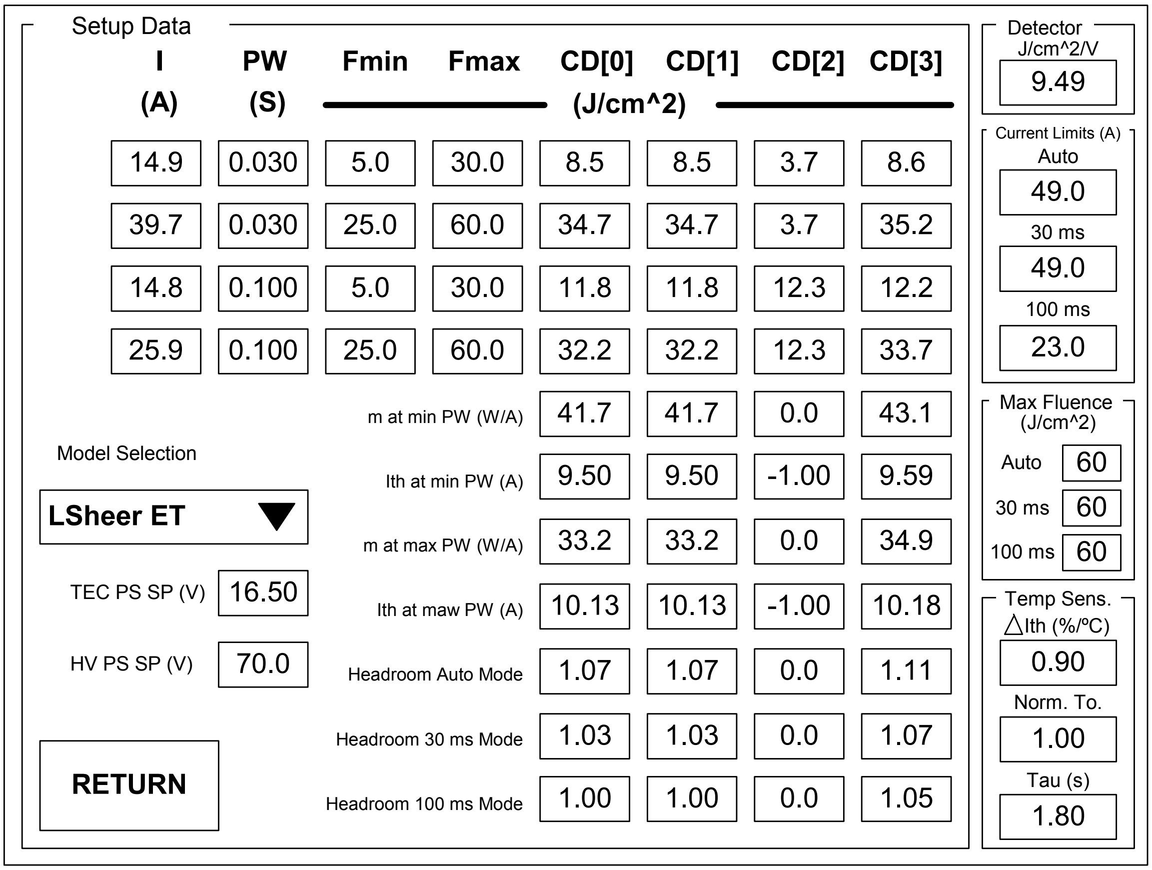


Figure 4-14: Setup Screen

4.4.4.2 Set-Up Screen

The Setup screen is accessible only after activation of Service mode. To enter the Set-up screen, press the "S" that is displayed on the upper right of the display. The end user will not have access to this screen.

The Setup screen allows the service engineer to set the current, pulse width, minimum fluence and maximum fluence for each of the 4 calibration shots. These settings are saved in NVRAM. These settings can be changed only when supervisory mode is enabled.

The service engineer can set the detector sensitivity (in units of Joules/cm^2/Volt) as determined by the calibration procedure.

In supervisory mode, the maximum currents in Auto, 30ms and 100ms pulse modes, and the maximum fluences possible for pulse width settings of Auto, 30msec and 100msec are adjustable. See the settings chart below. These settings are saved in NVRAM.

Setting Description	ST	ET
Current Limit – Auto	49.0	49.0
Current Limit – 30msec	36.0	49.0
Current Limit – 100msec	20.0	23.0
Max Fluence – Auto	40	60
Max Fluence – 30msec	40	60
Max Fluence – 100msec	40	60
HV PS Set Point (V) – Lambda Power Supply	112.0	112.0
HV PS Set Point (V) – Martek Power Supply	70.0	70.0
TEC PS Set Point (V)	16.50	16.50

The service engineer can view the results of the previous calibration attempts, as indicated by column labels shown below:

- CD[0] - current calibration data.
- CD[1] - calibration data from most recent attempt to pass.
- CD[2] - calibration data from most recent attempt to pass after prompting the user to clean the window.
- CD[3] - calibration data that is used as a baseline for long-term degradation.

The service engineer can also view the efficiency and threshold current at the two pulse widths, the headroom in automode, the headroom in 30ms mode, and the headroom in 30ms mode, and the headroom in 100 ms mode for the four stored calibration results.

The service engineer can change the High Voltage and TEC power supply voltage set points. For systems with a Lambda Power Supply, set this value to 112.0. For systems with a Martek Power Supply, set this value to 70.0. For all systems, the TEC voltage set point is 16.50.

The service engineer can change the desktop, startup and screen saver graphics by using the “Model Selection” control. This control will also preset the appropriate setup values to those specific to the selected model. The control, when opened, displays a list of the available model settings. This control only has effect when supervisory mode is enabled.

The following controls can only be changed in supervisory mode:

- “Model Selection” control.
- The Fmax control.

The service engineer can change the values for Ith, Tau and Tnorm.

4.4.4.3 Diagnostic Screen

The Diagnostic screen is accessible only after activation of Service mode. To enter the Diagnostic screen, press the "D" that is displayed on the upper left of the display. The end user will not have access to this screen.

Faults that occur while in this Diagnostic screen are not displayed. Instead, when the user returns from this screen, the system enters Fault. Interlocks that occur while this screen is visible are not displayed. If the interlock still exists when the user returns from this screen, the system enters Interlock.

The Diagnostic screen displays the current digital and analog inputs and outputs. This display is updated continuously on "idle" events (<1 sec. interval).

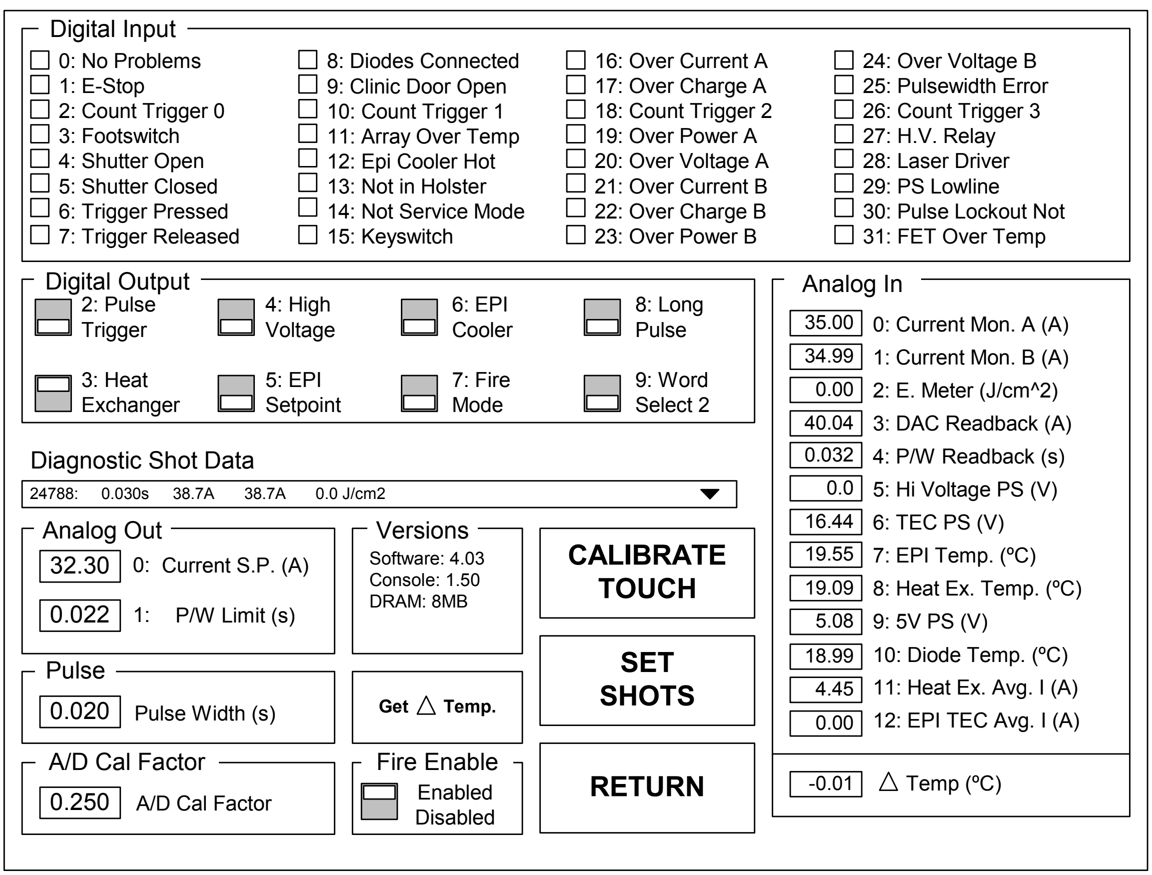


Figure 4-15: Diagnostic Screen

1. Digital Input = A check mark in the box indicates that the status bit is high, true, or active (Digital Input = 5V). No check mark in the box indicates that the status bit is low, false, or not active (Digital Input =0V.)

   Box colors: Red (if checked, the fault cleared only by cycling power), Violet (if checked, the fault set if any box checked and shutter is/was open), Yellow (not used), Green (user activated switch – not a fault), and Blue (status bit set by the computer).

Bit	Color	Signal Name	Active (checked) when…
0	Yellow	No Problems	N/A
1	Violet	E-Stop	Emergency Stop button is pressed.
2	Yellow	Count Trigger 0	N/A
3	Green	Foot Switch	Footswitch contact is closed (pedal down).
4	Green	Shutter Open	When Q11 is ON (if FET Driver Board installed) or when Martek power supply is enabled.
5	Green	Shutter Closed	When Q11 is OFF(if FET Driver Board installed) or when Martek power supply is disabled.
6	Green	Trigger Pressed	Handpiece trigger is pressed (pulled).
7	Green	Trigger Released	Handpiece trigger is not pressed (released).
8	Blue	Diodes Connected	The laser diode arrays are electrically connected to the system (must be disconnected for Demo Mode).
9	Violet	Clinic Door Open	Remote Interlock is open (treatment room door open).
10	Yellow	Count Trigger 1	N/A
11	Violet	Array Over Temp	Diode temp is greater than 31ºC
12	Blue	EPI Cooler Hot	EPI temp is greater than 9ºC
13	Blue	Not In Holster	Handpiece is not placed in the holster (pocket).
14	Blue	Not In Service Mode	SW1 on the Main Board is OFF (open).
15	Green	Key Switch	Key switch is fully clock-wise (in the RUN position).
16	Red	Over Current A	Current through the Diode Array A is 5% + 1.5A over requested setting.
17	Red	Over Charge A	
18	Yellow	Count Trigger 2	N/A
19	Red	Over Power A	
20	Red	Over Voltage A	If voltage across Diode Array A is greater than 75V (checked 4msec after pulse begins).
21	Red	Over Current B	Current through the Diode Array B is 5% + 1.5A over requested setting.
22	Red	Over Charge B	
23	Red	Over Power B	
24	Red	Over Voltage B	If voltage across Diode Array B is greater than 75V (checked 4msec after pulse begins).
25	Red	Pulse Width Error	Pulse width is detected to be 2msec longer

			than requested (up to 30msec selected) or if pulse width is detected to be 10% longer than requested (all selected pulses over 30msec).
27	Blue	High Voltage Enable	When key is ON in proper sequence
28	Red	Laser Driver	N/A
29	Red	P.S. Low Voltage	Either +5, +12, or –12 Vdc power supplies are detected to be out of range (5% low).
30	Blue	Pulse Lockout Not	Pulse begins, for 240msec from beginning of pulse.
31	Violet	FET Over Temp	FET temp is greater than 80ºC

Table 4-5: Digital Input Bit Definitions

2. Digital Output = A touch to the button toggles between OFF (center button down) and ON (center button up). The definitions of the digital output bits are shown in Table 4.6.

Digital Output Bit	Signal Name	When ON…
2	Pulse Trigger	Not used.
3	Heat Exchanger	Enables main heat exchanger pump, main heat exchanger TEC modules, turns on cooling fan, and enables handpiece cooling functions
4	High Voltage	Enables the shutter logic gate
5	EPI Set Point	Sets the EPI cooler temperature setting to 5ºC
6	EPI Cooler (EPI_ENB)	Enables EPI cooler (current to Lockhart/TEC module).
7	Fire Mode	Not used.
8	Long Pulse	Not used.
9	Word Select 2	Not used.

Table 4-6: Digital Output Bit Definitions

3. Diagnostic Shot Data

   The drop down list box shows the cumulative shot counter, pulse width, measured current on the A array, the measured current on the B array, and the energy meter reading for each shot that occurs. For shots made within the diagnostic screen, the raw meter voltage is also displayed. The list box has 10 lines. The top line (which is always visible) displays the information from the most recent shot. When pulled open, the list box displays an additional nine lines showing the last nine shots.

4. Analog Out

   The Field Service Engineer can set the current set point and pulse width limit analog outputs in the appropriate units.

   0: Current S.P. (A)

   1: P/W Limit (s)

5. Pulse

   The Field Service Engineer can set the pulse width between 1 and 500 ms.

6. A/D Cal Factor

   The Field Service Engineer can adjust this value to make the ADC read +5.00Vdc on the 5 volt power supply input and Analog In Channel #9.

7. Versions

   - The LightSheer software version number is displayed on this screen.
   - The amount of installed DRAM is displayed.
   - The console electronics version is displayed.

8. The "Get Δ Temp" button allows the service engineer to capture the difference between the laser diode temperature and the heat exchanger temperature. This value is stored in NVRAM. The difference is displayed at the bottom right in a field labeled "Δ Temp (ºC)".

9. The "Fire Enable" control, when disabled (OFF), set to disabled, the status of the trigger and footswitch are displayed, and the shutter is opened when the required conditions are met (i.e., high voltage enabled. shutter open, key on), but no shot is fired. When this control is set to enabled, with the key on, the footswitch pressed, firing is allowed when the trigger is pressed. This control defaults to "OFF" when the diagnostic screen is displayed.

10. The CALIBRATE TOUCH button allows the service engineer to adjust the calibration of the touch screen. When pressed, a screen is displayed that prompts the technician to touch a cross hair in the upper left and lower right corners of the screen. The touch must be done with a clean, dry finger. No fingernails or pencil erasers.

11. The SET SHOTS button allows the service engineer to set the value of the cumulative shot counter. Pressing this button opens the ENTER SHOTS screen.

12. The RETURN button allows the service engineer to return to the originating screen.

13. Analog Inputs: The numbers in the boxes indicate the measured or calculated analog values respectively. The analog input bit descriptions are shown in Figure 4.7.

Channel	Signal Name	Nominal Level [conversion factor]
0	Current Mon. A (A)	Varies depending upon requested fluence. 10A/V.
1	Current Mon B (A)	Varies depending upon requested fluence. 10A/V.
2	E. Meter (J/cm^2)	Varies depending upon diode current. 0.1V/J.
3	DAC Read back (A)	
4	P/W Read back (s)	Varies depending upon requested pulse width. 1 mV/msec
5	Hi Voltage PS (V)	
6	TEC PS (V)	
7	EPI Temp (ºC)	Normally regulated to 5ºC if enabled but not firing, or 2ºC if enabled and firing. 0.1V/ºC
8	Heat Ex. Temp (ºC)	Normally regulated to 19ºC. 0.1V/ºC
9	5VDC PS (V)	5.00Vdc [±0.10Vdc]
10	Diode Temp. (ºC)	May be slightly higher than 19ºC. 0.1V/ºC
11	Heat Ex. Avg. I (A)	Varies depending upon heat load from diodes and ChillTip loop (and FETs if installed). 2A/V.
12	EPI TEC Avg. I (A)	Varies depending upon heat load at ChillTip. 2A/V.

Table 4-7: Analog Input Channels

4.4.5 Demo Software

To operate the laser in Demo software, gain access to the Service mode screen as described in Section 4.4.3.1 and select “Demo” mode. The laser diodes must be disconnected from the FET Driver Board or the Power Supply. Operating the system in demo mode simulates complete operation of the system without firing the lasers. The system will turn on normally and prompt the user to perform the normal start up sequence. The system will perform a mock 4-point calibration without laser emission and mock treatments can be made. When finished, return the system to Normal Operation (User software) as described in Section 4.4.4.1.

4.5 Electrical Connection and Conditioning

In order to energize the LightSheer ET/ST Diode Laser System, an electrical connection from the facility outlet to the console is required and must be switched on. This is accomplished via facility electrical outlet, a removable power cord, the power input module (power cord connection point), and a main power switch (a circuit breaker). The power cord is connected (friction fit) to the power input module at the rear of the console.

Warning: Electrical Shock Hazard

Electrical shock hazard when system cover is removed. Potentially lethal voltages are present inside the LightSheer ET/ST Diode Laser System whenever connected via power cord to a live electrical circuit. Do not attempt to remove the top cover without first disconnecting the system power cord. Service is to be performed by certified technicians only.

4.5.1 Facility Electrical Outlet

The customer's electrician is responsible for providing an electrical connection in the treatment room(s) where the LightSheer ET/ST Diode Laser System will be used. Furthermore, for international (non-USA) installations, the customer is responsible for supplying a mating plug for the receptacle to be used with the LightSheer ET/ST Diode Laser System.

The electrical requirements for the LightSheer ET/ST Diode Laser System are 100-240Vac, 50/60 Hz, 1Ø, 12A (12A max. for 100V installations), with earth ground.

4.5.2 Power Cord

One of two different power cords is shipped with the LightSheer ET/ST Diode Laser System. The power cords are rated for 1250W service, and both power cords are 2.4m (8 feet) in length. The cords contain 3 wires: hot (brown), neutral (blue), and earth ground (green w/yellow stripe).

For systems shipped to the United States, the power cord is connectorized with a NEMA 5-15 plug that will connect to most standard 3-wire outlets. For systems shipped outside the United States, the plug end of the power cord is bare and ready to accept a plug approved for use in the destination country. The customer is required to supply a plug that mates with their specific outlet configuration.

4.5.3 Power Input Module

Electrical power is applied to the LightSheer ET/ST Diode Laser System at the power input module located at the rear of the console. The power cord must be firmly pushed into the module, and is held in place by friction.

The alternating current enters the module, and then goes directly to the LINE side of the main power switch (circuit breaker). The center wire, earth ground, is connected to the console base plate.

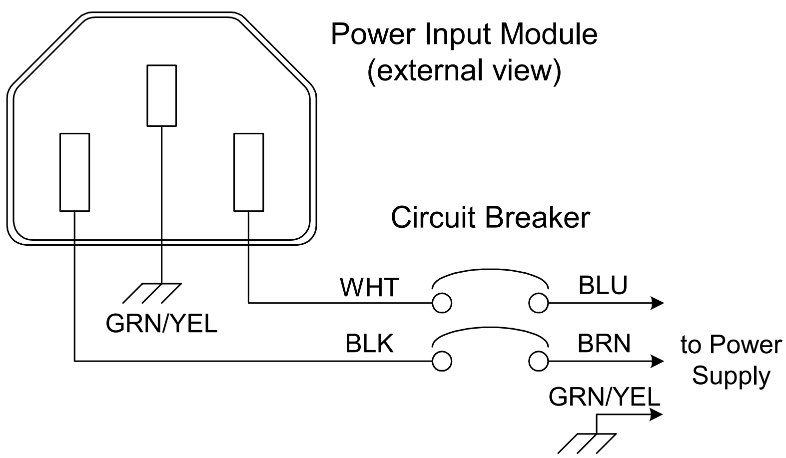


Figure 4-16: Power Input Module, Circuit Breaker

4.5.4 Main Power Switch (Circuit Breaker)

The main power switch, located on the console rear panel just above the power input module, is a circuit breaker rated at 250V/12A. The circuit breaker switches both the hot and neutral lines to the Power Supply.

Anytime the LightSheer ET/ST Diode Laser System is connected via power cord to a live electrical source, high voltage is present inside the console at least up to the circuit breaker. If the circuit breaker is on (closed), the power supply is connected to high voltage. Do not attempt to remove the top cover without first disconnecting the system power cord.

4.6 Power Supply and Laser Drive

4.6.1 Lambda, and FET (Diode Current Control)

Refer to the block diagram in Figure 4-17.

The power supply consists of three output blocks: 1) the Low Voltage power supply outputs for the control electronics, 2) the TEC power supply output for the heat exchanger (cooling system), and 3) the Laser power supply (current supply to drive the laser diodes).

The Low Voltage power supply outputs consists of separate direct current (DC) outputs of +5Vdc @ 2.5A, +12Vdc @ 300mA, and –12Vdc @ 300mA. The outputs are not adjustable.

The TEC power supply outputs consists of a direct current (DC) output of +16.5Vdc @ 22A. The output is not adjustable.

The output of the Laser power supply generates 112 VDC @ 2.7A (negative with respect to ground) which is stored in the capacitor bank. A balanced series capacitor bank is utilized to store current for the laser diode array. The capacitor bank consists of two 180,000µF series capacitors, which combine for a total capacitance of 90,000µF. The capacitors are charged anytime the system is plugged into an AC power source and the circuit breaker turned ON. The energy stored on the capacitor bank can be calculated using the following formula:

$E = \frac{1}{2} C (V^2)$, where $C = 0.09$ and $V = 112$. Therefore, $E = 565$ Joules.

The function of the high voltage relay is to deliver energy stored on the capacitor bank to the FET assembly by way of the FET Driver Board. The FETs are mounted to a heat sink assembly, which is then mounted to the main heat exchanger. A control voltage to the Gate of each FET will allow a measured amount of current to pass from the Drain to the Source. Current passes from the capacitor bank, through the FET to the laser diode array. The length of time that the FET is ON (passing current) defines the width of the laser pulse.

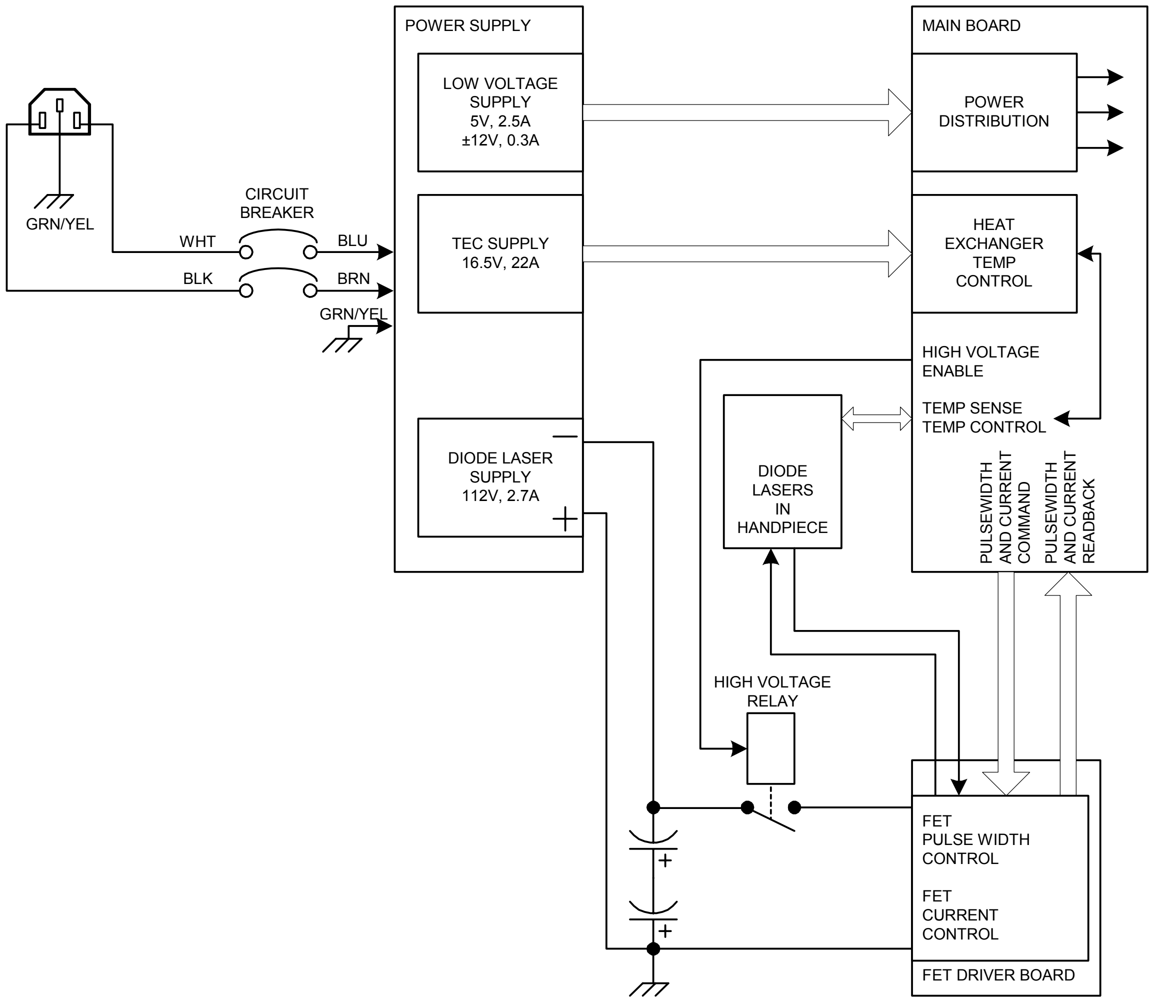


Figure 4-17: Lambda Power Supply and FET Driver Board Block Diagram

For systems with FETs, a detailed connection diagram is shown in Figure 4-18.

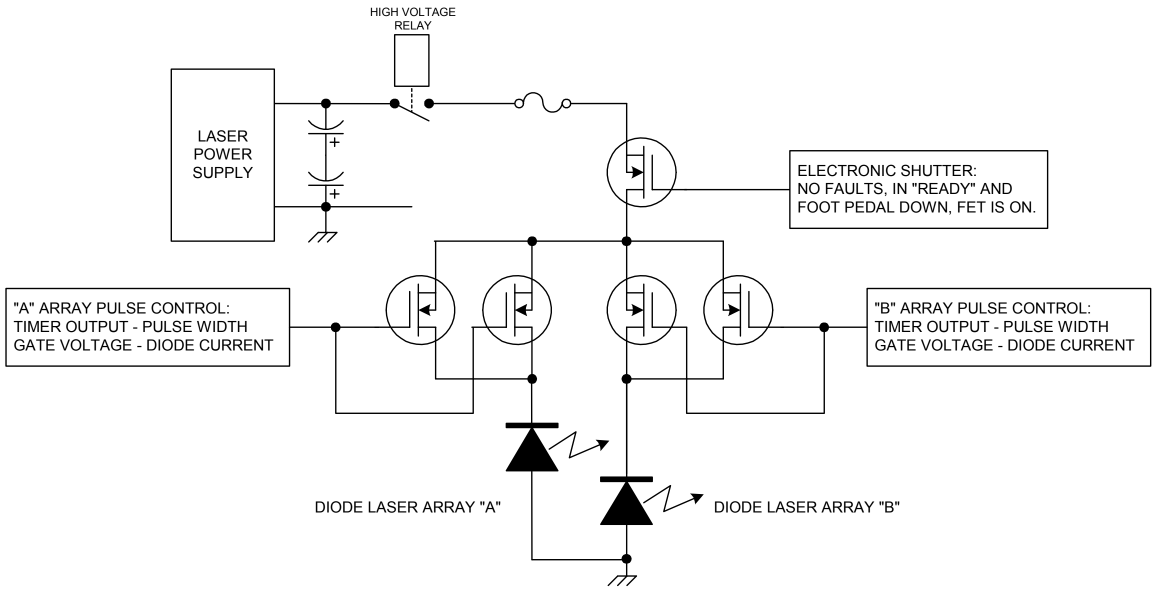


Figure 4-18: FET Connection Diagram

4.6.2 Martek Power Supply

Refer to the block diagram in Figure 4-19.

The Martek power supply, like Lambda, provides three output blocks: 1) the Low Voltage power supply outputs for the control electronics, 2) the TEC power supply output for the heat exchanger (cooling system), and 3) the Laser power supply (current supply to drive the laser diodes), as well as perform the diode laser drive functions (FET Driver Board and FETs).

The Low Voltage power supply outputs consists of separate direct current (DC) outputs of +5.1Vdc @ 2.5A, +12Vdc @ 300mA, and –12Vdc @ 300mA. The outputs are not adjustable.

The TEC power supply outputs consists of a direct current (DC) output of +16.5Vdc @ 22A. The output is not adjustable.

The output of the Laser power supply generates a pulsed 70Vdc @ 50A (negative with respect to ground) which is stored in a capacitor bank. The capacitor bank is utilized to store current for the laser diode array.

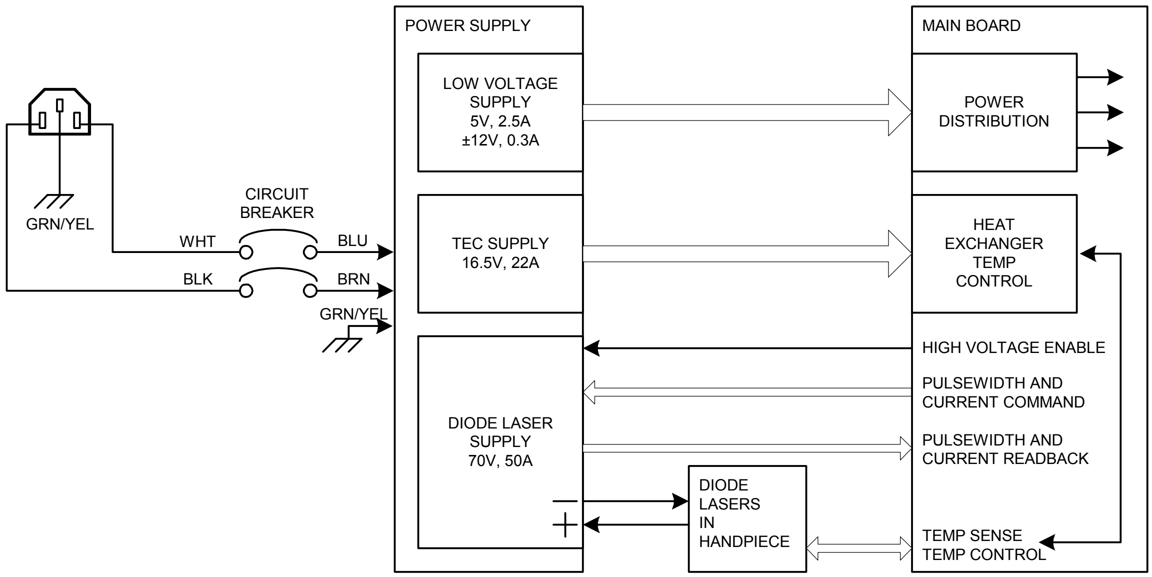


Figure 4-19: Martek Power Supply Block Diagram

4.7 Control Electronics

The control electronics consist of a CPU Controller Board and the Main Board that interface together to distribute power and signals to allow the system to function in both user and service (diagnostic) modes. Refer to the System Block Diagram in Section 8.

4.7.1 CPU Controller Board

The CPU Controller Board is the control center of the system as it runs the system program (software), responds to user inputs, monitors specific analog and digital signals, monitors safety signals, and controls the treatment exposure dosimetry by Field Effect Transistor (diode current flow for fluence control and on-time for treatment pulse width).

The operations described above are controlled by a Motorola ColdFire microprocessor, U13, which runs specific software routines from memory. The controller board flash memory, U15, is programmed with software at the factory, while system specific (changeable and updateable) data is stored in non-volatile memory and RTC (Real Time Clock) device, U18. Up to 32 megabytes of RAM are installed in SIMM connector J15 for computer use.

The board contains a video controller chip, U12, which handles the LDC screen (pixel data) communications between the LCD screen (pixel data) and the controller board. Additionally, the board provides the display panel electronics with power, and a dual channel UART (Universal Asynchronous Receiver/Transmitter) is used as the serial port communications with the touch sensitive overlay.

The CPU Controller Board also contains data latches for handling digital data, and ADC (analog-to-digital converter) and DAC (digital-to-analog converter) devices for handling analog command and read-back signals.

4.7.2 PWA Main Board

The PWA Main Board is the primary interface board in the LightSheer ET/ST Diode Laser System. All signals, except display communication, are routed through this board. The Main Board distributes the low voltage direct current from the power supply to all system circuits. Also, the circuitry on the Main Board receives, processes and transmits analog and digital signals between the CPU Controller Board and the attached electro-optical, electrical, thermal and mechanical components.

4.7.3 Energy Meter Board

The Energy Meter Board is mounted from the inside, directly to the top cover at the calibration port. The energy meter board contains a photodiode (energy meter) and connectors to pass the energy signal to the PWA Main Board.

4.7.4 PWA Fan Filter Board

The Fan Filter Board is located at the console right rear and mounted to the cooling fan assembly near the Footswitch/Interlock panel. The filter board is designed to eliminate acoustic noise generated by the fan.

4.8 Cooling System

4.8.1 Overview

Refer to the drawing in Figure 4-20.

The purpose of the cooling system is two-fold: 1) to remove heat produced by the laser diode arrays, and 2) to remove heat produced by the Lockhart/TEC Module (which cools the ChillTip). The coolant, when forced through the hot sides of micro-channel copper heat exchangers in the handpiece (the back plane heat exchanger and the Lockhart/TEC Module) absorbs heat. The heated coolant is forced through the cold side of another micro-channel copper heat exchanger, the main heat exchanger, where heat is transferred through the TEC modules to the heat sink. Room temperature air blown through the heat sink transfers the heat from the heat sink to the exhaust air.

The coolant loop, a closed loop, forced air heat exchanger, contains approximately 475ml (⅛ gal.) of anti-freeze (1 part ethylene glycol to 4 parts distilled water). A temperature dependent speed controlled fan blows air into the system and exits out of the base plate then dispersed into the treatment room. If the system is equipped with the Lambda power supply, FET Driver Board and FETs, the cooling system also provides cooling of the power FETs. The FETs will be mounted directly to the Main Heat Exchanger.

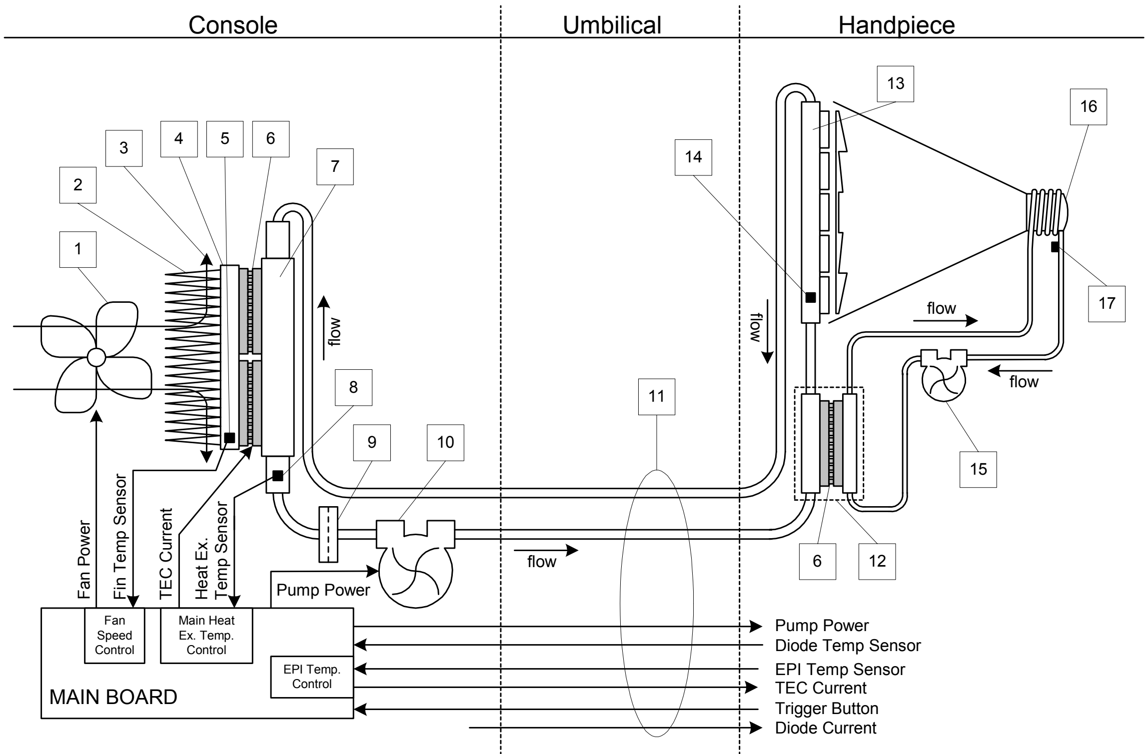


Figure 4-20: Cooling System Diagram

4.8.2 Components

The components in the cooling system (main console and handpiece) are identified by the numbered boxes shown in Figure 4-20. The cooling system consists of the following components:

Coolant - A mixture of 1 part ethylene glycol (anti-freeze) to 4 parts distilled (de-ionized) water is used in both the main heat exchanger cooling loop and the

ChillTip cooling loop. Combined, both cooling loops use approximately 475ml (⅛ gal.) of coolant.

Main Heat Exchanger Temperature Controller - The main heat exchanger temperature controller is incorporated into the Main Board. The function of this controller is to maintain a coolant temperature of 19ºC in the main heat exchanger loop. The coolant temperature is monitored so that if the coolant gets warmer than the set point temperature of 19ºC, it allows current to flow through the TEC devices. A greater temperature difference (above the 19ºC set point) allows more current to flow through the TEC devices.

EPI (ChillTip) Temperature Controller - The EPI (ChillTip) temperature controller is incorporated into the Main Board. The function of this controller is to maintain a coolant temperature of 5ºC (if ChillTip is ON but not firing) or 2ºC (if ChillTip is ON and firing) in the EPI loop. The coolant temperature is monitored so that if the coolant gets warmer than the set point temperature of 5ºC, it allows more current to flow through the TEC devices in the Lockhart assembly. The temperature of the ChillTip is controlled to be 5ºC ± 2ºC (if ChillTip is ON but not firing) or 2ºC ± 2ºC (if ChillTip is ON and firing).

Fan Speed Controller - The fan speed controller is incorporated into the Main Board. The function of this controller is to affect the rotational speed of the fan. The heat sink temperature is monitored so that as the heat sink gets warmer, the fan speed increases, forcing more air through the heat sink. A higher heat sink temperature allows more current to flow through the fan motor.

1 **Fan -** The large fan, mounted to the cooling fins of the heat sink at the rear of the chassis, draws ambient air in through the rear console vent, forcing it through the cooling fins on the heat sink, and out of the console through slots in the bottom base plate. The speed of the fan is proportional to the temperature of the heat sink.

2 **Cooling Fins (part of Heat Sink) -**The purpose of the cooling fins is to increase the surface area of the heat sink, allowing the heat sink to radiate heat more efficiently to the ambient air that is forced into the cooling fans by the action of the fan.

3 **Ambient Air -** Ambient air is drawn into the console from the vent at the rear of the console, forced through the cooling fins on the heat sink, and exhausts out the vents in the console base plate. To prevent the system from overheating, the ambient air must be between 15ºC and 27ºC (60ºF and 80ºF), and the temperature must be above the dew point.

4 **Heat Sink -** The heat sink is mounted to the hot side of the TEC modules of the main heat exchanger assembly. Heat from the TEC modules is absorbed by the heat sink, which dissipates heat as the fan draws air across the cooling fins.

5 **Heat Sink Temperature Sensor -** A two-terminal integrated circuit temperature transducer (thermistor) is used to measure the heat sink temperature. The device produces an output current proportional to absolute temperature, 1µA/ºK. The temperature of the heat sink is monitored and used to control the speed of the fan.

6 **Thermo-Electric Cooler (TEC) -** The Thermo-Electric Cooler, a Peltier device, is a two-sided ceramic component where one side gets hot, drawing heat from the other side (which gets cold) as current is passed through it. A number of TEC devices are used in the main heat exchanger as well as in the Lockhart/TEC Module. In the Main Heat Exchanger, the hot side of the TEC is mounted to the heat sink (with attached cooling fins), while the cold side is mounted to the heat exchanger. In the Lockhart/TEC Module, the hot side of the TEC is in-line with the Diode Back Plane Heat Exchanger, while the cold side is mounted to the heat exchanger.

7 **Main Heat Exchanger -** The coolant flows through a micro-channel copper heat exchanger where heat in the coolant is absorbed by the metal and is transferred to (removed by) the cold side of the TEC devices.

8 **Main Heat Exchanger Temperature Sensor -** A two-terminal integrated circuit temperature transducer (Analog Devices AD590) is used to measure the heat sink temperature. The device produces an output current proportional to absolute temperature, 1µA/ºK. Originally, this device was mounted in contact with the heat exchanger, just below the output fitting. Currently, and as a field replacement part, an in-line temperature sensor is installed on the output fitting, placing the probe in direct contact with the coolant.

9 **Coolant Particle Filter -** The coolant flows through a fine mesh screen particle filter to trap large particles. The filter may be split in two (unscrew the sides) to inspect and clean the screen.

10 **Main Pump -** A constant flow rate pump is used to force coolant (at 19ºC) from the main heat exchanger out to the handpiece and back. The flow rate is approximately 1.1 liter/min. at 1 bar (0.3 gal/min. at 15 psi).

11 **Umbilical Cord -** The connection between the console and the handpiece, the umbilical cord contains wires and tubing to transfer electrical signals and coolant between the two assemblies.

12 **Lockhart/TEC Module -** The Lockhart/TEC Module is a micro-channel liquid-to-liquid heat exchanger located in the handpiece. Coolant at 19ºC from the main heat absorbs heat from the hot side of the TEC module before going to the diode array back plane heat exchanger. The cold side of the Lockhart/TEC module removes heat from the EPI (ChillTip) loop.

13 **Diode Array Back Plane Heat Exchanger -** The coolant flows through a micro-channel metal heat exchanger where heat, generated by the laser diodes, is absorbed by the metal and transferred to the coolant as it is forced through. Heated coolant travels from the diode array back plane heat exchanger to the main heat exchanger in the console.

14 **Diode Temperature Sensor -** A two-terminal integrated circuit temperature transducer (thermistor) is used to measure the diode back plane heat exchanger temperature. The device produces an output current proportional to absolute temperature, 1µA/ºK.

15 **Handpiece Pump -** A constant flow rate pump is used to force coolant from the ChillTip to the Lockhart/TEC Module and back.

16 **ChillTip -** The ChillTip is the optical element (sapphire) that transmits the laser light generated by the diodes from the handpiece to the skin. The sapphire optic is wrapped in a blanket of cold coolant, and when in contact with the skin, removes heat from the skin (cools the skin) to provide pain relief and to prevent thermal damage to the skin. The temperature of the ChillTip is controlled to be 5ºC ± 2ºC (if ChillTip is ON but not firing) or 2ºC ± 2ºC (if ChillTip is ON and firing).

17 **EPI Temperature Sensor -** A two-terminal integrated circuit temperature transducer (Analog Devices AD590) is used to measure the EPI (ChillTip) temperature. The device produces an output current proportional to absolute temperature, 1µA/ºK.

4.9 Delivery System (Optical)

4.9.1 Overview

The delivery system (handpiece) consists of the diode laser array (multiple semiconductor lasers), an optical path to deliver the laser light to the patient skin, a cooling system (as described in Section 4.8), and a trigger switch to activate laser pulses. All of the components are contained within the handpiece covers. The handpiece is connected to the system console via the umbilical cord. Refer to Figure 4-21.

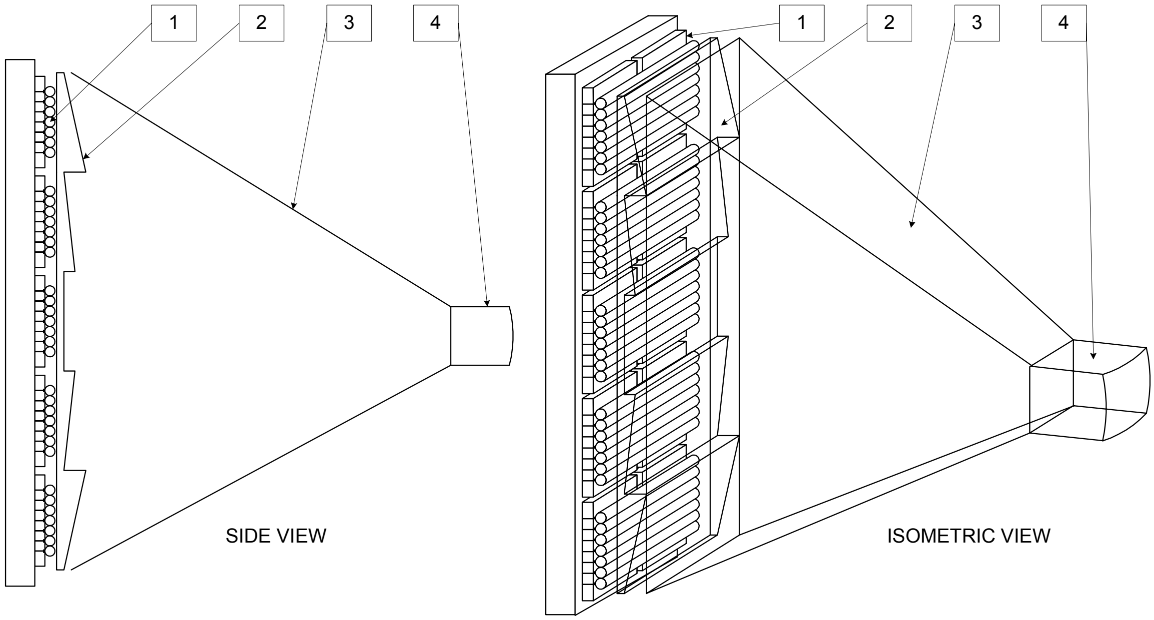


Figure 4-21: Handpiece Optical System

4.9.2 Components

1 **Laser Diode Array -** The laser diode array consists of two parallel columns (side A and side B) of five banks of diode modules each. Each diode module consists of 7 diode bars, and each diode bar contains approximately 19 diode junctions. A cylindrical micro lens to collimate the laser beams covers each diode bar. The array is mounted to the Diode Back Plane Heat Exchanger (see Section 4.8), which removes the heat generated by the current flow through the diode bars.

2 **Fresnel Lens -** A Fresnel lens is used to collect the collimated laser light from the laser diode array and direct the light towards the sapphire tip.

3 **Condenser -** a highly polished gold plated wave-guide (condenser) is used to direct any scattered laser light from the Fresnel lens and deliver it to the sapphire tip.

4 **Sapphire Tip (EPI or ChillTip) -** A 9mm x 9mm (cubic) sapphire lens acts as the output aperture of the Handpiece. Laser light is transmitted through the sapphire tip to the patient skin. The tip is wrapped in the EPI cooling loop to maintain a tip temperature in the range of 5ºC ± 2ºC (or 2ºC ± 2ºC while the handpiece trigger is depressed).

Trigger - A trigger switch is pulled to command the laser to fire. The trigger switch is active whenever the laser is in the Treatment Mode and the shutter is open (footswitch down). The trigger remains active as long as no faults are detected and the footswitch remains depressed.

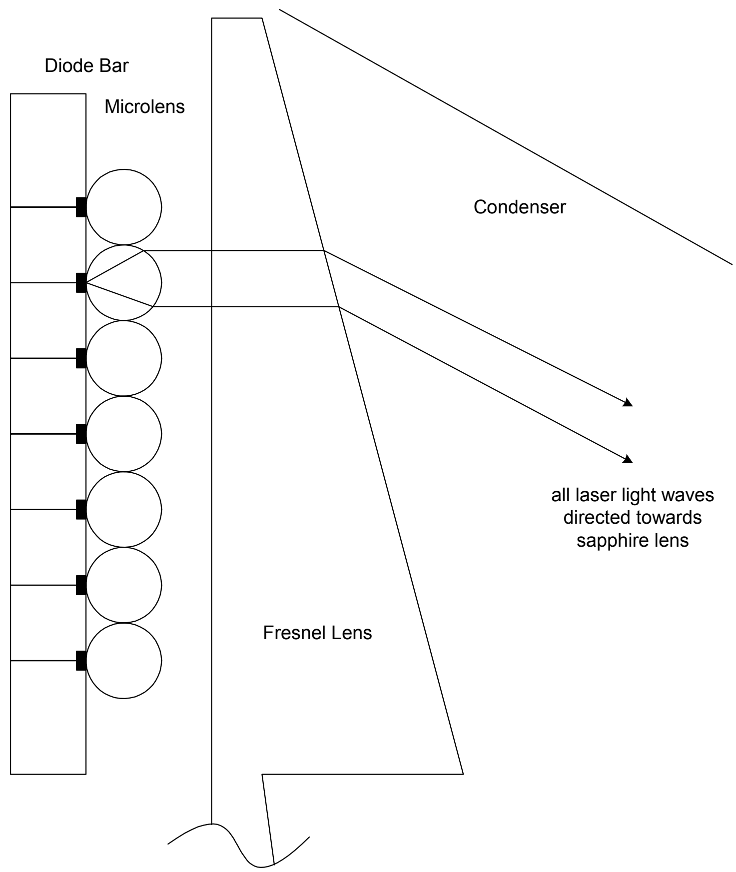


Figure 4-22: Laser Diode Array Optical Detail

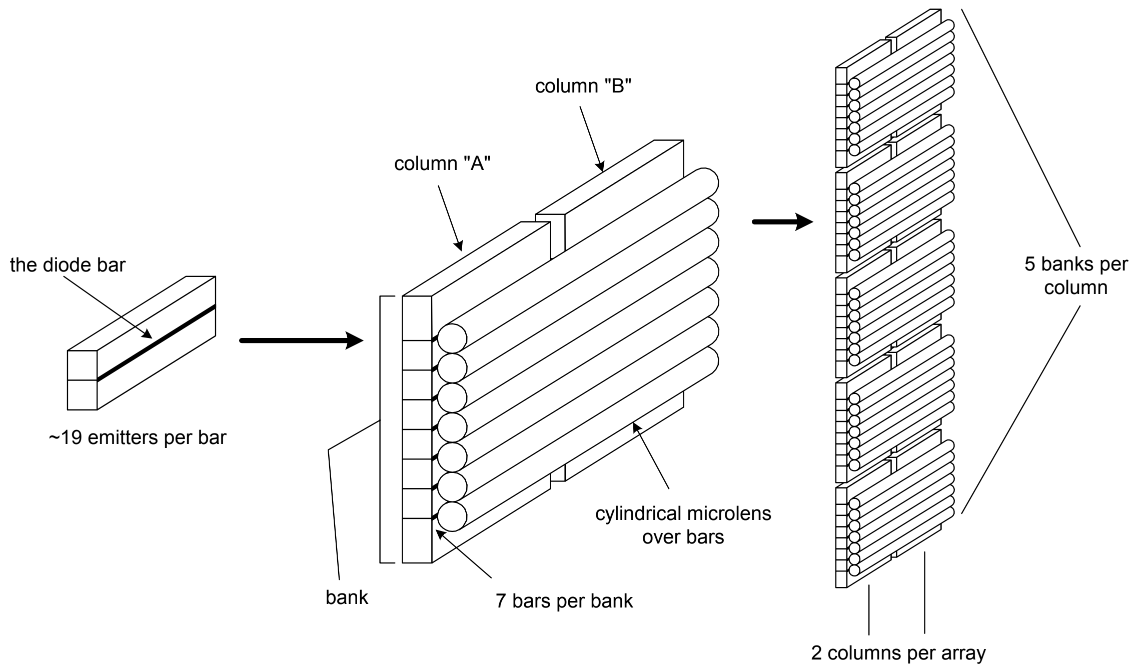


Figure 4-23: Laser Diode Array Construction Detail

5.0 Troubleshooting

5.1 Overview

The LightSheer ET/ST Diode Laser System is designed to require little adjustment or calibration, and to detect and report malfunctions by error or fault code or message as displayed on the color display panel. Replacing high-level assemblies generally repairs system failures. Most assemblies are built specifically to support a field repair, and consist of a part or group of parts determined to be suitable for field replacement. When a part fails that is a part of an assembly, the assembly is replaced, not the individual part.

A service engineer or technician who has completed the Lumenis certification service training for the LightSheer ET/ST Diode Laser System system must perform corrective or preventive maintenance.

Field calibration, adjustment and test procedures are covered in detail in Section 3. Special purpose tools and test equipment are required to repair and maintain the LightSheer ET/ST Diode Laser System operation at factory specifications. The tools are listed in Section 6.

The entire optical path is sealed within the handpiece assembly. Removing the covers and disassembling the handpiece exposes the interior contents to foreign matter (i.e. dust, humidity, contaminants, etc.), and therefore should be done only in a clean environment and only by specially trained service technicians. In general, do not remove the covers or disassemble the handpiece. If it is absolutely necessary to remove any cover, minimize exposure by removing the cover only in an environmentally controlled clean room. Use a clean plastic sheet to cover the system while the cover is off, and replace all covers as soon as the repair or service activity is complete.

After power up, and before the LightSheer ET/ST Diode Laser System moves to its standby condition, the software performs a series of self-tests. Self-test failures result in fault codes or messages displayed on the color display panel. These error or fault codes provide an indication of what malfunction was detected, which should point to a specific area of the system for further investigation. Explanations of the fault codes are included in this section. Most hardware malfunctions will be detected at this time.

During normal operation the software continues to monitor for system malfunctions, and to report any detected malfunctions by fault code on the color display panel.

The LightSheer ET/ST Diode Laser System has service software diagnostic routines available to the service engineer. The diagnostic routines are contained in the EPROM and are activated by switching closing the service switch, SW1 on the PWA Main Board, or by touching the display panel in the manner described in Section 4.4.4.1. The diagnostic routines facilitate calibration and troubleshooting. The diagnostic routines are described in detail in Section 4.4.4.

5.2 Interior Access and Part Location

Note: There are no User serviceable parts within the system console.

Warning: Electrical Shock Hazard

Electrical shock hazard when system cover is removed. Potentially lethal voltages are present inside the LightSheer ET/ST Diode Laser System whenever connected via power cord to a live electrical circuit. Do not attempt to remove the top cover without first disconnecting the system power cord. Service is to be performed by certified technicians only.

Warning: Electrical Shock Hazard

Electrical shock hazard when system cover is removed. A lethal electrical charge may be present on the storage capacitors of the power supply. Do not attempt to service the power supply. When working on or around the power supply, verify that the capacitor charge has dissipated before performing service to the system. Service is to be performed by certified technicians only.

Access to the console interior is gained by removing the top cover. Always disconnect the console from electrical service before removing the cover.

5.2.1 Top Cover Removal

Refer to Figure 5-1.

To remove the top cover, disconnect the power cord. Position the front of the console over the edge of a desk or workbench and remove two front bottom screws (that come up through the base into the top cover). Carefully reposition the console on the desk to locate and remove the side bottom screw, toward the rear, one screw at each side. Once the screws have been removed, carefully lift the top cover straight up and away from the console. If necessary, disconnect the cables that connect the PWA Energy Meter Board and the Proximity Sensor (part of the top cover) and the PWA Main Board. Set the top cover aside.

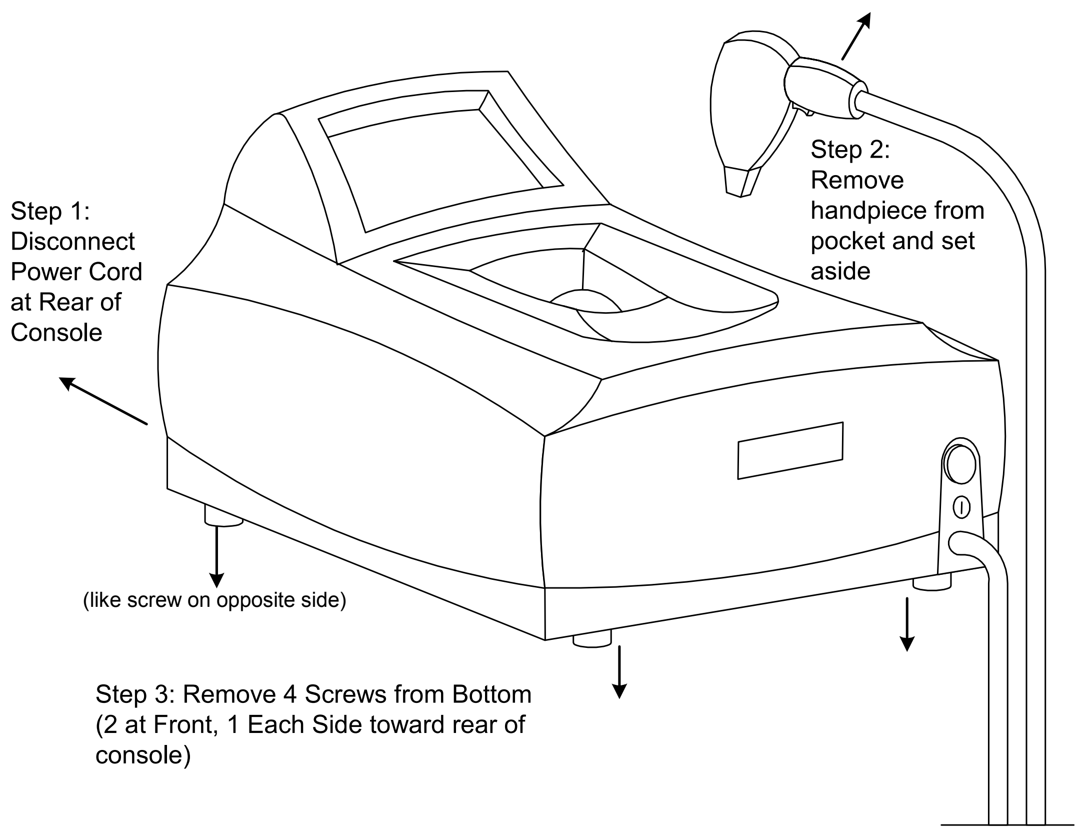


Figure 5-1: Top Cover Removal

5.2.2 Component Location, Base Plate

Please refer to the illustration in Figure 5-2. The following components or assemblies are mounted to the base plate:

- Power Supply, Martek (see note below)
    - Note: If Lambda Power Supply, include FET on Heat Exchanger Assembly and PWA FET Driver Board

- Heat Exchanger Assembly (TEC modules, heat sink with cooling fins, temperature sensors, plumbing fittings)
- Pump Assembly
- Fan
- Fan Filter Board
- CPU Controller Board (under a metal cover)
- PWA Main Board
- Display Assembly
- Key switch and Emergency Stop Switch Assembly
- Remote Interlock and Footswitch Connector Assembly
- Power Input Module and Main Power Switch (Circuit Breaker) Assembly
- Rubber Feet (underneath)

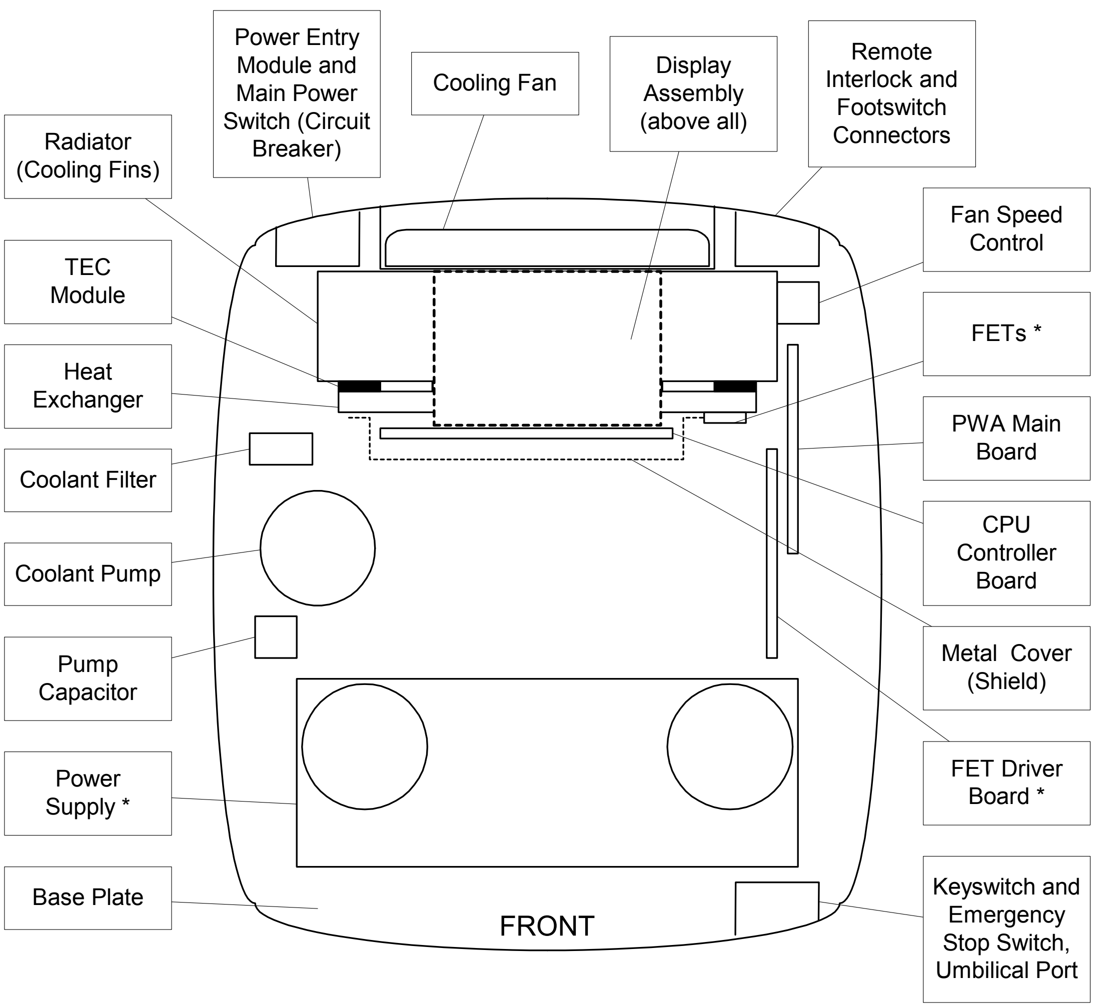


Figure 5-2: Component Location, Base Plate

5.2.3 Component Location, Top Cover

Please refer to the illustration in Figure 5-3. The following components or assemblies are mounted to the top cover:

- Handpiece Pocket
- Calibration Port Glass
- Proximity Sensor
- Energy Meter Board (includes energy detector)

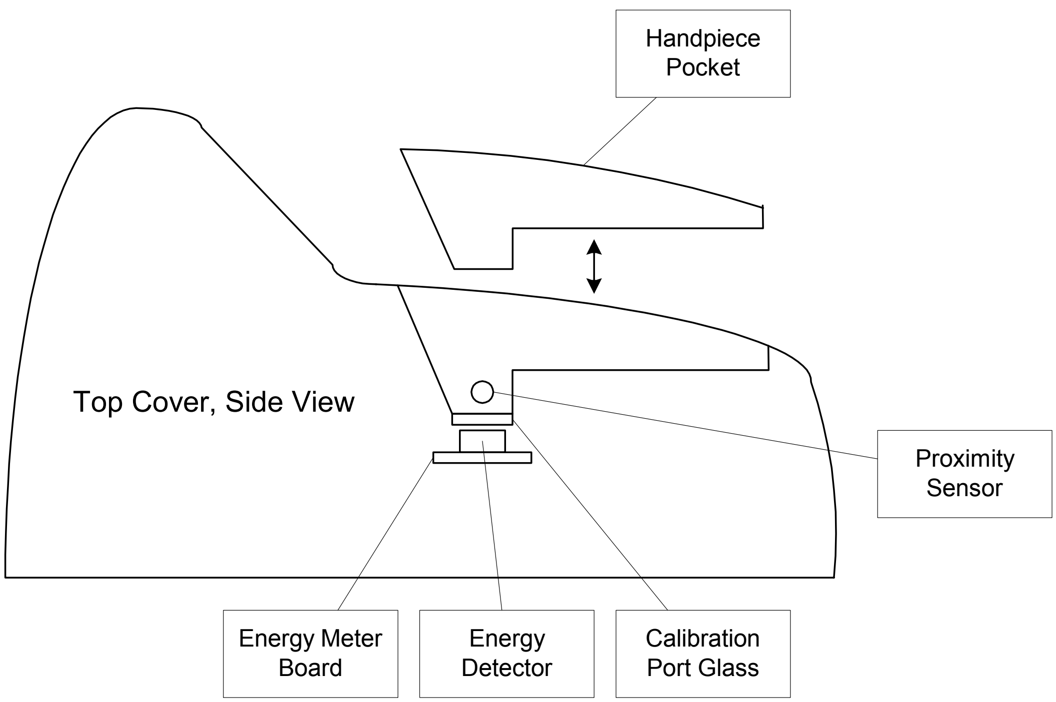


Figure 5-3: Component Location, Top Cover

5.2.4 Component Location, Handpiece Assembly

Please refer to the photographs in Figures 5-4, 5-5, and 5-6. The following components or assemblies are a part of the Handpiece Assembly:

- Umbilical Cord
- Handpiece Covers
- Condenser Assembly (includes Fresnel lens and ChillTip)

- Diode Back Plane Assembly (includes diode arrays mounted to back plane heat exchanger, and micro lens assembly)
- Lockhart/TEC Module
- EPI (ChillTip) Loop Pump
- Trigger button, trigger switch

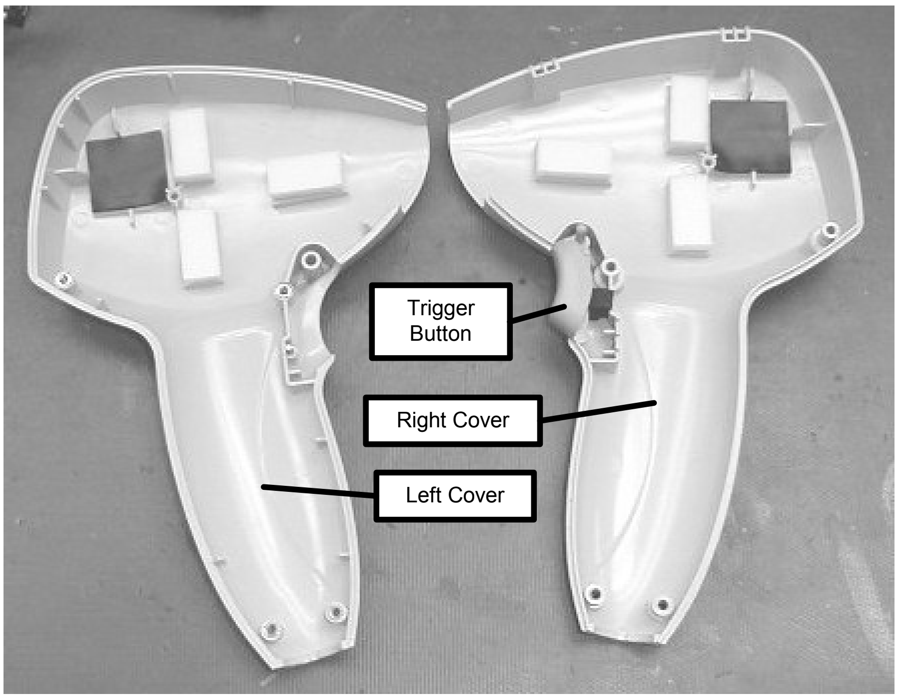


Figure 5-4: Handpiece Covers

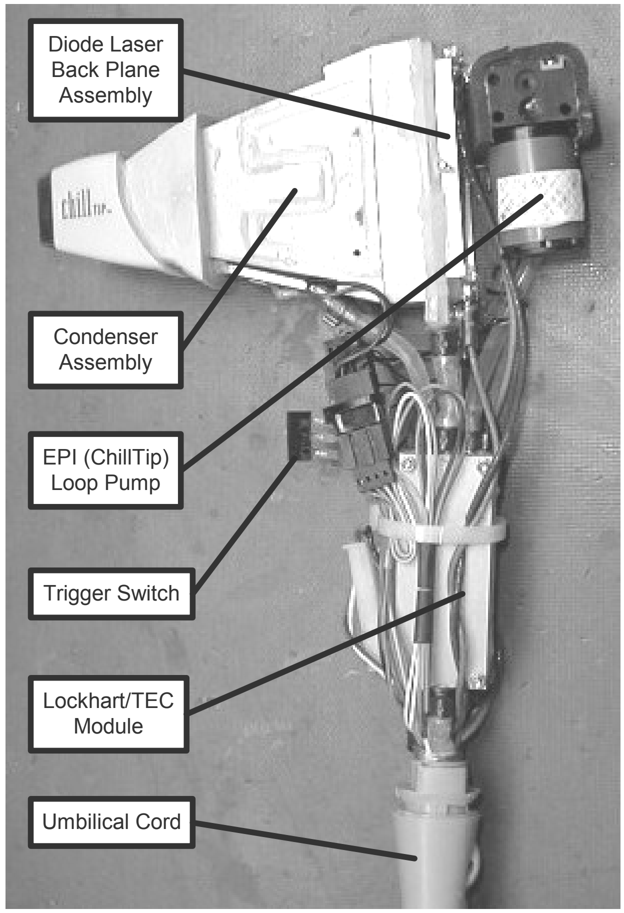


Figure 5-5: Component Location, Handpiece Assembly, Left Side View

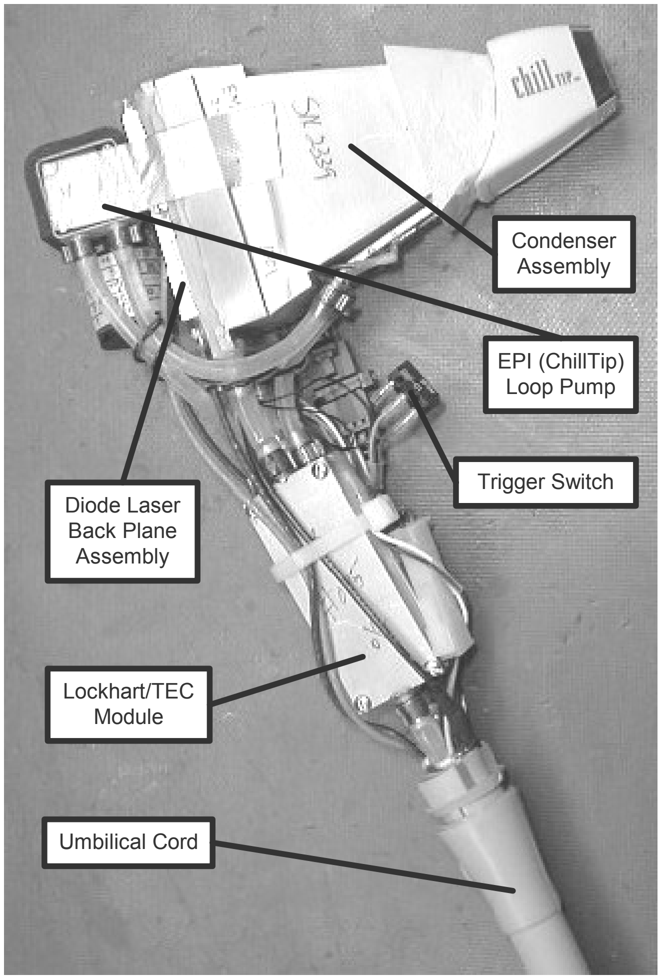


Figure 5-6: Component Location, Handpiece Assembly, Right Side View

5.3 Error Codes and Fault Isolation

The safety circuits monitor system functions to detect and report various conditions within the system and connected accessories. Typically when an error is detected, the microprocessor closes the shutter, places the system into the Standby state, and displays the error condition with a numeric error code and/or icon on the display panel.

The following Error Code Table lists error (fault) codes that are reported by system software. The error codes may change with software revision. Contact Technical Support for verification of error codes.

Some error conditions may be cleared by rotating the key switch, while others require cycling system power (or by resetting the microprocessor). Always verify that power supply outputs are correct and that all cables and connectors are properly terminated.

Code	Description of Fault or Condition	Likely Cause	Possible Remedy
E1	BIOS Memory Failure	Corrupted software or error on CPU Controller Board.	Cycle power. If problem persists, replace CPU Controller Board.
E2	I/O Board (Computer Failure)	Corrupted software or error on CPU Controller Board.	Cycle power. If problem persists, replace CPU Controller Board.
E3	Persistent Memory (NVRAM) Failure	Corrupted software or error on CPU Controller Board.	Cycle power. If problem persists, replace CPU Controller Board.
E4	Shutter (Electronic) Failed to Open	Laser Power Supply error (incorrect output voltage), or bad FET, Q11 (voltage across FET Q11 is >50V, indicating FET is off, when it should be ON) or bad FET Driver Board (only on systems with Lambda Power Supply.	If Lambda Power Supply is installed: check output voltage (measure -112V on both sides of F1 on FET Driver Board), also check FET Q11. If Martek Power Supply is installed: read output voltage on Diagnostic Screen (High Voltage = ON, foot pedal down).
E5	Shutter (Electronic) Failed to Close	If Lambda Power Supply is installed: Voltage across FET Q11 is <50V (indicating FET is on, when it should be OFF).	For either Lambda or Martek power supply: if the foot pedal is used as the fire trigger (instead of the handpiece trigger button), the software may intermittently display this fault. Advise the User to hold the foot pedal down, and request fire pulses with the handpiece trigger button.

E6	Over Current A Array	Current through diode array A exceeded the limit	Check for any loose wire connection. Of Lambda power supply installed, check FETs Q9, Q10, Q11, Q12 and Q13. Replace FET(s) as required. Verify all FET screws are tight. Also check R1 and R13 for 0.05 Ohms. Replace resistors or FET Driver Board as required. If Martek power supply installed, check connections. If all of the above check out OK, replace the Handpiece.
E7	Over Current B Array	Current through diode array B exceeded the limit.	Same as E6.
E8	Over Charge A Array	The integrated current pulse through the diode array A exceeded the set point.	Same as E6. Then, check whether the voltage of the fluence set limit at TP108 is set to +4.96V (for ET) and +3.51V (for ST). If not, adjust R79 for proper voltage.
E9	Over Charge B Array	The integrated current pulse through the diode array B exceeded the set point.	Same as E6. Then, check whether the voltage of the fluence set limit at TP108 is set to +4.96V (for ET) and +3.51V (for ST). If not, adjust R79 for proper voltage.
E10	Over Power A Array	Only on systems with Lambda Power Supply.	Check FET operation and verify 2.5V and 7.5V reference sources. Make sure that the drive pulse /DRV_PLS at TP22 goes high for the duration of the laser pulse. Verify pulse width reference voltage at TP19. Verify diode voltage (1V/10V at D11 and D14 cathode). Verify laser current feedback is correct (1V/10A). Verify operation of Q16 and correct comparator voltages at U28C and U28D.
E11	Over Power B Array	Only on systems with Lambda Power	Same as E10.

		Supply.	
E12	Over Voltage A Array	The voltage across the A Array is greater than 75V, either because of bad FETs, or too many shorted diodes (in the handpiece).	Check the FETs for proper operation (see FET Test section). Verify 7.5V reference at U27-5 and U27-7.
E13	Over Voltage B Array	Same as E12, but with respect to B Array components.	See E 12 above, but substitute B Array signals/components.
E14	Pulse Width Error	The actual laser pulse (on time) has exceeded the hardware limits of 36ms for short pulse mode or 120ms for long pulse mode, or the actual laser pulse (on time) was more than 2ms longer than requested.	Verify hardware limits of 6.67V (long pulse) and 5.85V (short pulse) at TP29. Verify hardware limits of –1.63V (long pulse) and –0.53V (short pulse) at TP19. With oscilloscope, measure laser current pulse (on time) at TP13 (A Array) or TP21 (B Array).
E15	Calibration Out of Range	The diode laser light output as measured by the console energy detector is too high or too low.	1. Clean handpiece sapphire tip and cal port glass, then recalibrate (User calibration). 2. Verify actual handpiece output with an external energy meter. Remember to multiply the Requested Fluence by 0.81 to get pulse energy. 3. Verify that Fmin and Fmax values in Setup screen are properly set.
E16	Calibration Energy Value Not Expected	The diode laser light output as measured by the console energy	1. This error code means that the energy of shot 2 < shot 1 or when the energy of shot 4 < shot 3. 2. Same as E15. May be drifting

		detector is not as expected (should be: increased current = increased light output = increased detector signal).	energy. Verify the energy with an external energy monitor. 3. Compare the energy with an external energy monitor, as there may be intermittent contacts.
E17	Calibration Inconsistent	(4th shot ~10% out - same as E15)	Clean handpiece output lens and cal port glass, then recalibrate.
E18	Current Tolerance Error	Current through either Diode Array is measured to be greater then 5% + 1.5A over the commanded current.	Check all connections and verify they are properly seated. Check all screws on the FET Drive Board and on FETs. Check power supply outputs. Set the current to 10A, PW to 20mS and the Pulse Limit to 22mS, then verify that the current is within the specified tolerance (Iset ±5% ±1.5A).
E19	Pulse Energy Too Low	Pulse Energy Too Low	Same troubleshooting procedure as described in E15. If Lambda power supply installed, the FETs might be leaking, check and replace FET(s) as required. Verify laser diode electrical connections to Martek power supply. Replace the handpiece if no problems are found in the electronics.
E20	DAC Read-back Port A Error	The DAC is not functioning properly.	Bad CPU Controller Board. Replace the CPU Controller Board.
E21	DAC Read-back Port B Error	[NOT USED]	N/A
E22	Pulse Width Limit Error	The pulse width limit command (a voltage that	Cycle power. If problem persists, replace CPU Controller Board.

		represents 2ms over the requested pulse width) cannot be read back by the ADC.	
E23	Laser Driver (/DRV CW) Bit Error	The FET Driver Board or Martek Power Supply indicated that the pulse width was greater than 300msec.	If Lambda Power Supply, replace FET Driver Board. If Martek Power Supply, replace supply.
E24	Low Voltage Power Supply Voltage Out of Tolerance	The computer has detected one or more of the low voltage power supplies is out of range (±12Vdc or +5Vdc).	Measure power supply voltages (+12V at TP9, -12V at TP5, and +5V at TP3) on Main Board. Check operation of Q13 and U20E.
E25	Laser Diodes Connected in Demo Mode	The computer has detected that the laser diodes are connected while in Demo Mode.	If using the laser in Demo Mode, disconnect the laser diodes from either the FET Driver Board or from the Martek Power Supply.
E26	High Voltage Power Supply Voltage Out of Tolerance	The computer detects that the high voltage (laser) power supply output is ±15V out of the set point range (as entered in Setup Screen HV PS SP (V).	In Setup Screen, verify HV PS set point of 112V for Lambda Power Supply, or 70V for Martek Power Supply. Read power supply output read back (Analog In #5) when laser is Enabled and foot pedal is down.
E27	TEC Power Supply Voltage Out of Tolerance	The computer detects that the TEC power supply voltage is ±2V out of set-point range (as entered in Setup	In Setup Screen, verify TEC PS set point of 16V. Read power supply output read back (Analog In #6) when Heat Exchanger, EPI Set-point, and EPI

		Screen TEC PS SP (V).	Cooler are turned on.
E28	Pulse Lockout Error	The software detected inadequate delay time between pulses (the forced delay time is 240ms between pulses).	Using the positive edge of DRV_PLS signal (TP22 on Main Board) as oscilloscope trigger, verify U19-12 (/Q output) strobes HI and does not go LO for 240 milliseconds.
E29	Board Revision (Compatibility) Error	The software and hardware are not compatible (according to voltage reading).	Verify +1.5V at TP20 on Main Board. If present, check connection between Main Board (J1) and CPU Controller Board (J16). If voltage is present at ADC U27-21 on CPU Controller Board, replace CPU Controller Board. If +1.5V is not present at TP20 on Main Board, verify negative (- 12V) power supply, then measure -2.5V reference voltage at Zener Diode U40-4.
System Stuck in "Cooling" Mode	The ChillTip is too hot.	The EPI (ChillTip) temperature is 5ºC over the set point temperature.	ChillTip set point temperatures: EPI OFF = 25ºC, EPI ON = 5ºC, EPI ON and Trigger Pressed = 2ºC. Verify adequate coolant flow between Lockhart/TEC Module, ChillTip and pump in handpiece. Calibrate cooling loop (see Section 3).
P1	Emergency Stop Button is Depressed	The emergency stop button is active (pressed).	Reset (rotate) the emergency stop switch, check switch electrical, and confirm connection of J12 on PWA Main Board.
P2	Door Interlock (Remote Interlock) is Open	The remote interlock plug is missing, not fully inserted, or if connected to a treatment room door switch, the	Verify connector (plug or wires from door switch) is inserted at rear of console. Check J13 connection on PWA Main Board.

		door is not fully closed or the door switch is faulty.	
P3	Laser Diode Too Hot (Over Temperature)	The temperature of the Diode Laser Back Plane Heat Exchanger is greater then 31ºC as measured by the computer.	Verify adequate coolant flow to Lockhart/TEC Module in handpiece. Calibrate cooling loop (see Section 3). If Main Heat Ex. Temp. and Diode Temp. are both over 31ºC, check the fan. If Main Heat Ex. Temp. is normal, but Diode Temp. is above 31ºC, suspect pump or kink in umbilical cord.
P4	FET Too Hot (Over Temperature)	The temperature of the FET is measured to be greater then 52ºC as measured by the computer.	Verify cooling fan operation. Verify cooling fins on heat sink are clean and unobstructed. Calibrate cooling loop (see Section 3).
P5	$\lvert\Delta T\rvert > 4°C$ (Main Heat Exchanger Temp. and Diode Temp.)	The temperature difference between the Main Heat Exchanger and the Diode Laser Back Plane Heat Exchanger is greater than 4ºC.	Verify adequate coolant flow between Main Heat Exchanger and handpiece. Calibrate cooling loop (see Section 3).

Table 5-1: Error Code Troubleshooting Table

5.4 Test Point and Connector Arrangement

The following information is provided as a guide for troubleshooting purposes only. Always verify part locations, connection designators, test point numbers and jumper configuration by schematics in Section 8.

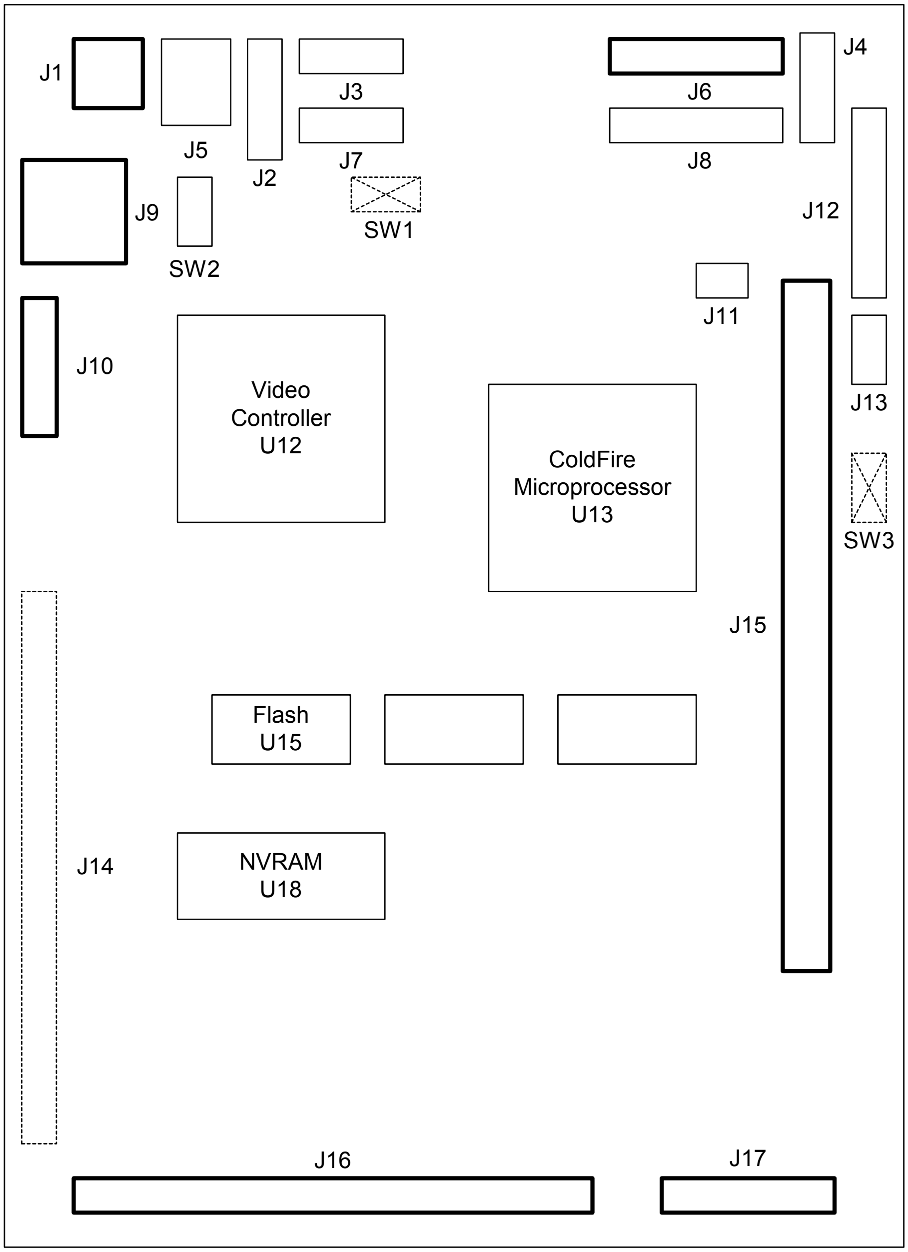


Figure 5-7: CPU Controller Board

CPU Controller Board			
Connectors	J1	2-pin twisted pair carries +5Vdc and ground.	Connects to ELO (display) interface board.
	J2	8-pin connector carrying VGA display signals.	Not Used.

	J3	6-pin single-row ribbon cable, for SX BDM serial port.	Not Used.
	J4	8-pin connector for test only.	Not Used.
	J5	2-pin twisted pair carries +12Vdc and ground.	Not Used.
	J6	10-pin dual-row ribbon cable carrying transmit, receive, power and ground.	Connects to ELO (display) interface board.
	J7	6-pin single-row ribbon cable, for COM1 serial port.	Not Used.
	J8	10-pin dual-row ribbon cable carrying transmit, receive, power and ground.	Not Used.
	J9	6-pin connector carrying power and ground for display backlight.	Connects to Display Backlight.
	J10	31-pin connection carrying pixel data to display panel.	Connects to LCD Display Panel.
	J11	4-pin connector for IC2 bus.	Not Used.
	J12	26-pin dual-row connector for programming FLASH and NVRAM data.	Connects to FACTORY PROGRAMMER.
	J13	5-pin connector.	Not Used.
	J14	72-pin expansion connector.	Not Used.
	J15	72-pin SIMM connector.	8 Megabyte RAM board installed.
	J16	50-pin ribbon cable carrying analog and digital signals, and ground.	Connects to J1 on PWA Main Board.
	J17	8 pin Molex connector carrying low voltage power and ground.	Connects to J3 on PWA Main Board.
Switches	SW1	U2 Master CLEAR	Not installed.
	SW2	LCD Display Image Rotation Control	See note on schematic.
	SW3	RESET	Not installed.

Table 5-2: Controller Board Component Identifiers

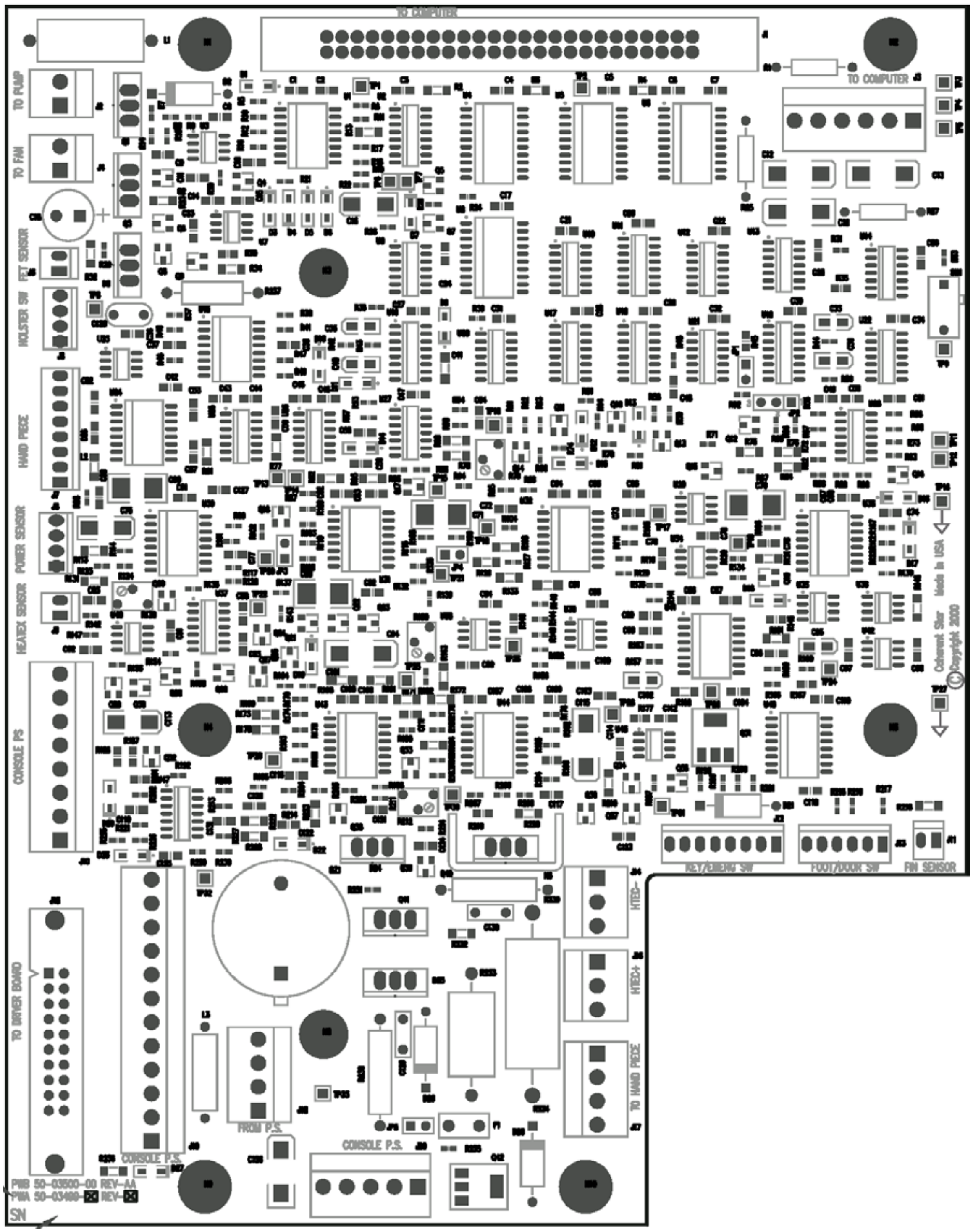


Figure 5-8: PWA Main Board

PWA Main Board			
Connectors	J1	50-pin ribbon cable carrying analog and digital signals, and ground.	Connects to J16 on CPU Controller Board.
	J2	2-pin Molex connector carrying power to the main heat exchanger pump.	Connects to coolant pump.
	J3	6-pin Molex connector carrying low voltage power and ground to CPU Controller Board.	Connects to J17 on CPU Controller Board.
	J4	2-pin Molex connector carrying power to the cooling fan filter board.	Connects to Fan Filter Board.
	J5	FET Temp Sensor (not used)	
	J6	4-pin AMP connector carrying power to and signal from the holster proximity sensor.	Connects to Proximity Sensor in top cover.
	J7	8-pin AMP connector carrying handpiece trigger signals, diode temperature and EPI temperature signals.	Connects to the Umbilical Cord (handpiece).
	J8	4-pin AMP connector carrying signal from external energy meter	Connects to the Energy Meter Board on the top cover.
	J9	2-pin AMP connector carrying temperature signal from the Main Heat Exchanger heat sink.	Connects to a thermistor on the Heat Sink just below coolant outlet fitting, or in-line at the outlet fitting.
	J10	8-pin Molex connector carrying low voltage direct current power from the power supply to the Main Board.	Connects to Power Supply (and High Voltage Relay on systems with the Lambda Power Supply).
	J11	2-pin AMP connector carrying temperature signal from the heat sink cooling fins.	Connects to a thermistor on the Heat Sink cooling fins.
	J12	8-pin AMP connector connection carrying key switch and emergency stop switch signals.	Connects to key and emergency stop switches at front panel.
	J13	6-pin AMP connector connection carrying remote interlock and footswitch signals.	Connects to interlock and foot switch at rear panel.
	J14	3-pin Molex connector carrying power for the Main Heat Exchanger TEC Modules.	Connects to the Negative side of the Main Heat Exchanger TEC Modules.
	J15	20-pin ribbon cable carrying signal to the FET Drive Board.	Connects to J4 on FET Driver Board. (not used on systems with the Martek Power Supply).
	J16	3-pin Molex connector carrying power for the Main Heat	Connects to the Positive side of the Main Heat Exchanger TEC Modules.

		Exchanger TEC Modules.	
	J17	4-pin Molex connector carrying power for the Lockhart/TEC Module and the EPI loop coolant pump.	Connects to the Umbilical Cord (handpiece).
	J18	4-pin Molex connector carrying power from the Power Supply to the Main Board.	Connects the 16V TEC Power Supply to the Main Board.
	J19	12-pin Molex connector – power bus.	Jumper installed.
	J20	4-pin Molex connector carrying power from the Power Supply to the Main Board.	Connects the 16Vdc TEC Power Supply to the Main Board.
Test Points	TP1	LSR_I_CMD	Laser Current Command from CPU Controller Board, 0.1V/A.
	TP2	/PULSE	Pulse Width Command from CPU Controller Board, 30ms or selected.
	TP3	+5V Filtered	+5Vdc Filtered
	TP4	+5V Unfiltered	+5Vdc Unfiltered
	TP5	-12V Filtered	-12Vdc Filtered
	TP6	Ground	Can be shorted to J7 to reset all flip-flops (reset latched faults).
	TP7	/RESET	Low for short time at power up to reset all flip-flops (reset latched faults).
	TP8	+7.5V Reference Voltage	+7.5Vdc Reference Voltage
	TP9	+12V Filtered	+12Vdc Filtered
	TP10	Fluence Limit Set Point (R79)	
	TP11	Electronic Shutter Comparator Limit (jump to TP12 during test)	5V on comparator means 50Vdc on FETs with Lambda Power Supply (or 50Vdc in Martek Power Supply). If voltage across FET is >50Vdc, laser diodes are OFF, and shutter is CLOSED.
	TP12	4.7kΩ resistor to ground in voltage divider, for Shutter Test.	Jump to TP11 during Shutter test. Test simulates ~20Vdc across FETs, making it appear that the laser diodes are ON, and the shutter is OPEN.
	TP13	Input to A Array Laser Current Integrator Amplifier	0.1V/A
	TP14	A Array Integrated Laser Energy	V/J
	TP15	B Array Integrated Laser Energy	V/J
	TP16	Ground	Ground
	TP17	ARY_TMP	Temperature of Diode Laser Back Plane Heat Exchanger, 0.1V/ºC.
	TP18	EPI_TMP	Actual EPI (ChillTip) Temperature, 0.1Vdc/ºC.
	TP19	Pulse Width Reference Signal	-0.53Vdc for Long Pulse, -1.63Vdc for Short Pulse.
	TP20	BOARD_REV	1.50Vdc for 4th Generation Main Board.
	TP21	Input to B Array Laser Current Integrator Amplifier	0.1V/A

	TP22	DRV_PLS	Laser Pulse Drive Signal to FET Board (if Lambda Power Supply) or Martek Power Supply.
	TP23	EPI (ChillTip) Temperature Offset	0.03Vdc/ºC (nominally –8.2Vdc).
	TP24	HEATX_I	Average DC voltage proportional to the Current in the Main Heat Exchanger TEC Modules, 2A/V.
	TP25	EPI (ChillTip) Temperature Offset	0.03Vdc/ºC (nominally –8.2Vdc).
	TP26	EPI_I	Average DC voltage proportional to the Current in the Lockhart/TEC Module, 2A/V.
	TP27	Ground	Ground
	TP28	HEATX_TMP	Actual Coolant Temperature as measured by the thermistor either at the Main Heat Exchanger coolant output fitting, or in-line in the coolant output fitting, 0.1Vdc/ºC.
	TP29	Pulse Width Hardware Limit	6.67Vdc for Long Pulse, 5.85Vdc for Short Pulse.
	TP30	Heat Exchanger Temperature Offset	0.03Vdc/ºC (nominally –8.2Vdc).
	TP31	HEATX_DRV	DC Voltage proportional to PWM controller (the higher the temp from 19ºC, the higher the drive voltage).
	TP32	Buzzer Drive	Buzzer generates two tones.
	TP33	TEC_PS2_+	16Vdc
Potentiometers	R79	Fluence Limit Set Point	See Section 3.
	R138	Energy Detector (photodiode) Gain	See Section 3.
	R150	EPI (ChillTip) Temperature Offset Calibration	See Section 3.
	R212	Heat Exchanger Temperature Offset Calibration	See Section 3.
Switches	SW1	Service Switch	Normally Open. Close to activate Service Mode software.
Jumpers	JP1	Drive Pulse Logic (default = jumped)	Pins 1-2 jumped = EPI Temp Shift when trigger pressed (5ºC → 2ºC) Pins 1-2 open = No Shift
	JP2	Pulse Width Control (default = Computer Control)	Pins 1-2 jumped = Computer Control Pins 2-3 jumped = Long Pulse Pins 1-2-3 open = Short Pulse
	JP3	Single/Dual Diode Stack Select (default = dual stack)	Pins 1-2 jumped = dual stack Pins 1-2 open = single stack
	JP4	Single/Dual Diode Stack Select (default = dual stack)	Pins 1-2 jumped = dual stack Pins 1-2 open = single stack

Table 5-3: PWA Main Board Component Identifiers

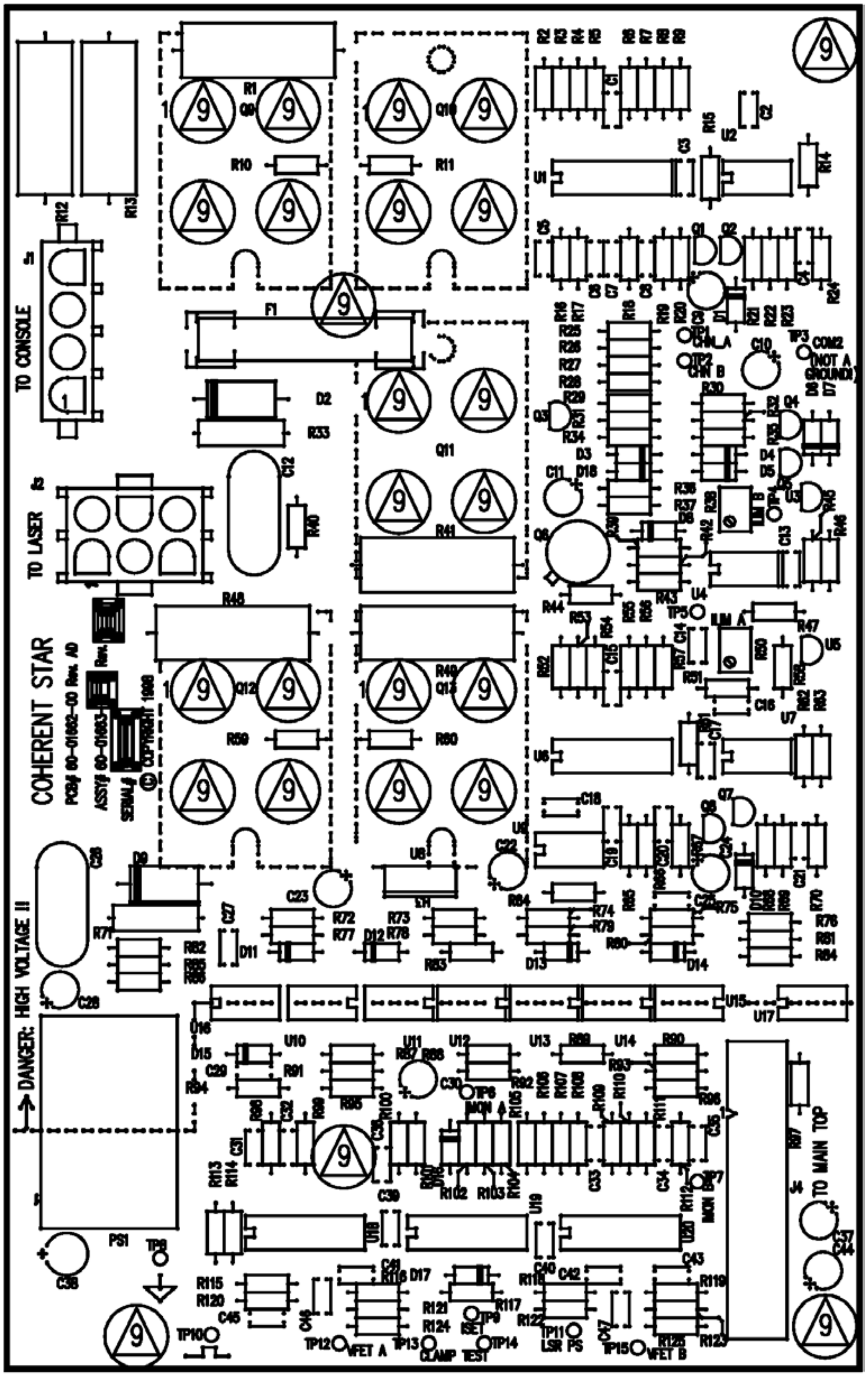


Figure 5-9: PWA FET Driver Board

FET Driver Board		Note: this board not used on systems with the Martek Power Supply.	
Connectors	J1	4-pin AMP connector carrying power from the Capacitor Bank on the Laser Power Supply.	For Lambda Power Supply.
	J2	6-pin AMP connector carrying current to the laser diode arrays in the handpiece.	Connected to the Umbilical Cord (handpiece).
	J4	20-pin ribbon cable carrying signal from Main Board.	Connected to J1 on the PWA Main Board.

Table 5-4: FET Driver Board Component Identifiers

Note: FETs not used on systems with the Martek Power Supply.

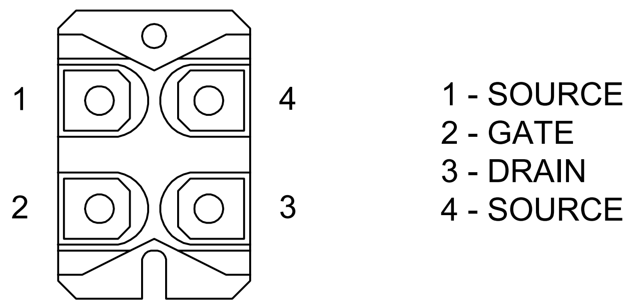


Figure 5-10: FET Connections

FET Troubleshooting

a. Remove the electrical connections of the FET under test (Q9, Q10, Q11, Q12, Q13)
b. Connect a jumper between terminals 1 & 4 (Source) on the FET.
c. Set the DVM for diode measurement and connect the leads to terminals 1 & 4 (Source).
d. Move the negative lead of the DVM to terminal 2 (Gate). The reading on the DVM should read ∞Ω or OL (Open Loop = infinite impedance).
e. Move the negative lead of the DVM to terminal 3 (Drain). The reading on the DVM should read approximately 0.380 kΩ.
f. Connect the negative lead of the DVM to terminal 1 or 4 (Source).
g. Connect the DVM positive lead to terminal 2 (Gate). The DVM reading should read ∞Ω or OL (Open Loop = infinite impedance).
h. Move the positive lead to terminal 3 (Drain). The DVM should read <1 MΩ.
i. Check the impedance between terminal 2 & 3. The DVM should ∞Ω or OL (Open Loop = infinite impedance) regardless of probe polarity.

Figure 5-11: Lambda Power Supply

Lambda Power Supply			
Connectors	Mains In	White - NEUTRAL Black – HOT (Line) Grn/Yel – EARTH (GND)	Blue and Brown wires from Circuit Breaker (Main Power Switch), Green/Yellow wire from chassis ground stud.
	High Voltage	Pins 1 & 2 Pins 3 & 4	-112V Return -112Vdc Connects to Capacitors
	TEC4	Pins 1 & 2 Pins 3 & 4	+16.5Vdc +16.5V Return Connects to J18 on PWA Main Board.
	TEC5	Pins 2 & 3 Pins 4 & 5	+16.5Vdc +16.5V Return Connects to J20 on PWA Main Board.
	Logic	Pins 1 & 2 Pins 3 & 4 Pin 5 Pin 6 Pin 7 Pin 8	+5Vdc +5 and ±12Vdc Return +12Vdc -12Vdc HV Relay Relay Return Connects to J10 on PWA Main Board, and HV Relay.

Table 5-6: Lambda Power Supply Signal List

Figure 5-12: Martek Power Supply

Martek Power Supply			
Connectors	Mains In	Blu - NEUTRAL Brn - HOT Grn/Yel – EARTH GROUND	Blue and Brown wires from Circuit Breaker (Main Power Switch), Green/Yellow wire from chassis ground stud.
	Low Volts Cable Out	+5Vdc, ±12Vdc and Ground	Connects to J10, J18 and J20 of PWA Main Board.
	Laser Drive (at front of power supply)	6-pin cable to Laser Diodes	Connects to Umbilical Cord (handpiece).
	Control Cable (ribbon cable pin numbers follow)		
	1, 3	Power input to present linear regulator	+ 5V Input, Internally connected.
	5	Power input to present linear regulator	+ 12V Input, Internally connected.
	7	Power input to present linear regulator	- 12V Input, Internally connected.
	2, 4, 6	Common	Common. Do not connect to Ground.

	17	DRV PLS	Input Gate for Laser Pulse Trigger.
	18	/SHUTOFF	Input, Inhibit PS. Low or open shuts off laser PS.
	8	DRV ENB	Enable PS. High enables laser driver.
	16	ISETPOINT	Input, 20A/V Gated Current Control. Current Command.
	15	VFET_Mon_B	10V/V output.
	14	Laser_PS	20V/V output.
	19	I-Mon_A	10A/V output. Current Monitor for the A Laser Stack.
	20	I_Mon_B	10A/V output. Current Monitor for the B Laser Stack.
	9	Laser Con	Output, Connector status. High indicates laser is connected to the laser PS.
	11	/Drive_CW	Output, Excess current request. Not used. Set to logic high.
	10	/ILIM_A	Output Over-current Channel A.
	12	/ILIM_B	Over-current Channel B Output.
	13	VFET_MON_A	10V/V output.

Table 5-7: Martek Power Supply Signal List

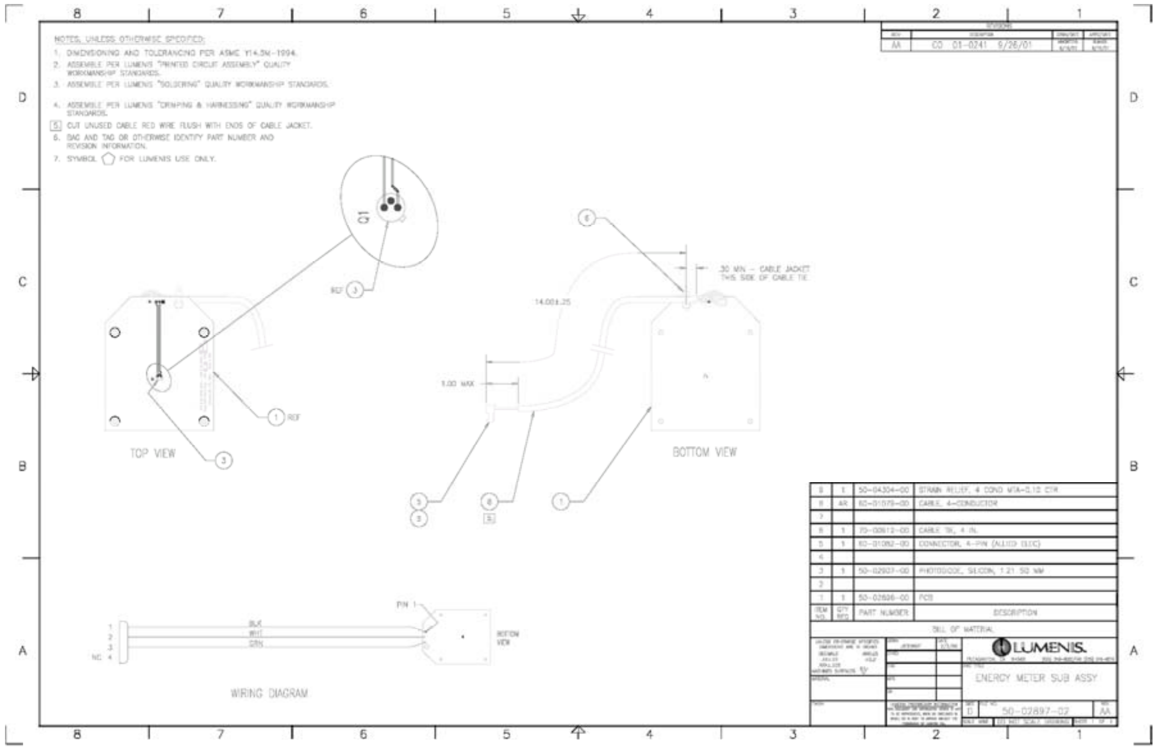


Figure 5-13: Energy Meter Board

Energy Meter Board			
Connectors	J1 (4-pin)	Energy Signal and Shield	To J8 on PWA Main Board.
Component	Q1	Photodiode Energy Detector	To J1 cable (on same board).

Table 5-8: Energy Meter Board Component Identifiers

5.5 Remove and Replace Procedures

Please contact Lumenis Product Technical Support for more information.

(this page intentionally blank)

LUMENIS
Enhancing Life. Advancing Technology.

6.0 Selected Part Numbers

The Bill-of-Materials for the LightSheer ET/ST Diode Laser System is maintained under document control by Lumenis Inc., and is subject to change without notice. The following list of parts is provided for convenience. Always confirm the part number for a given part with Product Technical Support before ordering.

Note: The ITEM NUMERS listed on the following pages are for parts identification only! When ordering, always specify the PART NUMBER, in the format xx-xxxxx-xx.

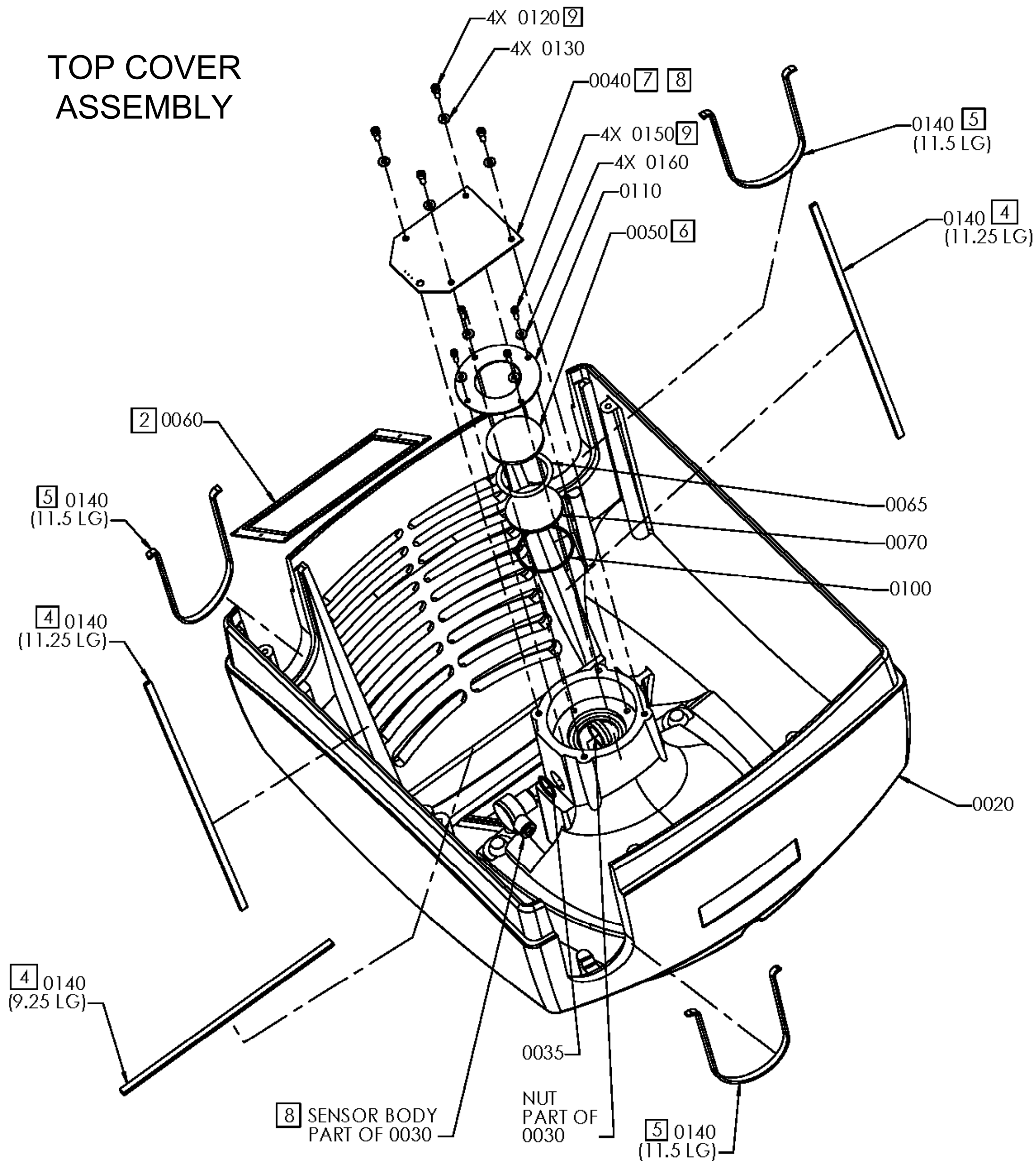
6.1 Miscellaneous Parts / Quick List

Laser Safety Eyewear (Glasses)	50-04082-00
Patient Eye Shields	50-04083-00
790-830nm Laser Danger Door Sign	10-01600-02
LightSheer ET/ST User Manual, English	10-03656-04
Accessory Cart, USA (Ships to US Addresses Only)	50-03929-xx
Accessory Cart, Int'l (unassembled, needs Drawer & Tray)	50-03600-xx
Drawer & Tray (for International Accessory Cart)	50-03931-xx
Foot Pedal Assembly, Pneumatic	50-02807-xx
Interlock Plug	60-00805-xx
Key	50-04192-xx
Power Cord, 125V	50-03607-xx
Power Cord, 250V	50-03606-xx

6.2 Special Service Tools

Service Manual	10-04719-00
Power Meter Adapter Fixture	90-04658-00
Digital Voltmeter with true RMS capability (such as Fluke 77)	supply locally
Temperature Probe (such as Fluke 80TK with surface sensor)	supply locally
Current Probe (such as Fluke 80i-110s)	supply locally
Firing Box (an enclosed space to trap laser light)	call Technical Support
Dummy Load (handpiece simulation)	call Technical Support
Frequency Counter (such as HP 5315A)	supply locally
20 MHz Oscilloscope with probes	supply locally
Laser Energy Measurement Equipment: Coherent FieldMaster GS with LM-40J Detector Head, or Molectron EPM-1000 with PM150-50C Head	supply locally

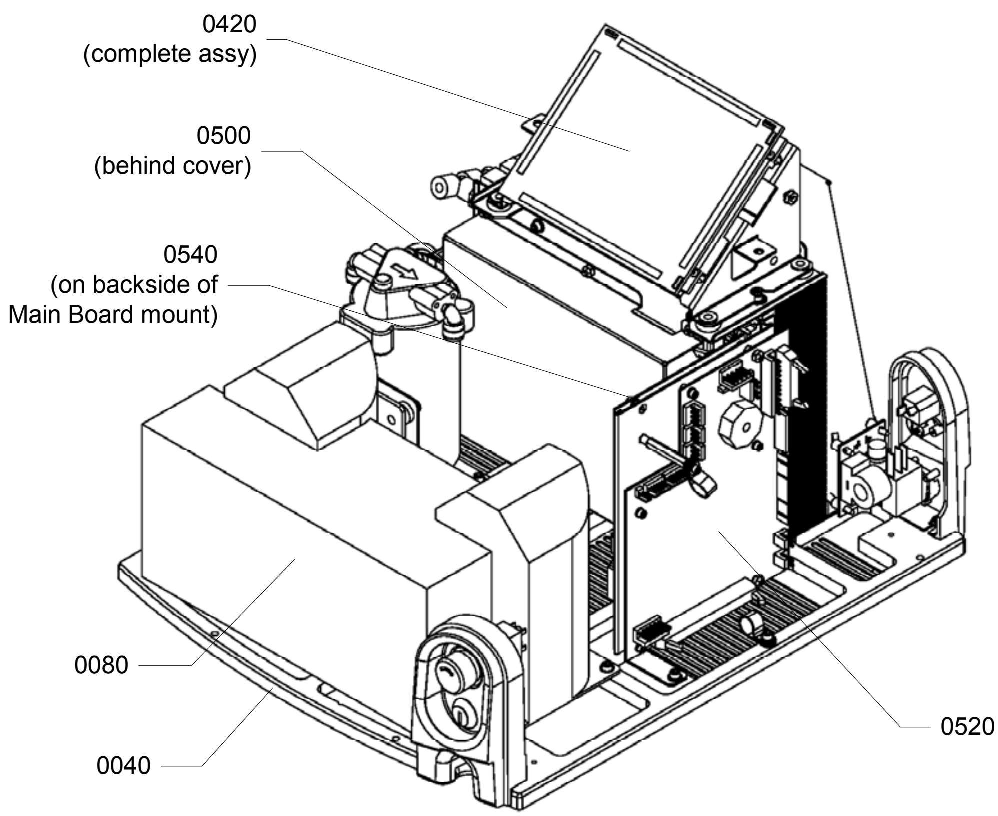
6.3 Top Cover Assembly

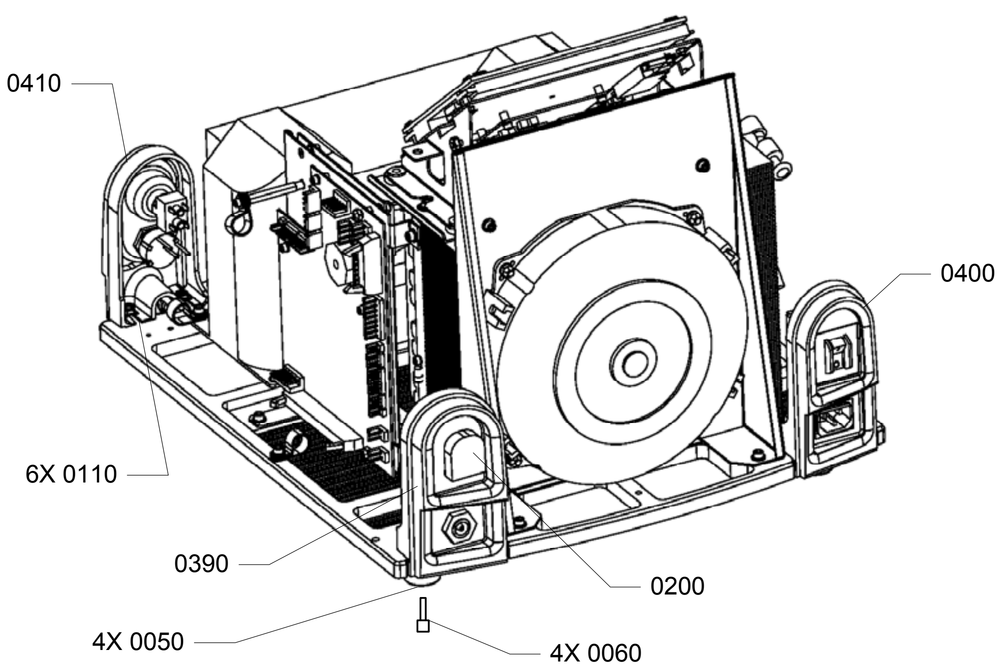


Part List for Top Cover Assembly

Item No.	Qty (unit)	Description	Part Number	Note(s)
0020*	1 (each)	Cover, Gray	50-03738-00	See below for matching labels
0020*	1 (each)	Cover, Green	50-03740-00	See below for matching labels
0020*	1 (each)	Cover, Red	50-03739-00	See below for matching labels
0020*	1 (each)	Cover, Blue	50-03741-00	See below for matching labels
0030	1 (each)	Sensor, Proximity	50-03612-01	
0035	1 (each)	O-Ring	50-03737-00	
0040	1 (each)	Energy Meter Assembly	50-03646-00	With Filter
0050	1 (each)	Window, Glass, Opal	50-03613-00	Opalized surface to be in contact with spacer
0060	1 (each)	Gasket, Window	50-03501-00	
0065	1 (each)	Spacer	50-03810-00	
0070	1 (each)	Glass, Borosilicate	50-03614-00	
0100	1 (each)	O-Ring	50-01907-00	
0110	1 (each)	Ring, Retaining	50-03616-00	
0120	4 (each)	Screw, SHC 8-32x0.375" SS	50-02514-00	
0130		Washer, Flat #8 SS	70-00310-00	
0140	8 (foot)	Foam Strip, Self-Adhesive, 0.25" x 0.125"	50-03753-00	
0150	4 (each)	Screw, SHC 6-32x0.375" SS	50-02563-00	
0160	4 (each)	Washer, Flat #6 SS	70-01106-00	
N/A	1 (each)	Calibration Port Pocket	50-02820-01	Only comes in the gray color
N/A	1 (each)	ET Label, Gray	10-03729-00	
N/A	1 (each)	ET Label, Green	13-03729-00	
N/A	1 (each)	ET Label, Mauve	11-03729-00	For Red Cover
N/A	1 (each)	ET Label, Blue	12-03729-00	
N/A	1 (each)	ST Label, Gray	10-03599-00	
N/A	1 (each)	ST Label, Green	13-03599-00	
N/A	1 (each)	ST Label, Mauve	11-03599-00	For Red Cover
N/A	1 (each)	ST Label, Blue	12-03599-00	
N/A	1 (each)	Label, Rear Panel	10-03643-02	

6.4 Base Plate Assembly

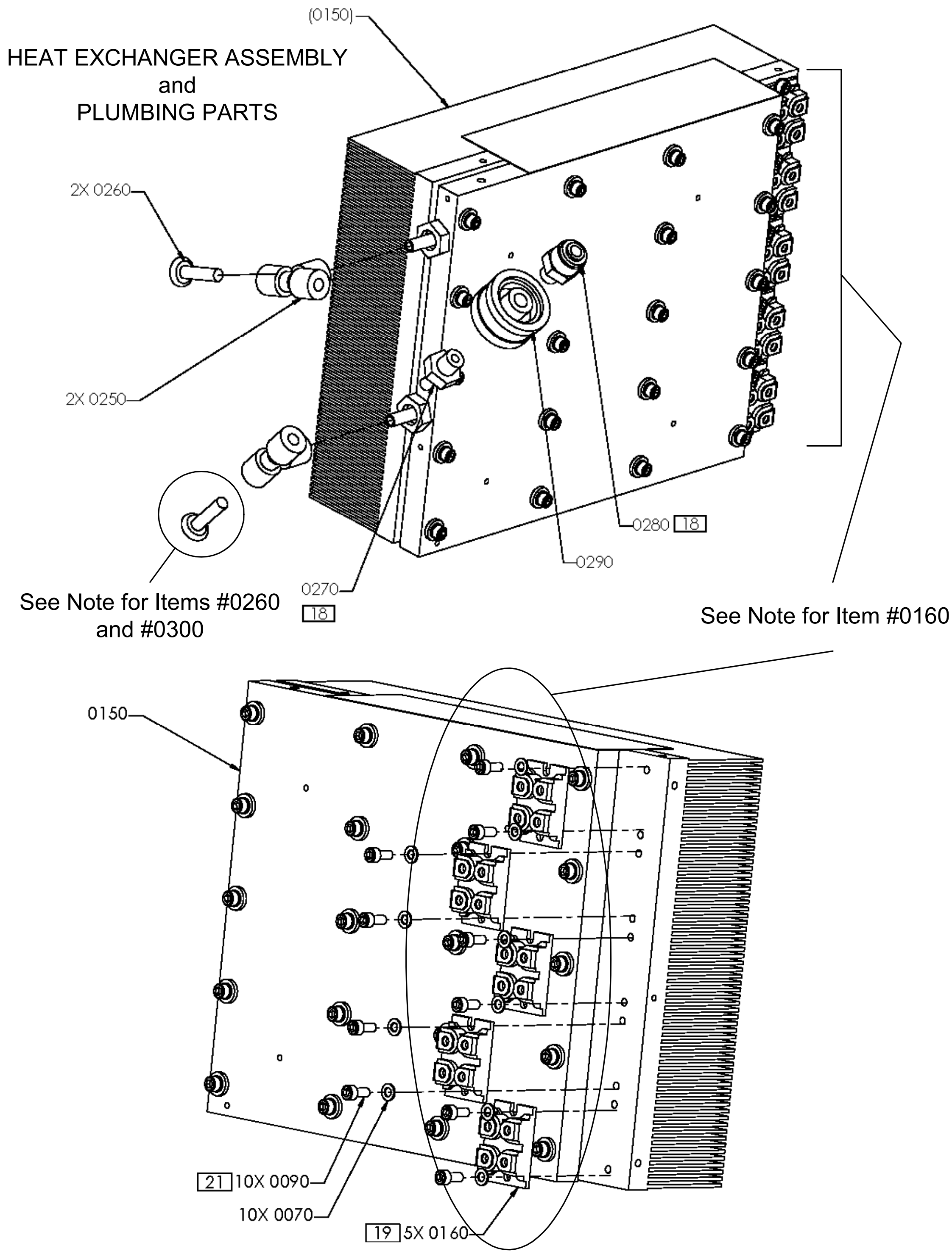




Part List for Base Plate Assembly

Item No.	Qty (unit)	Description	Part Number	Note(s)
0040	1 (each)	Base Plate	50-03618-00	
0050	4 (each)	Foot, Rubber	50-03619-00	
0060	4 (each)	Screw, 8-32x1" SS	70-01559-00	
0080*	1 (each)	Power Supply, Martek	50-03892-00	Replaces Lambda Power Supply, See FSB#1274
0080*	0	Power Supply, Lambda	obsolete	Replaced by Martek Power Supply, See FSB#1274
0110	6 (each)	Screw, FlangeHead	50-03812-00	Secures Items 0390, 0400, and 0410 to Base Plate
0200	1 (each)	Interlock Plug	60-00805-00	Part of Item #0390
0390	1 (each)	Assembly, Interlock/Footswitch	50-03996-00	Panel with Interlock Jack and Footswitch (includes Item 0200)
0400	1 (each)	Assembly, Power Input/Circuit Breaker	50-03623-00	Panel with Input Power Module and 12A Circuit Breaker
0410	1 (each)	Assembly, Key/E-Stop	50-03625-00	Panel with Key switch and E-Stop Button (no key)
0420	1 (each)	Assembly, Display, Touch	50-03626-01	Complete Assembly with Touch Panel, LCD Display, Backlight and Electronics
0500*	1 (each)	PWA CPU Controller Board, ST	50-04262-00	Pre-programmed to work on ST class systems
0500*	1 (each)	PWA CPU Controller Board, ET	50-04261-00	Pre-programmed to work on ET class systems
0520	1 (each)	PWA Main Board	50-03499-05	
0540	1 (each)	PWA FET Driver Board	60-01663-02	Only used on systems with Lambda Power Supply
N/A	1 (each)	Key	50-04192-00	A126 key with Lumenis logo

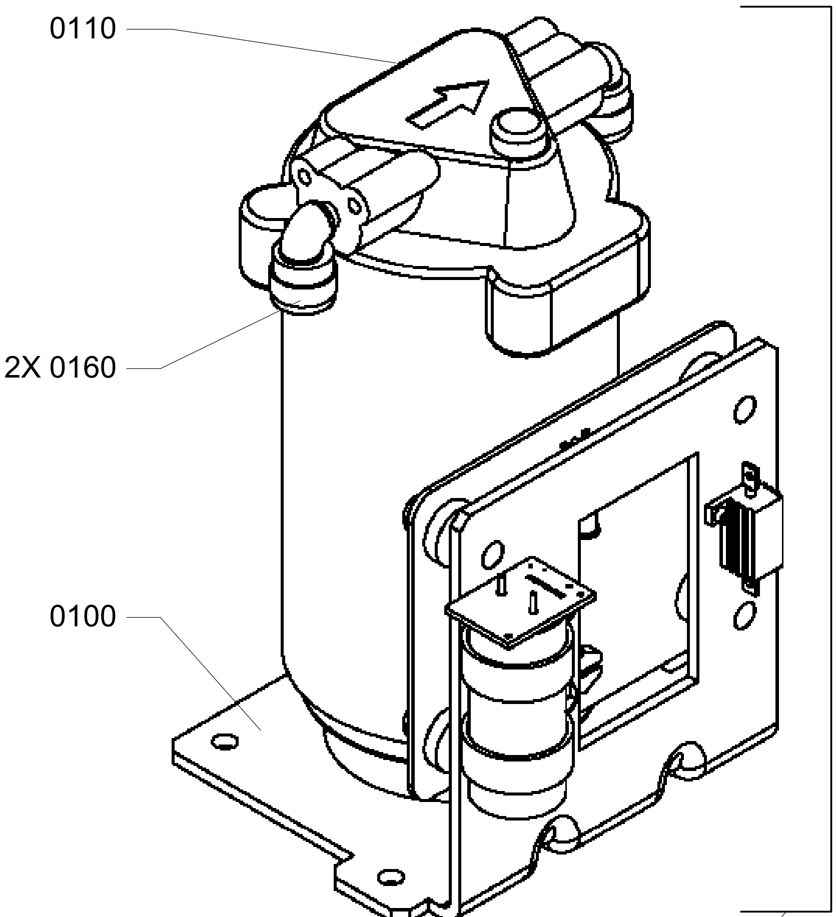
6.5 Heat Exchanger Assembly and Attached Parts



Part List for Heat Exchanger Assembly and Plumbing Parts

Item No.	Qty (unit)	Description	Part Number	Note(s)
0070	10 (each)	Washer, Flat #8 SS	70-01707-00	Only used at this location on systems with FETs
0090	10 (each)	Screw, SHC 8-32X0.375" SS	50-02516-00	Only used at this location on systems with FETs
0150	1 (each)	Heat Exchanger Assembly (also called 240W Chiller Assy w/Offset Heatsink)	50-03620-01	Consists of Heat Sink, TEC Modules, Cold Plate, In/Out Fittings, Temp Sensor (Fan), and Temp Sensor (Heat Exchanger)
0160	5 (each)	FET	60-00618-00	FETs only used on systems with the Lambda Power Supply
0250	2 (each)	Tee, Union, ¼"	50-02192-00	
0260	2* (each)	Plug, ¼"	50-02191-00	*See Note for Item #0300
0270	1 (each)	Fitting, Adapter	70-01126-00	
0280	1 (each)	Fitting, Adapter	50-03751-00	
0290	1 (each)	Filter, Particle, In-line	70-01537-00	
0300 (not shown)	1 (each)	Sensor, Temp, In-line	50-04699-01	May be used in place of Item #0260 at Cold Plate Output Fitting (the bottom one) – See TN#1278

6.6 Pump Assembly

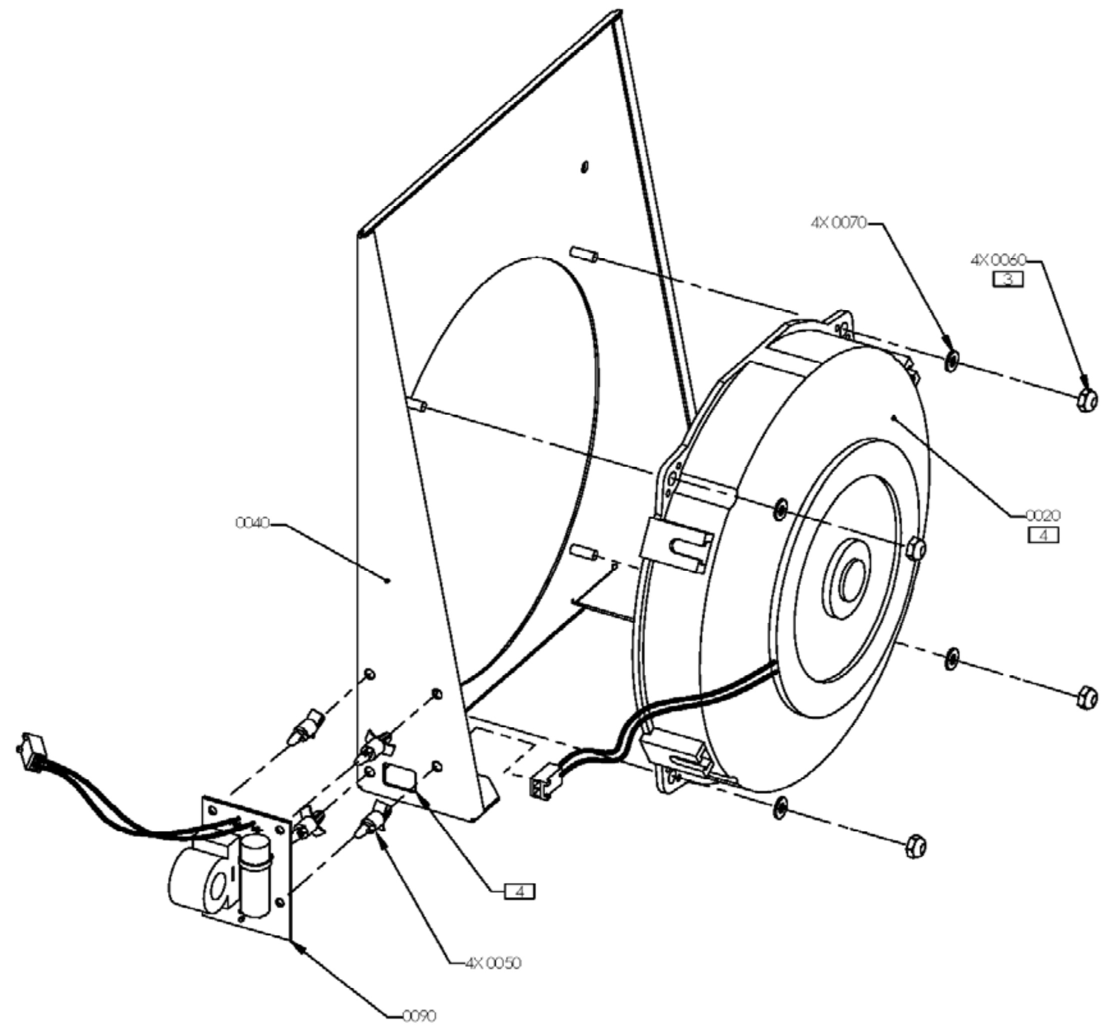


50-03644-00
Pump Assembly, Complete
(wires not shown)

Part List for Pump Assembly

Item No.	Qty (unit)	Description	Part Number	Note(s)
N/A	1 (each)	Pump Assembly, Complete	50-03644-00	Assembly is completely wired, with connectors
0100	1 (each)	Bracket, Mounting, Pump	50-03424-00	
0110	1 (each)	Pump	50-03187-00	
0160	2 (each)	Fitting, Elbow, ¼”	50-03752-00	

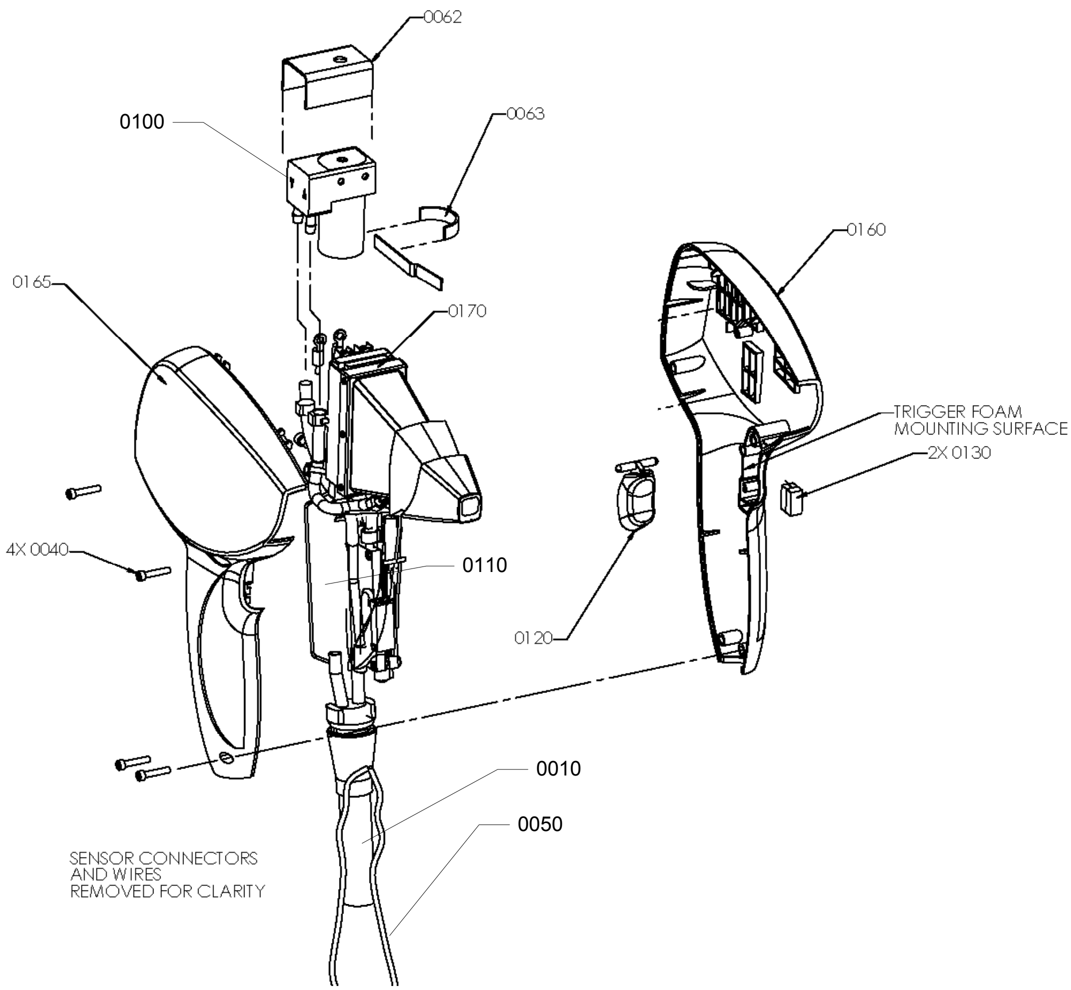
6.7 Cooling Fan Assembly



Part List for Cooling Fan Assembly

Item No.	Qty (unit)	Description	Part Number	Note(s)
N/A	1 (each)	Assembly, Fan, Cooling	50-03633-00	Complete as shown, wired with connector
0020	1 (each)	Fan	50-03634-00	
0040	1 (each)	Bracket, Fan	50-03428-00	
0050	4 (each)	Stand-off	50-03765-00	
0060	4 (each)	Nut, Lock, 8-32 SS	70-00599-00	
0070	4 (each)	Washer, Flat #8 SS	70-00310-00	
0090	1 (each)	PWA Fan Filter Board	50-04091-00	

6.8 Handpiece Assembly



Part List for Handpiece Assembly

Item No.	Qty (unit)	Description	Part Number	Note(s)
N/A	1 (each)	Assembly, Handpiece, 9x9mm, 70 bar	50-03439-00	Complete Handpiece with Umbilical Cord (does not include wrist strap)
0010	1 (each)	Cord, Umbilical	50-03505-xx	
0040	4 (each)	Screw, 4-40x0.5625” SS	70-01610-00	
0050	1 (each)	Strap, Wrist	70-01429-00	
0062	1 (each)	Foam, Pump	50-01884-00	
0063	1 (yard)	Tape, High Bond	70-01377-00	
0100	1 (each)	Pump	50-03181-xx	
0110	1 (each)	Module, Lockhart/TEC	50-03442-xx	
0120	1 (each)	Button, Trigger	50-02822-00	
0130	2 (each)	Foam, Trigger	70-00181-00	
0160	1 (each)	Covers, Handpiece	50-02355-00	Includes #0165
0170	1 (each)	Assembly, Back Plane/Laser Diode/Condenser/Tip	50-03440-01	

Part List for Handpiece Repair Kit (not shown)

Item No.	Qty (unit)	Description	Part Number	Note(s)
N/A	1 (each)	Handpiece Repair Kit	50-03777-00	Complete Kit that contains the parts listed below
N/A	1 (foot)	Tubing, Plastic	50-02385-00	
N/A	4 (each)	Clamp, Hose, 6.6mm	70-00982-00	
N/A	6 (each)	Clamp, Hose, 6.1mm	70-00991-00	
N/A	6 (each)	Clamp, Hose, 7.0mm	70-01269-00	
N/A	1 (each)	Foam, Pump	50-01884-00	
N/A	2 (each)	Foam, Pump, Top & Bottom	50-01915-00	
N/A	6 (each)	Foam, Condensor, Lower	70-00179-00	
N/A	2 (each)	Foam, Trigger	70-00181-00	
N/A	2 (each)	Conn, Crimp, Male	50-01908-00	
N/A	1 (each)	Strap, Wrist	70-01429-00	
N/A	1 (each)	Cover, Handpiece, Left	50-02584-00	
N/A	1 (each)	Cover, Handpiece, Right	50-02585-00	
N/A	1 (each)	Button, Trigger	50-02822-00	
N/A	1 (each)	Plunger, Pin, Switch	80-01220-00	
N/A	1 (each)	Syringe, Disposable, 5cc	70-01381-00	
N/A	1 (each)	Tip, Syringe, 0.023	40-03057-00	

6.9 Cable Harnesses

Item No.	Qty (unit)	Description	Part Number	Note(s)
N/A	1 (each)	Video Display Cable	50-03629-00	
N/A	1 (each)	10 Conductor Ribbon Cable	50-03845-00	
N/A	1 (each)	Display Inverter Power Cable	50-03628-00	
N/A	1 (each)	50 Conductor Ribbon Cable	50-03640-00	
N/A	1 (each)	Computer Power Cable	50-03621-00	
N/A	1 (each)	20 Conductor Ribbon Cable	50-03844-00	

7.0 Technical Notes and Field Change Orders

7.1 Introduction

The information in this Service Manual is subject to change without notice. As required, Lumenis Technical Support Department releases Technical Notes (TN) and/or Field Change Orders (FCO) to update its Field Service personnel on technical issues concerning this product. At their release, the documents become part of the Service Manual.

Place all distributed Technical Notes and Field Change Orders for this product Service Manual behind this page.

7.2 Index of TN and FCO

TN/FCO Number	Date	Description	Systems Affected
1274	27-Aug-2002	Martek Power Supply Upgrade	Any ET and ST system with original Lambda Power Supply
1277	11-Mar-2003	Martek Power Supply Information	Any ET and ST system with Martek Power Supply
1278	11-Mar-2003	New Main Heat Exchanger Temp. Sensor	All ET, ST and XC systems
1279	11-Mar-2003	Handpiece Desiccant Information	All LightSheer models

(this page intentionally blank)

Lumenis Pleasanton, Field Service Bulletin # 1274

August 27, 2002

Martek Power Supply

Objective:

The objective of this bulletin is to instruct trained and certified Field Service Engineers (FSE) on how to replace the Lambda power supply with the Martek power supply when required.

This procedure must be performed under ISO safety regulations and in an ESD controlled environment.

Brief Background:

The Martek power supply will be used to replace the Lambda power supply on the tabletop models (ST/ET). The Driver PC board is incorporated into the Martek. The FSE will need to remove the existing Driver board from the system during this procedure.

Ordering Information:

Use your existing procedures on ordering the parts as required.
Martek Power Supply: P/N: 50-03892

Required Tools:

1 X Standard Tool Kit

Reference Documents:

Lightsheer ST/ET Service Manual P/N: TBD

Affected Equipment:

Lambda Power Supply: P/N: 50-03183
Driver PC Boards: P/N: 60-01663

Instructions:

1. Turn off and unplug system.
2. Remove top cover from system and set aside.
3. Disconnect all wires from the Lambda supply to the system.
4. Disconnect all wires from the Driver Board to the system.
5. Remove the Lambda supply from the system and set aside.
6. Remove the Driver Board and set aside.
   -If this is a known good Driver Board, save it in an ESD shielded bag for future use.
7. Install the new Martek supply in system.
8. Connect the cables from the Martek supply to the system.
9. Connect the AC input from the Martek supply to the AC circuit breaker.

10. Re-install the Top Panel on the system.

Calibration Procedures:

1. Chiller Test:	30-04790-xx
2. Verification Test:	30-01436-xx
3. Final Test:	30-01563-xx

END OF PROCEDURE

Disposition of removed parts:

1. Send Lambda Power Supply to Depot for disposition.
2. If the Driver Board is good, keep in an ESD Shielded bag for future use, otherwise, return to Depot.

Contacts:

If you have any questions regarding the above bulletin, please contact the Product Support Department at Lumenis Pleasanton.

Phone Number: 1-925-249-8000

Fax Number: 1-925-249-8010

END OF PROCEDURE

Issued By: Terry Heaton Signature: ____________________ Date: _____________

Approved By: Signature: ____________________ Date: _____________

LUMENIS
Enhancing Life. Advancing Technology.

Pleasanton Service Department
1249 Quarry Lane, Suite 100
Pleasanton, CA 94566 USA
Tel: 925.249.8005
Fax: 925.249.8010

TN	Technical Note Proprietary and Confidential			Product LightSheer ET/ST
TN No. 1277	Issue Date March 11, 2003	Rev. A	Rev. Date March 11, 2003	Written By: Derek Rutz Signature:
Page 1 of 1	TN Title **Martek Power Supply Information**			Approved By: Keir Eilersen Signature:

Purpose

The purpose of this Technical Note is to inform the field depot repair technicians of two fault conditions that may be encountered on any LightSheer ET or ST Diode Laser System having a Martek 600W Power Supply (Part No. 50-03892) installed.

Background

The Martek 600W Power Supply (Part No. 50-03892) has been installed on all production systems and has been available as a field replacement part for the original Lambda Power Supply since Q3 2002 (see FSB#1274).

Recently, two intermittent fault conditions have been discovered which have been associated with the Martek Power Supply:

1. Intermittent Over Current Faults (E6/E7)

   The intermittent Over Current fault is caused by a part tolerance problem within some of the Martek Power Supplies. The manufacturer has since corrected the problem, and all in-house power supplies have been upgraded. For field systems, see corrective procedure below.

2. Intermittent Shutter Error – Failed to Close (E5)

   The intermittent Shutter Error is caused by using the foot pedal as a trigger (instead of holding the pedal down and using the handpiece switch as the trigger). The voltage measurement of the Martek Power Supply is done differently by system electronics which can make it appear that the power supply is causing the generation of this fault.

Procedure

For the Over Current fault, check the manufacturers label on the power supply chassis (a black and white label on the top left). Power supplies that have been already corrected are labeled: "PS2303 Rev.B", or later revision. If Rev. A or A1 *and* the system experiences intermittent Over Current faults, change the supply, p/n 50-03892.

For the Shutter Error – Failed to Close fault, kindly instruct the User to cycle power to clear the fault, then press *and* hold the foot pedal down and to fire the laser using the handpiece button. The foot pedal is configured as the fire enable/disable signal, and the handpiece button is configured as the fire pulse trigger. Remember: the foot pedal is not the trigger! No parts required.

Pleasanton Service Department
1249 Quarry Lane, Suite 100
Pleasanton, CA 94566 USA
Tel: 925.249.8005
Fax: 925.249.8010

TN	Technical Note Proprietary and Confidential			Product LightSheer ET/ST and XC
TN No. **1278**	Issue Date March 11, 2003	Rev. A	Rev. Date March 11, 2003	Written By: Derek Rutz Signature:
Page 1 of 2	TN Title **In-line Main Heat Exchanger Temp. Sensor**			Approved By: Keir Eilersen Signature:

Purpose

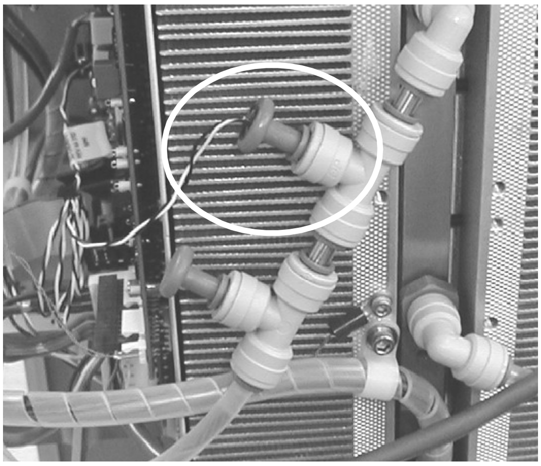
The purpose of this Technical Note is to inform the field depot repair technicians of a new in-line Main Heat Exchanger temperature sensor and associated temperature calibration procedure now used on LightSheer ET/ST and XC Diode Laser Systems.

Background

The original Main Heat Exchanger temperature sensor was contact mounted just below the cold plate outlet fitting. It was discovered that the temperature measurement was not accurate as some warm exhaust air (from the cooling fan) was warming the sensor. In an attempt to eliminate the warm air from artificially being measured as warm coolant, the sensor was covered with RTV. In any case, some systems would still exhibit P5 Errors.

To more accurately measure the coolant temperature, an In-line Temperature Sensor is now being used in production and is available as a field replacement part. The part number for the In-Line Temperature Sensor is 50-04649-01. Installation and calibration instructions are provided below.

on ET/ST systems



on XC system

Procedure

Installation of In-line Temperature Sensor

Pleasanton Service Department
1249 Quarry Lane, Suite 100
Pleasanton, CA 94566 USA
Tel: 925.249.8005
Fax: 925.249.8010

1. Disconnect the twisted pair wire from the existing temperature sensor at the Main Boar, J9.

2. Remove the plug at the heat exchanger output coolant fitting Tee and insert the new in-line temperature sensor into the fitting. If any coolant is spilled, refill the coolant loop per standard procedure.

3. Route the new twisted pair wire harness from the in-line temperature sensor to the Main Board, and connect to J9.

4. Calibrate the new in-line temperature sensor (see next section).

Calibrate Heat Exchanger Temperature Controller

1. From Diagnostic screen, verify that EPI Cooler and EPI set-point are OFF. Turn the Heat Exchanger ON, and wait 5 minutes for the handpiece temperature to stabilize.

2. Slowly adjust potentiometer R212 on the Main Board until the Heat Ex. Temp. and Diode Temp. are within 0.5º C of each other. Recheck after 2 minutes and readjust R212 as necessary.

3. Allow the heat exchanger to operate an additional 3 minutes. Make no potentiometer adjustments during this period. Confirm that the Heat Ex. Temp. and Diode Temp. remain within 0.5º C of each other.

4. With an external temperature probe, measure the heat exchanger temperature at the coolant outlet fitting (the bottom fitting on the ET/ST systems, or the top fitting on the XC system). Confirm that the external temperature measurement is within 2.0º C of the Heat Ex. Temp.

   Note: The external temperature measurement may yield false readings if the probe is not closely coupled to the heat exchanger near the coolant output fitting. It may be necessary to make a few readings to find an average temperature. Since the in-line sensor is bathed in the coolant, it is far more sensitive to immediate changes in coolant temperature than the heat exchanger itself. Step 4 is really just a confirmation that the heat exchanger temperature measurement is close to reality.

5. If software version is 4.01 or higher, double touch "Get Δ Temp" on the Diagnostic screen. The temperature difference must be 0 ± 1.0º C.

End.

Pleasanton Service Department
1249 Quarry Lane, Suite 100
Pleasanton, CA 94566 USA
Tel: 925.249.8005
Fax: 925.249.8010

TN	Technical Note Proprietary and Confidential			Product(s) all LightSheer
TN No. **1279**	Issue Date March 13, 2003	Rev. A	Rev. Date March 13, 2003	Written By: Derek Rutz Signature:
Page 1 of 1	TN Title **Handpiece Desiccant Information**			Approved By: Keir Eilersen Signature:

Purpose

The purpose of this Technical Note is to inform field depot repair technicians of an engineering change order that calls for the discontinued of use of desiccant blocks in all new and refurbished handpieces as well as parts contained within the handpiece repair kit.

Background

Desiccant blocks were originally installed in all handpieces with the purpose of eliminating moisture (condensation) within the handpiece as a result of cold ChillTip temperatures (temperatures below dew-point). Engineering tests comparing handpieces with and without desiccant blocks have shown that diode lifetime and output energy are unaffected by the presence or absence of the desiccant. As such, the desiccant blocks have been removed from all handpiece assemblies and repair kits as of Q4 2002. The following points justify desiccant removal.

- When installed, the desiccant looses effectiveness (becomes saturated) in 3 to 11 months (depending upon local environment). Once saturated, the desiccant contributes more to the moisture level (and subsequent condensation) within the handpiece than if it were not there. Studies have found minimal (negligible) reduction in output energy in handpieces with internal condensation.

- Desiccant takes up space within the handpiece making assembly difficult and increasing the likelihood of damage to other components during re-assembly. In some cases, the desiccant becomes loose, requiring the handpiece to be repaired for rattling noises. Dust released from loose desiccant blocks has proven to cause diode failure.

- During the test period spanning 2001 and 2002, over 130 handpieces (of both 9mm and 12mm size tips) were built without desiccant. Upon examination, no failures in these handpieces were caused by lack of desiccant.

There may be an increase in the number of doctors who complain of condensation inside the tip, even though the condensation may be present with or without desiccant. Please use the information given above to show the advantages of desiccant removal.

Procedure

In handpiece kit instructions, remove all references to desiccant. No parts required.

8.0 Drawings and Schematics

This section includes a selection of schematic diagrams and drawings for the LightSheer ET/ST Diode Laser System as produced at the release of this manual.

Name	Drawing Number	Page Number(s)
Simplified System Block Diagram	N/A	8-3
System Wiring Diagram (for systems with Lambda Power Supply)	25-03610-01	8-4
System Wiring Diagram (for systems with Martek Power Supply)	50-04650-01	8-5
CPU Controller Board Schematic	50-02784-05	8-6 through 8-13
PWA Main Board Schematic	50-03499-04	8-14 through 8-16
PWA FET Driver Board Schematic (for systems with Lambda Power Supply)	60-01662-00	8-17
PWA Energy Meter Board Schematic	50-02897-02	8-18
PWA Fan Filter Schematic	50-04091-00	8-19

(this page intentionally blank)